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Frequency of feeding behaviour on growth performance of walking cat fish, *Clarias batrachus* (Linn.)

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Abstract:- The study was conducted to assess the impact of different feeding rates and feeding frequencies on the growth and survival of fingerlings of Walking Cat Fish, *Clarias batrachus*. Test fish were fed with supplementary diet composed of rice bran, mustard meal, meat cum bone meal & soyabean meal (in equal proportions) @10% (FR_1), 20% (FR_2), 30% (FR_3), 40% (FR_4), 50% (FR_5) & 60% (FR_6) body weight. The experiment was conducted for 90 days where 20% feeding rate (FR_2) resulted in maximum body weight gain of fish, when reared indoor. Likewise, specific growth rate as well as protein efficiency ratio was also significantly ($P<0.05$) highest in (FR_2). In another experiment, the fishes were fed at three feeding frequencies *viz*. Once a day (FF_1), twice a day (FF_2) and thrice a week (FF_3) for 60 days. The maximum increase in body weight was observed in (FF_1). No significant differences were observed for water quality parameters with respect to different feeding rates and feeding frequencies. It was concluded that feeding rate @ 20% body weight once a day is appropriate for indoor rearing of fingerlings of *Clarias batrachus* (LINN).

Key words: Specific Growth Rate, Protein Efficiency Ratio, *Clarias batrachus*, Feeding Rate, Feeding Frequency.

INTRODUCTION

The Walking Cat Fish (*Clarias batrachus*) is one of the most preferred food fish for the consumers of Asian Countries due to its good taste, less bones and medicinal qualities. Feed is the most expensive part of Cat Fish production. As feed cost occupies 60-70% of the total operational expenditure in fish culture, Selection of appropriate feeding rate as well as frequency are of primary concern, which otherwise increase the input cost to significant level. For the larval stages, live food mainly Zooplanktons are given along with supplementary feed to satiation[1] and for later stages (fry and fingerlings), the feed amount as well as the feeding frequency needs to be defined so as to avoid wastage. Time of feeding, feeding rate and feeding frequency affect the feed intake and its utilization and growth performance of different fish species such as *Heteropneustes fossilis*, Channel Cat Fish, *Ictalurus punctatus*[2] *Chanos chanos*[3] and African Cat Fish, *Clarias gariepinus*[4]. The present study was conducted to evaluate the effect of different feeding rates and feeding frequencies on growth of fingerlings of *Clarias batrachus*.

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MATERIALS & METHODS

Experiments were conducted in plastic pools of 960 capacities. The pools were prepared 2-3 days before the

stocking of fish. At the base, 4cm thick soil bed was provided and water level of 20 cm was maintained. Azolla (An aquatic weed) was spread on the surface and small pieces of PVC pipes were placed in each pool to provide shelter for fish. The fish were reared in stored ground water and aeration was provided through aerator stones, attached to a compressor.

For both the experiments, pools were manured with cow dung and poultry dropping @ 300g (1:1) and initial doze and 40g (1:1) at fortnightly intervals. Manure was soaked for 2 hours in water and then the filtered water was added to the ponds.

The supplementary feed used in both the experiments was composed of deoiled rice bran, deoiled mustard cake, meat cum bone meal and soyabean meal (each and equal proportion). The finely ground ingredients and prepared diets were analyzed for proximate composition and gross energy[5,6] (Table-1).

For first experiments, ten fingerlings of *Clarias batrachus* were stocked in each pool. The average length and weight of fish was 8.29±0.64 cm and 4.75±0.90g respectively, at the time of stocking. Fish were fed with supplementary diet @ 10%(FR_1), 20% (FR_2), 30%(FR_3), 40%(FR_4), 50%(FR_5) & 60%(FR_6) body weight of fish once a day at 16:00-16:30 hr.

In second experiments, ten fingerlings of *Clarias batrachus* were stocked in each pool. The average length and weight of fish was 18.17±0.5 cm and 48.03±7.90g respectively, at the time of stocking. Fish were fed with supplementary feed @ 20% fish body weight once a day (FF_1) at 4:00pm, twice a day (FF_2) at 10:00am and 4:00pm and thrice a week (FF_3) at 4:00pm.

Water of pools was replaced every week after taking the water samples of analysis.

Water samples were taken in the morning hours (9:00am) and were analyzed for temperature, pH and dissolved oxygen[7].

Survival of fish was determined by comparing the number of fish present at the termination of experiment with that of fish stocked. Growth of fish was assessed by measuring body length and weight. Data was analyzed for the following parameters:

A.	Increase in length (%)	=	final body length - initial body length/initial body length x 100
B.	Increase in weight (%)	=	final body weight - initial body length/initial body weight x 100
C.	Specific growth rate	=	1_n final body weight - 1_n initial body weight/culture period x 100
D.	Protein efficiency ratio	=	fish weight gain(g)/protein intake(g)

Data was analyzed statistically using one way ANOVA

RESULTS AND DISCUSSION

Experiment.1

The effects of different feeding rates were observed on the water quality parameters, Survival and growth of *Clarias batrachus*. The studied revealed that during the experimental period no significant differences in different treatment were observed for water quality parameters. Survival of fish recorded was 100% in all the treatments. Percent increase in weight was maximum in FR_2 followed by FR_4, FR_6, FR_1, FR_5 and FR_3 both SGR (1.24) and per (0.29) were observed maximum in FR_2 (Table-2)

Experiment.2:

Effects of different feeding frequencies were observed in water quality parameters, Survival and growth of *Clarias batrachus*. The studied revealed that during the experimental period no significant differences in different treatment were observed for water quality parameters. Hundred percent survivals of fish were recorded in all the treatments. Percentage increase in weight was maximum in FF_1 followed by FF_3 and FF_2 respectively. PER was observed maximum (0.17) in FF_1 (Table-3).

Table-1: Proximate composition of different ingredients and prepared diet (on dry weight basis)

Ingredients	Crude protein (%)	Ether extract (%)	Nitrogen free extract (%)	Energy (k cal/g)	P/E ratio (mg protein/cal)
Rice bran*	22.75	2.87	39.01	2.94	77.38
Mustard meal*	42.0	7.09	33.56	4.09	102.68
Meat cum bone meal	20.34	6.90	38.86	3.19	63.76
Soyabean meal*	4.29	1.75	37.411	4.10	120.21
Prepared diet	33.60	4.66	37.28	3.59	93.59

**Solvent Extracted*

Table-2: Water quality parameters and growth of *Clarias batrachus* fed at different feeding rates.

Parameters	FR_1	FR_2	FR_3	FR_4	FR_5	FR_6
Temp. (^oC)	$20.60^a \pm 1.16$	$20.72^a \pm 1.15$	$20.84^a \pm 1.14$	$20.96^a \pm 1.19$	$21.08^a \pm 1.19$	$21.17^a \pm 1.20$
pH	$10.01^a \pm 0.12$	$9.61^{ab} \pm 1.16$	$9.26^a \pm 1.10$	$9.25^b \pm 0.11$	$9.19^b \pm 0.12$	$9.31^b \pm 0.09$
D.O (mgl^{-1})	$5.97^a \pm 0.57$	$5.50^a \pm 0.58$	$2.53^b \pm 0.36$	$2.46^b \pm 0.23$	$2.48^b \pm 0.29$	$2.81^b \pm 0.36$
Initial body length	$8.22^a \pm 0.64$	$8.40^a \pm 0.57$	$8.10^a \pm 0.73$	$8.44^a \pm 0.48$	$8.26^a \pm 0.47$	$8.34^a \pm 0.73$
Final body length	$10.28^{ab} \pm 0.62$	$12.26^a \pm 0.15$	$9.96^b \pm 0.43$	$10.92^{ab} \pm 0.53$	$10.60^{ab} \pm 0.59$	$10.54^{ab} \pm 0.54$
% Change in length	25.06	45.95	22.96	29.38	28.32	26.37
Initial body length	$4.63^a \pm 1.04$	$4.56^a \pm 0.89$	$4.89^b \pm 0.07$	$4.74^a \pm 0.69$	$4.83^a \pm 0.85$	$4.87^a \pm 1.07$
Final body length	$8.72^{ab} \pm 1.41$	$13.46^a \pm 0.74$	$8.47^b \pm 0.27$	$10.89^{ab} \pm 1.40$	$10.12^{ab} \pm 1.85$	$10.08^{ab} \pm 1.21$
% Change in length	88.33	195.17	73.21	129.74	83.00	106.98
Protein Efficiency rate	0.034	0.036	0.011	0.016	0.008	0.007
Specific growth rate	0.49	1.24	0.46	0.58	0.54	0.54

Table-3: Water quality parameters and growth of *Clarias batrachus* fed at different feeding frequencies.

Parameters	FF_1	FF_2	FF_3
Temp.(^oC)	24.20 ± 1.07^a	24.20 ± 1.07^a	24.20 ± 1.07^a
pH	7.64 ± 0.07^a	7.42 ± 0.04^a	7.45 ± 0.07^a
D.O ($mg\ l^{-1}$)	1.86 ± 0.56^a	1.44 ± 0.67^a	1.78 ± 0.38^a
Total alkalinity ($mg\ l^{-1}$)	305.20 ± 65.42^a	385.20 ± 60.33^a	3418.40 ± 18.34^a
Initial body length	17.53 ± 1.58^a	18.73 ± 0.78^a	18.87 ± 0.73^a
Final body length	18.56 ± 1.18^a	18.43 ± 0.78^a	18.93 ± 0.64^a
% Change in length	5.6	1.4	0.3
Initial body length	47.03 ± 12.28^a	47.10 ± 4.69^a	49.96 ± 5.01^a
Final body length	52.24 ± 10.87^a	47.20 ± 6.30^a	50.55 ± 7.68^a
% Change in length	11.07	0.21	1.18
Protein Efficiency rate	0.173	0.003	0.019

The present study revealed that feeding @20% FBW once a day resulted in higher growth of *Clarias batrachus*, when reared indoor. Feeding at lower (10%) and higher (>20%) rates resulted in reduced growth. Not much work has been done on the feeding rates of *Clarias batrachus*. Ng *et.al.* (2000)[8] reported feeding rates of 2.5% FWB peer day for bagrid Cat Fish (*Mystus numerous*) at the best.

There was decrease in growth parameters when the frequency was increased from once a day to twice a day. The growth was also very low when the frequency was decreased from once a day to thrice a week. These results are in contrast with that of channel Cat Fish, *Ictalurus punctatus* where feeding once a day was found to be similar in terms of growth when fed twice a day[9]. The study by Jarboe and Grant (1997)[10] indicated that feeding small fingerlings of channel Cat Fish once daily and grow out channel Cat Fish twice daily gave the best results, in term of growth.

For different fish species, different results observed with respect to feeding frequency[3]. Feeding once a day was found to be appropriate in stinging Cat Fish, *Heteropneustes fossilis*[11]; twice per day for channel Cat Fish, *Ictalurus punctatus*[12]; four times a day for Himri Barbel, *Barbus luteus* Heckel[13]. For *Clarias batrachus*, it could be concluded that feeding @ 20% FBW once a day resulted in best growth in indoor rearing experiments.

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Interspecific variations in biochemical parameters of *Clarias batrachus* (Linnaeus, 1758) and *Clarias gariepinus* (Burchell, 1822): A comparative study

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Abstract: *Clarias batrachus* (Linnaeus, 1758) commonly called as 'Magur' is Indian native (endogenous) species has similar morphological features as *Clarias gariepinus* (Burchell,1822), which is deadly carnivorous exotic African species introduced to India through various trade means. The indigenous species *C. batrachus* is highly preferred by Indian and other Asiatic consumers because of its high content of arginine and lysine in its flesh and medicinal value. The exotic species *C. gariepinus* has undergone tremendous adaptation to the Indian physicochemical condition resulting in mass substitution of the native species *C. batrachus* from the Indian fish market. The aim of the present study was to investigate and establish reference ranges by comparing biochemical values of two fresh water air breathing species of family Clariidae i.e. *C. batrachus* and *C. gariepinus* from the state Bihar. The study was aimed to investigate serum biochemical parameters like Liver Function Test (LFT) including Serum Glutamic Pyruvate Transaminase (SGPT), Serum Glutamic Oxaloacetate Transaminase (SGOT), Serum Alkaline Phosphatase (ALP), serum bilirubin, and Kidney Function Test (KFT) including serum Total Protein (TP), albumen, globulin, A/G ratio and serum creatinine and blood urea etc. The mean ± SEM and range were established. In the present study, the estimated biochemical values were compared between and within the two species of cat fish using One-Way Analysis of Variance (ANOVA). Nearly all most all parameters differ significantly ($p<0.05$) between *C. batrachus* and *C. gariepinus*. The present study will lead to the proper characterization of LFT and KFT parameters of these two closely allied species of family Clariidae and later on can be used as a sensitive index to monitor various physiological and pathological changes in fish due to xenobiotic stress. It will also help in indicating the health status of the fish and pollution status of the aquatic bodies. Further studies are needed to trace the genetic basis of the significant variation in the biochemical parameters of these two closely allied fresh water air breathing species.

Key words: Biochemical analysis, Blood, *Clarias batrachus*, *Clarias gariepinus*, KFT, LFT, Serum.

INTRODUCTION

Fisheries and aquaculture is a fast-growing sector in India, providing nutrition and food security to large population of country apart from generating income and employment to more than 14.5 million people (The Economic Times, Agriculture, July 04, 2019). Globally, India stands second in culture fisheries production. The fisheries sector plays an important role in the Indian economy and up-liftment of socio-economic status of fishing communities of state and foreign exchange earnings[1]. The fisheries resources in the state of Bihar is chiefly comprised of ponds, tanks, small reservoirs, rivers and water logged areas like ox-bow lakes and chaurs[2-3]. Around 65,000 ha of water areas are covered by ponds and tanks and nearly 35,000 ha of water areas consist of

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ox-bow lakes and chaurs[2]. The family Clariidae represents specialized group of air breathing fishes with high content of arginine and lysine in their flesh, high Vitamin D content, low level of omega-3 fatty acids and a much higher proportion of omega-6 fatty acids, prolific breeding potential and high therapeutic values. They are easy to handle in laboratory, leading to inexpensive and easy rearing demands. In Asiatic countries three species of *Clarias* (family Clariidae) are known to be cultivated by fish farmers as- *Clarias batrachus* (Linnaeus, 1758) in India[4]; *Clarias macrocephalus* in South- East Asia[5] and *Clarias fuscus* (Lucep'de,1803) in Taiwan and Hawaii[6-7].

Clarias batrachus, commonly known as 'magur' is a favourite edible fish with high medicinal value in India and other Asian countries including Bangladesh, Thailand, Vietnam, Malaysia and Indonesia as it is an endogenous Asian species[8-10]. The fish finds its origin in India and later has been shown to spread in the different parts of Globe[11-12]. Even four sub species of *Clarias batrachus* have been reported from South East Asia, Java and India as *C. batrachus; C. aff. batrachus "Indochina"; C. aff. batrachus "Sundaland" and Clarias magur* (Hamilton, 1822)[13]. In India declining trend in the population of the *C. batrachus* is partly contributed by habitat degradation and overfishing, but the major cause is the introduction of a morphologically alike and deadly carnivores African species *Clarias gariepinus* (*Burchell,*1822)[14]. This exotic species have been reported to feed on the native species and also hybridize naturally with them[15]. Due to excessive predation on the native species, this exotic species have generated a potential threat for the survival of *Clarias batrachus* and accordingly the Ministry of agriculture in India put ban on the culture and import of *C. gariepinus* in 1997[16]. But due to prolific breeding ability, excessive acclimatization in the Indian condition and simplicity in rearing, the fisherman in different states of India preferably culture this fish, setting aside the ban and restriction imposed by the ministry of agriculture, Government of India.[17]

Blood parameters analysis has been used widely to determine the systematic relationship among species of fish[18]. Blood parameters have also been considered as indicators of the physiological condition or sub-lethal stress response in fish exposed to internal or external factors[19]. The physiological status of fish can be evaluated authentically by hematological indices. Changes in hematological parameters depend upon the aquatic biotope, fish species, age, and sexual maturity and health status[20].

Blood of fishes comprises 1.3-7% of the total body weight and that is why it is considered as one of the mostactive components that contribute to metabolic processes by ensuring gas exchange between the organism and the environment. They serve to transport all kinds of nutrients, enzymes, hormones, neurotransmitters and all metabolic intermediaries in the different parts of body, apart from helping in transporting nitrogenous wastes outside the body of the fish.

The principle objective of the present study is to characterize a baseline concrete biochemical parameters of *Clarias batrachus* and *Clarias gariepinus,* which can be later used as reference point. It will also generate an authenticated base line data, which can be applied for monitoring of health and disease status, reproduction and ecotoxicological studies. The finding of the present study can be correlated with the advanced mitochondrial genome analysis of these two closely allied species of family *Clariidae.*

MATERIALS AND METHODS

Biological sampling and Acclimatization: Adult specimen of *C. batrachus* ranging from 60-110 gm and size between 4.6"-7.1" and *C.gariepinus* ranging from 120-240gm and size between 5.2"-9.2"were collected from two different fish farms from Purnia, Bihar, India. Fishes were brought to laboratory and disinfected with 0.1% $KMnO_4$ solution. They were segregated as per their size and species in different sized large plexi glass aquaria (capacity 50L, 80 L and 100L) having dechlorinated, aerated tap water at normal temperature and pressure. Fishes were acclimatized to the ideal laboratory condition for 15 days as per standard method[21]. Fishes were fed *ad libitum* and then fresh water was changed regularly in the morning hours of every day.

Collection of Blood Sample: The peripheral blood was collected by puncturing the genital opening with heparin treated 21 gaugex 0.5 inch needle fitted with 2 ml syringe. The collected blood samples were quickly transferred into collection tubes containing EDTA and were gently mixed by carefully turning it upside down. The collection tubes were then kept in the sampling boxes containing ice.

Biochemical Analysis: Blood samples were centrifuged in Remi centrifuge at 3000 rpm for 15 minutes in order to get the serum. Then serum was carefully siphoned out from centrifuged blood of both the species. The serum was used for estimation of Liver Function Test (LFT) and Kidney Function Test (KFT) respectively.

Liver function test: For liver function test, the determination of Serum glutamate pyruvate transaminase (SGPT), Serum Glutamate Oxaloacetate Transaminase (SGOT) was done as per standard protocol[22]. Serum Alkaline Phosphate (ALP) was estimated by specific method[23] and Serum bilirubin was assessed as per standard protocol[24].

Kidney function test: For Kidney function test the determination of Total Protein (TP) was estimated by standard Biurette method[25-26], Serum Albumen (SA), Serum Globulin (SG) &, A/G ratio was estimated by standard BCG Method[27]. Serum creatinine and serum uric acid were estimated by standard Jeff method[28].

Statistical Analysis: For each blood samples, the data of each parameter was analyzed and presented as mean and standard error of mean. One way Analysis of Variance (ANOVA) test was done to observe the variation in different parameters between two species or even within species. All statistical analysis was done using SPSS.

RESULT

A slight morphological variation has been marked in both these species. The body colour of *Clarias batrachus* was dark greyish to greyish black, while in *C. gariepinus* it is silvery greyish black. The demarcating morphological feature was the shape of the occipital, which was broad and blunted in *C batrachus* while narrow and somewhat pointed in *C gariepinus*. Apart from that majority of the morphometric characteristics in both the species were nearly same as per the standard protocol of taxonomic identification of freshwater fish species[29-30].

Results of Liver Function Test

The value of various parameters of Liver function test has been represented in Table- I.

Table I: Various parameters of Liver Function Test of *Clarias batrachus* and *Clarias gariepinus*

Parameters	Sample size	*Clarias batrachus*		*Clarias gariepinus*		Calculated F value	Table F -value
		Mean	SEM	Mean	SEM		
Serum Glutamate Pyruvate Transaminase (SGPT = U/L)	20	92.6	12.58	47	7.99	9.35	5.31
Serum Glutamate Oxaloacetate Transaminase (SGOT = U/L)	20	167.6	24.31	34.8	7.28	27.3	5.31
Serum Alkaline Phosphate (ALP = U/L)	20	18.8	3.55	61.8	5.40	44.14	5.31
Serum Bilirubin (mg/dl)	20	2.02	0.53	1.39	0.22	1.15	5.31

Figures in each parameters have been shown as Mean ± SEM, the sample size in each in case is n = 20, significant at p < 0.05. One way analysis of Variance was done. Calculated value has been compared with table value.

Results of Kidney Function Test

The value of various parameters of Kidney function test has been represented in Table – II.

Table II: Various parameters of Kidney Function Test of *Clarias batrachus* and *Clarias gariepinus.*

Parameters	Sample size	*Clarias batrachus*		*Clarias gariepinus*		Calculated F value	Table F -value
		Mean	SEM	Mean	SEM		
Total Protein (gm/dl)	20	3.91	0.38	3.20	0.16	2.82	5.31
Serum Albumen (gm/dl))	20	2.02	0.53	1.39	0.22	1.15	5.31
Serum Globulin (gm/dl)	20	1.40	0.19	2.10	0.17	6.90	5.31
Serum Creatinine (mg/dl)	20	0.50	0.083	0.62	0.037	1.71	5.31
Serum Uric Acid (mg/dl)	20	2.46	0.30	2.99	0.18	2.15	5.31

Figures in each parameters have been shown as Mean ± SEM, the sample size in each in case is n = 20, significant at p < 0.05. One way analysis of Variance was done. Calculated value has been compared with table value

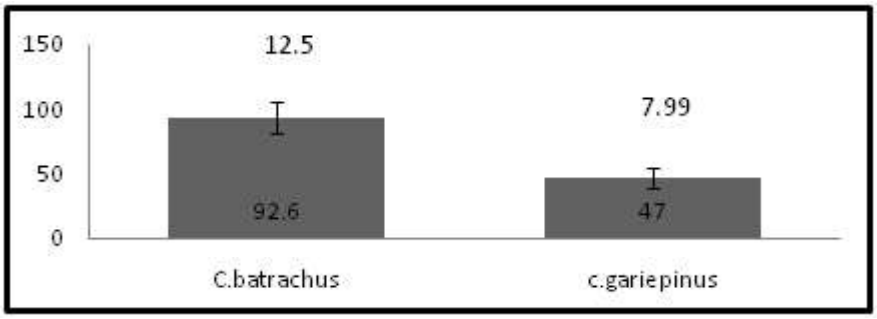


Fig.1: Histogram showing interspecific variation in SGPT (U/L) of *Clarias batrachus* and *Clarias gariepinus*

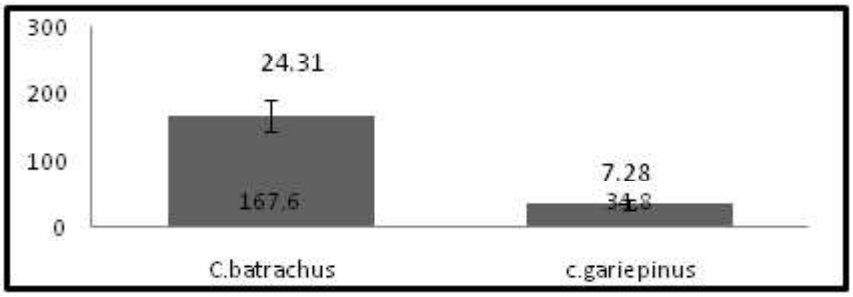


Fig.2: Histogram showing interspecific variation in SGOT (U/L) of *Clarias batrachus* and *Clarias gariepinus*



Fig.3: Histogram showing interspecific variation in ALP (U/L) of *Clarias batrachus* and *Clarias gariepinus*



Fig.4: Histogram showing interspecific variation in Serum bilirubin (mg/dl) of *Clarias batrachus* and *Clarias gariepinus*



Fig. 5: Histogram showing interspecific variation in Total protein (gm/dl) of *Clarias batrachus* and *Clarias gariepinus*

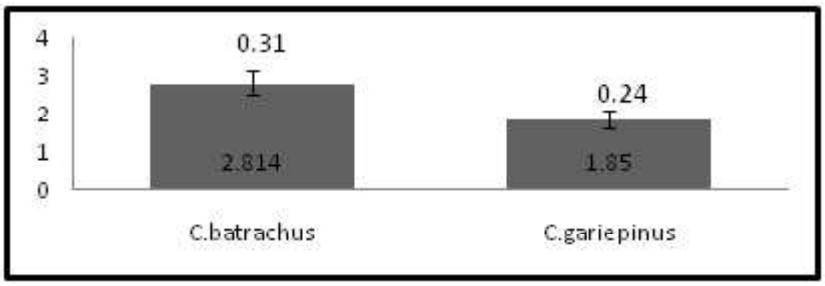


Fig. 6: Histogram showing variation in Serum albumin (gm/dl) of *Clarias batrachus* and *Clarias gariepinus*



Fig.7: Histogram showing variation in Serum globulin (gm/dl) of *Clarias batrachus* and *Clarias gariepinus*

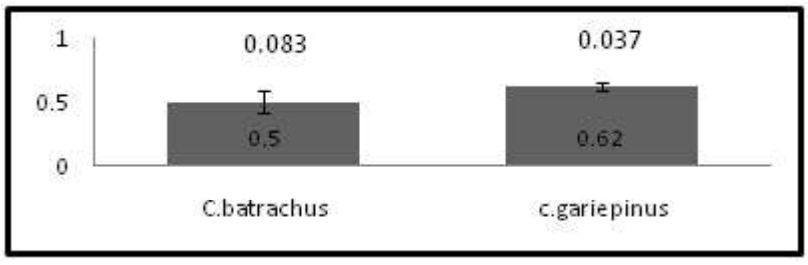


Fig. 8: Histogram showing variation in Serum creatinine (mg/dl) *of Clarias batrachus* and *Clarias gariepinus*

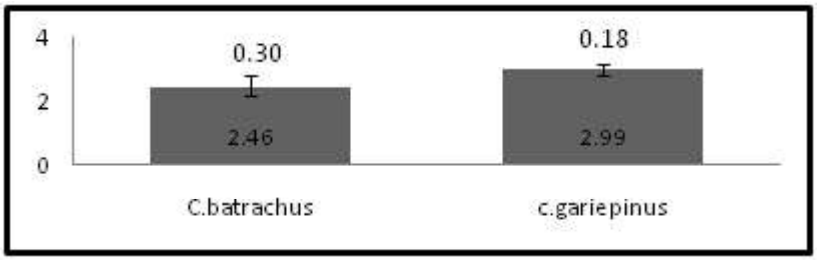


Fig. 9: Histogram showing variation in Serum uric acid (mg/dl) of *Clarias batrachus* and *Clarias gariepinus*

DISCUSSION

The central theme of the present investigation is to establish a baseline reference value of biochemical parameters of two closely allied species of family Clariidae i.e. *Clarias batrachus,* the indigenous species and *Clarias gariepinus*, the exotic species with special reference to Liver function test and kidney function test profile. The present study also deals with the biochemical variations in between these two species.

The blood of the fish is highly sensitive and truly reflects the health status of the fish. It is widely used to study the vital biological process taking place in a fish species[31]. Fish blood is being consistently been used in toxicological researches and environmental monitoring as a probable indicator of pathophysiological changes in effective fishery management[32]. Liver function test includes a battery of serological tests done for the initial detection and management of liver diseases. LFT serves three basic purposes- **a.** Assessing the potential of liver to transport organic anions and to metabolize drugs (including LFT parameters like serum bilirubin, urine bilirubin, urobilinogen etc.; **b.** Assessing the injury to hepatocytes (including LFT parameters like aminotransferases, alkaline phosphatase, glutamyltranspeptidase, 5 nucleotidase, leucineaminopeptidase etc.). **c.** Assessing the biosynthetic capacity of the liver (including LFT parameters like serum protein, albumen, prealbumen, serum ceruloplasmin, procollagen III peptide, α 1 antitrypsin, α feto protein etc.). In the present investigation SGPT, SGOT ALP and serum bilirubin has been considered under LFT profile.

Transamination represents one of the principal metabolic pathway for the synthesis and deamination of amino acids. Glutamic pyruvic transaminase (SGPT) or alanine amino transferases (ALT) catalyse the transfer of an amino group from L-alanine to α-ketoglutarate, the products of this reversible transamination reaction being pyruvate and L-glutamate. It helps the liver in converting food into energy. In the present study the level of SGPT in control *C. batrachus* and *C. gariepinus* have been recorded as 92.6±12.58 and 47±7.99 respectively. The calculated F value is 9.35, which is much higher than the table value *i.e.* 5.31. It clearly depicts that the liver metabolism and related hepatic impairment is higher in *C. gariepinus*. The transaminase level in different fishes has been earlier reported[33-34].

Aspartate Transaminase (AST) or aspartate amino transferase, also known as AspAT/ASAT/AAT or (serum) glutamic oxaloacetic transaminase (GOT, SGOT), is a pyridoxal phosphate (PLP)-dependent transaminase enzyme that catalyzes the reversible transfer of an α-amino group between aspartate and glutamate and, as such, is an important enzyme in amino acid metabolism. The level of SGOT in control *C. batrachus* and *C. gariepinus* have been recorded as 167.6 ± 24.31 and 34.8 ± 7.28 respectively. The calculated F value is 27.3, which is much higher than the table value i.e. 5.31. The difference is highly significant ($p<0.05$).

ALP is an enzyme found throughout the body, but it is mostly found in the liver, bones, kidneys, and digestive system. It plays an integral role in metabolism within the liver and development within the skeleton. When the liver is damaged, ALP may leak into the blood stream. High levels of ALP can indicate liver disease or bone disorders. The level of ALP in control *C. batrachus* and *C. gariepinus* have been recorded as 18.8±3.55 and 61.8±5.40 respectively. The calculated F value is 44.14, which is much higher than the table value i.e .5.31. The difference is highly significant ($p<0.05$).

Bilirubin (BR) is a yellow compound that occurs in the normal catabolic pathway that breaks down heme in vertebrates. This catabolism is a necessary process in the body's clearance of waste products that arise from the destruction of aged or abnormal red blood cells[35]. First the haemoglobin gets stripped of the heme molecule which thereafter passes through various processes of porphyrin catabolism, depending on the part of the body in which the breakdown occurs. The production of biliverdin from heme is the first major step in the catabolic pathway, after which the enzyme biliverdin reductase performs the second step, producing bilirubin from biliverdin[36-37]. The level of serum bilirubin (BR) in control *C. batrachus* and *C. gariepinus* have been recorded as 2.02±0.53 and 1.39±0.22 respectively. The calculated F value is 1.15, which is much lower than the table value i.e. 5.31.The difference is not significant at ($p<0.05$).

The Kidney Function Test (KFT) provides valuable information about the functional status of kidney and physiology of excretion and acid base balance. In general, nephrotoxicity can be generated by pre renal, renal and post renal causes. Serum total protein is estimated for

monitoring gross changes in protein levels marked in various pathological conditions. The accuracy of the test lies in association with the liver function test and protein electrophoresis. The level of serum total protein (TP) in control *C. batrachus* and *C. gariepinus* have been recorded as 3.91±0.38 and 3.20±0.16 respectively. The calculated F value is 2.82, which is much lower than the table value i.e.5.31. The difference is not significant at ($p<0.05$).The reference value of serum total protein in different group of teleost have been reported time to time as 2.02±2.37gm/dl in *Cyprinus carpio*[38], 4.55±0.13 gm/dl in *Clarias gariepinus*[39] and 3.99±0.99 gm/dl & 5.97±0.56 gm/dl in *Clarias batrachus*[40-41]. Plasma protein is the protein component of the blood and it increases with starvation and any other physiological stress[42]. Plasma protein gives an index of the health status of the brood fish[43] and serves as indicator of nutritional status[44].

Serum albumin is principally synthesized in the liver and maintains the osmotic pressure in the blood. It helps in transportation of lipids and general metabolism in fishes[45]. It also forms the part of amino acid pool. The rise in albumen concentration in animals due to loss through urine or faeces or through break down may result in impaired synthesis[46]. The level of serum albumin in control *C. batrachus* and *C. gariepinus* have been recorded as 2.02±0.53 and 1.39±0.22 respectively. The calculated F value is 1.15, which is much lower than the table value i.e.5.31. The difference is not significant at ($p<0.05$). The present finding is in agreement with earlier recorded reference value of serum albumen in different teleost as 1.083±0.82 gm/dl in *Clarias gariepinus*[39], 1.0±0.79 gm/dl in *Clarias batrachus*[41].

The globulin fraction includes hundreds of serum proteins including carrier proteins, enzymes, complement, and immunoglobulins. Most of these are synthesized in the liver, although the immunoglobulins are synthesized by plasma cells. Increase in the globulin fraction usually result from an increase in immunoglobulins and in some pathological condition. Malnutrition and congenital immune deficiency can cause a decrease in total globulins due to decreased synthesis, and nephrotic syndrome can cause a decrease due to protein loss through the kidney. The level of serum globulin in control *C. batrachus* and *C. gariepinus* have been recorded as 1.40±0.19gm/dl and 2.10±0.17gm/dl respectively. The calculated F value is 6.90, which is much higher than the table value i.e.5.31. It is significant at $p<0.05$.

Serum creatinine is an important indicator of kidney health because it is an easily measured byproduct of muscle metabolism that is excreted unchanged by the kidneys. Creatinine itself is produced via a biological system involving creatine, phosphocreatine and adenosine triphosphate (ATP) Creatine is synthesized primarily in the liver from the methylation of glycol-cyamine. It is then transported through blood to the other organs, muscle and brain, where through phosphorylation, it becomes the high-energy compound phosphocreatine. Creatine conversion to phosphocreatine is catalyzed by creatine kinase and in due course of reaction creatinine is formed spontaneously. The level of serum creatinine in control *C.batrachus* and *C. gariepinus* have been recorded as 0.50±0.083 mg/dl and 0.62±0.037mg/dl respectively. The calculated F value is 1.71, which is much lower than the table value *i.e.* 5.31. It is not significant at $p<0.05$.

Uric acid is the ultimate catabolite of purine metabolism. It is a weak organic acid that under physiologic conditions exists mainly as a monosodium salt. Serum uric acid reflects the interactions of four major processes:dietary purine intake, endogenous purine metabolism, urinary urate excretion, and intestinal uricolysis. Many additional factors, including exercise, diet, drugs, and state of hydration, may result in transient fluctuations of uric acid levels. The level of serum uric acid in control *C. batrachus* and *C. gariepinus* have been recorded as 2.46±0.30 mg/dl and 2.99±0.18mg/dl respectively. The calculated F value is 2.15, which is much lower than the table value i.e. 5.31 rendering it not significant at $p<0.05$.

CONCLUSION

The present study has generated the reference value of various biochemical parameters associated with Liver function test and Kidney function test of two closely allied species of cat fishes *Clarias batrachus* and *Clarias gariepinus* (Family:*Clariidae*). Statistical ANOVA test clearly reveals that apart from serum bilirubin all the parameters of LFT *viz.* SGPT, SGOT and ALP *etc.* significantly ($p<0.05$) vary between *C. batrachus* and *C. gariepinus.* Amongst KFT parameters except serum globulin (which is highly significant at $p<0.05$), all

parameters show non-significant difference between these two species. The biochemical parameters are considered as the basic valuable tool to monitor fish health and production parameters of the aquatic bodies. The base line reference data can be further used for studying any kind of xenobiotic stress in fish. It will help in making effective conservation strategies and successful productive management of the aquatic bodies. The findings of the present study can be further correlated with the advanced mitochondrial genome analysis of these two closely allied species of family *Clariidae.*

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Study of zooplanktonic communities present in the water of River Karamnasa at Buxar, Bihar, India

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Abstract: Zooplanktons are the heterotrophic, floating plankton that range from microscopic to large species. The zooplanktonic communities occupy an important place in food web of an aquatic ecosystem. It is also an integral part of the lentic aquatic ecosystem and significantly contributes in the productivity of a fresh water ecosystem. The diversity of zooplanktons is highly influenced by variation in season as well as physico-chemical parameters. The present paper deals with the zooplanktonic communities present in the water of river Karamnasa at Buxar, Bihar. For this purpose, the physico-chemical parameters of river Karamnasa were studied and the zooplanktonic samples were collected from the different sampling sites from the period January 2018 to December 2018. Quantitative analysis was done by dropping method of APHA (2005)[1]. The result of this finding concludes that 12 species of zooplanktonic communities have been found in the water of river Karamnasa. All the species belong to four major groups as Rotifera, Cladocera, Copepoda and Ostracoda. Cladocera was most dominant group while Ostracoda constitute the smallest group of zooplankton during the whole period of investigation.

Key words: Zooplanktons, Physico-chemical parameters, Karamnasa River, Buxar, Bihar.

INTRODUCTION

River body is the most important aquatic ecosystem and is characterized by its biotic constituents. Biotic constituents comprises of planktons and other macrofauna. Among planktons, zooplanktons are heterotrophic, floating plankton that range from microscopic to large species. Zooplankton plays an important role in determining the health of water body as well as influences all the functional aspects such as food chain, food web and flow of energy. The zooplanktons also serve as bioindicators and a tool for understanding the pollution status of the river.

The diversity among zooplankton exhibits a major biotic constituent of an aquatic ecosystem. The composition of zooplanktons varies according to the type of freshwater body and its physico-chemical parameters. Many workers have studied to describe the influence of physico-chemical parameters on zooplanktonic population as Battish S.K.(1992)[2], Trivedy R.K. (2000)[3], Lands V.M. (2004)[4], Patil *et al.* (2006)[5], Jayabhaye U.M. (2010)[6], Joshep *et al*.(2011)[7], Goswami *et al*. (2012)[8], Singh *et. al*.(2012)[9], Patel *et al.* (2013)[10], Gayathri *et al.*(2014)[11], Ramakrishna S.(2014)[12], A.N. Dede and A.L.Deshmukh (2015)[13], Khan *et al*.(2016)[14]. Pati and Sahu also found that Zooplanktons constitutes 13.2% of total planktonic population in a water body.

From the earlier work on zooplanktons it appears that various investigation have been done to carry out the

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zooplanktonic composition in river body. But no work has been carried out to find the zooplanktonic communities in river Karamnasa at Buxar, Bihar. Therefore, in the present research paper an assessment was done to find out the zooplanktonic communities present in the water of river Karamnasa.

MATERIALS & METHODS

River Karamnasa is a perennial and one of the largest rivers of Bihar. It originates at a height of 350 meter on the Northern face of Kaimur range near Sarodag in Kaimur district of Bihar. It flows in a North-Westernly direction through the plain of Mirzapur, then forms the boundary between U.P. and Bihar and finally joins with river Ganga at Chausa (Buxar). It has a total drainage area of approx. 4251 sq. miles. The water level of river Karamnasa becomes high in the month of July to March and less from April to June. River Karamnasa is a tributary of river Ganga and hence it shows various similarities in biological factors with River Ganga.

Many small and populous villages namely Banarpur, Khelafatpur, Kutubpur and Narbatpur are situated all along its length on the both side on the bank near Buxar. River Karamnasa carries the discharge of many canals, effluents of household, washing, bathing, cattle wallowing, organic and inorganic residues from agricultural field of the adjoining areas. However, the river has its own stock of organic matters which available to organisms as food.

Area of study: To find out the zooplanktonic communities, the three sampling sites were selected which stretches to a distance of 5 km covering the entire length of river passing through Buxar district upto the confluence with the river Ganga. Site I was situated near the village Banarpur. At this site, the river carries the discharge of domestic sewage, discharge of agricultural field as well as rice mill effluents. Site II was placed near the village Narbatpur. This site was 2 km downstream from site I and carries discharge of household, washing, bathing, cattle wallowing and agricultural effluents. Site III was situated near the confluence of river Karamnasa and river Ganga. This site was about 1 km downstream from site II.

All the above sites show a varied ecological conditions as well as physico-chemical characters and hence are helpful in analyzing biotic features like zooplankonic communities.

Collection and preservation: The water samples fir identifying zooplanktonic communities were collected monthly during the period January 2018 to December 2018 by filtering 50 litre of water of the river Karamnasa from considering sampling sites. For this purpose, a plankton net having mesh size 50 mm in used. The planktonic net was attached with a glass tube of 10 ml capacity at the bottom. The zooplanktonic samples were preserved in 4% (v/v) aqueous solution of formalin.

Analysis and Identification: The zooplanktons were identified with the help of keys given by Adoni *et al*. (1985). A key given by Edmondson (1959), Needham & Needham (1962), Michael (1973), Reddy (1994) and Dhanapati (2000) was also considered for this purpose.

Qualitative analysis of zooplanktonic population was done by the help of Sedgwick Rafter Planktonic Counting Cell as per standard method of APHA (1998). After this, zooplanktonic sample were examined by a compound microscope after preparing a stained temporary slides.

RESULT AND DISCUSSION

Zooplanktonic population of river body is influenced by the physico-chemical parameters. The various physico-chemical parameters of river Karamnasa were recorded in different months during the whole period of investigation are shown in following table-

Table 1: Showing the different physico-chemical parameters of Site-I

Parameters	Jan	Feb	Mar	Apl	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Temperature(^{o}C)	20.4	22.8	26.4	28.4	30.4	33.6	34.2	32.8	31.4	28.2	24.8	21.6
Turbidity(NTU)	82.5	80.8	81.6	78.8	74.2	58.8	20.2	10.8	10.2	17.6	38.4	76.6
pH	8.1	8.0	7.8	7.8	7.8	7.8	7.6	7.8	8.1	8.4	8.2	8.3
Free CO_2 (mg/l)	0.8	0.0	0.0	0.0	0.1	4.8	6.4	4.4	4.6	4.4	3.0	1.0
COD (mg/l)	5.7	7.2	7.8	8.1	9.0	12.4	11.8	10.0	10.6	9.2	8.7	7.0
BOD (mg/l)	1.9	2.2	2.4	3.0	3.8	7.2	6.4	6.0	5.4	5.1	4.8	4.7

Table 2: Showing the different physico-chemical parameters of Site-II

Parameters	Jan	Feb	Mar	Apl	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Temperature(^{o}C)	20.8	22.8	26.3	28.4	30.4	32.8	32.9	34.4	31.6	27.8	24.6	21.8
Turbidity(NTU)	80.6	78.4	80.2	78.8	72.5	52.8	20.0	10.8	10.6	18.4	38.2	66.9
pH	7.8	7.8	7.8	7.3	7.4	7.5	7.5	7.0	7.8	7.4	7.8	8.2
Free CO_2 (mg/l)	8.2	9.4	8.8	10.0	10.2	10.4	12.6	10.8	10.8	10.6	8.2	8.1
COD (mg/l)	5.8	7.3	7.8	8.2	9.2	12.5	11.9	10.6	10.8	9.3	8.8	7.1
BOD (mg/l)	2.0	2.4	2.6	3.1	3.9	8.6	6.8	6.2	5.5	5.2	4.9	4.4

Table 3: Showing the different physico-chemical parameters of Site-III

Parameters	Jan	Feb	Mar	Apl	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Temperature(^{o}C)	21.2	22.6	26.5	28.5	30.2	32.8	34.0	32.6	31.8	27.6	24.8	21.6
Turbidity (NTU)	82.9	78.4	82.7	78.6	74.4	58.1	20.9	9.8	8.6	18.1	40.9	68.2
pH	8.0	7.9	7.8	7.8	7.5	7.8	7.8	7.8	7.8	7.8	7.8	7.6
Free CO_2 (mg/l)	0.0	0.0	0.0	0.0	0.0	6.0	7.2	5.0	6.2	3.4	4.2	3.6
COD (mg/l)	5.9	7.5	7.9	8.4	9.9	12.6	12.0	11.7	10.9	9.4	9.2	7.2
BOD (mg/l)	2.6	2.8	3.1	4.1	4.9	8.9	7.9	6.1	6.7	6.2	5.8`	4.6

The above physico-chemical parameter influences the zooplanktonic population in the Karamnasa River. In the present study 12 species of zooplanktons have been identified from the samples collected from the three different sites. The dominant species of different groups of zooplanktons are tabulated below:-

Table 4: Showing the different Groups & their dominant species of Zooplankton

S.N.	Groups	Dominant Species
1.	Rotifera	*Keratella, Brachionus, Filinia,*
2.	Cladocera	*Daphnia, Sida, Moina,*
3.	Copepoda	*Cyclops, Nauplii, Diaptoma, Canthocamptus, Calanoids,*
4.	Ostracoda	*Cypris*

The population density of different groups of zooplanktons during the different months of investigation is shown in following table-

Table 5: Monthly variation of population density of different groups of Zooplanktons (Unit/L)

Month	Cladocera	Copepoda	Rotifera	Ostracoda	Mean	Standard Deviation
January	62.4	23.6	10.2	3.8	25.0	22.7
February	63.6	30.2	3.4	2.8	25.0	24.9
March	60.6	26.6	12.8	0.0	25.0	22.6
April	45.6	54.4	0.0	0.0	25.0	25.2
May	63.2	28.6	8.2	0.0	25.0	24.4
June	45.6	36.2	10.6	7.5	25.0	16.3
July	48.2	34.1	8.4	9.3	25.0	16.9
August	68.6	26.4	2.6	2.4	25.0	26.9
September	21.8	78.2	0.0	0.0	20.3	31.9
October	28.1	68.6	1.8	2.5	25.2	27.2
November	54.4	45.6	0.0	0.0	25.0	25.2
December	55.4	38.8	2.4	3.4	25.0	22.9

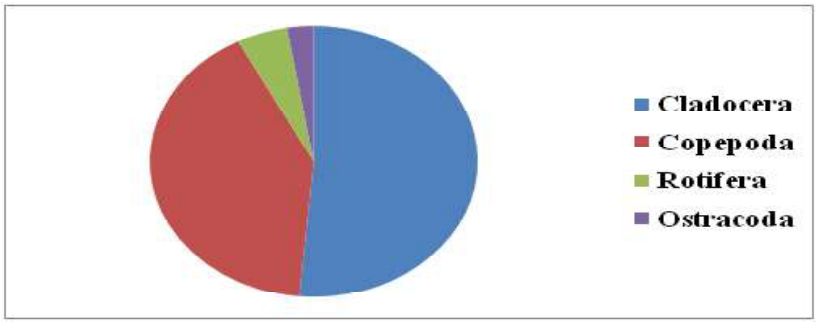


Fig- Showing annual distribution of zooplanktonic population in Karamnasa River

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Genetic divergence and phylogenetic analysis of fish digestive enzymes in carnivores, herbivores and omnivores

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Abstract: The present study has been designed to understand the dynamics of physiology of digestion in carnivorous, herbivorous and omnivorous fishes by assessing the activity of different digestive enzymes (lipase, amylase and trypsin) and gastric (pepsin). The assessment of the activity of digestive enzymes showed that enzymes involved in the digestion of proteins, lipids and carbohydrates were present in all the three groups of fishes with three different feeding habits. In the present study, we collected all enzymatic nucleotide sequences from National Center for Biotechnology Information (NCBI). Further multiple sequence alignment and phylogenetic relationship was established by using Clustal W software for considering the ontogenetic development of the digestive enzymes. Our results support the divergence of developing gastric gland and pancreatic gland. In carnivores lipase enzyme is more prominently active in comparison to herbivores and omnivores. Similarly, pepsin and trypsin are moderately active while amylase is less active in carnivore due to less consumption of carbohydrate rich feeding habit with respect to herbivores and omnivores. Considering the functional development of the digestive system in these fishes, the findings of the present study will be useful to improve the actual larval rearing techniques for different fish species from the Indian sub-continent. It will raise the status of aquaculture sector in India.

Keywords: Digestive enzymes, Fishes, NCBI, Clustal W, Phylogenetic tree.

INTRODUCTION

Digestive enzymes are one of the most important factors that influence the growth, efficiency of feed utilisation, and feed conversion in fish. Digestive processes in fishes are poorly understood in comparison to mammals; although the data obtained in fish so far, show that the digestive enzymes studied are qualitatively similar to those observed in other vertebrates. A very few comparative studies of the digestive enzymes in different fish species have been reported earlier[1-2]. The characterisation of digestive enzymes provides information regarding the digestive capacity of fish to hydrolyse carbohydrate, protein and lipid of feed ingredients. Enzymes enable metabolic and physiological processes to occur by reducing the activation energy of the chemical reaction thereby speed up the rate of the reaction. The study of digestive enzymes is an essential foremost step towards understanding the mechanism of digestion and the adaptation of fish to the changing nutritional environment[3].

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The specific activity of digestive enzymes along the gut changes in fish with the different feeding habits. Some of the enzymes like amylase play a key role in the catalytic breakdown of carbohydrates to provide energy sources for the fish. Assay of enzymes in the gut of a fish provides information about its nutritional physiology[4]. Protease, a major class of enzymes, with high activity and stability in a wide range, is associated naturally in gastrointestinal tracts of fishes and performs major role in digestive processes. They catalyse the hydrolysis of proteins into smaller fragment called peptides. These peptides enter the intestinal tract, where pancreatic and intestinal proteases are secreted and peptidases cleave peptides into amino acids[5]. Lipases are responsible for the breakdown of lipids into triglyceride molecule. Lipase activity in all fishes is very high because of digestibility of fat is higher in case of fish and also act as the main source of energy in aquatic organisms[6]. Dietary fibre generally refers to all indigestible plant matter, mainly cellulose and other complex polysaccharides. It has long been considered as an inert and insignificant part of an animal's diet, mainly because it was believed to contribute little nutritionally. Vegetable biomass is abundant in many freshwater environments and cellulose, being the main structural material of plants, is the most abundant carbohydrate in nature[7].

Studies on the digestive physiology of carps clearly indicate the existence of a correlation between the enzymes and the diet of the animals. It has been found that the amount of carbohydrate splitting enzymes was maximum in the predominately herbivorous fish and minimum in the carnivorous. In contrast, the concentration of proteases was found to be miximum in the carnivorous fishes. Pepsin, a low pH proteolysis enzyme, is mostly absent in herbivorous and omnivorous fishes, since they lack a true stomach. However, omnivorous species have amylase activities in the intestine, which is much higher in the other species[6].

The rationale behind this study is to compare the digestive enzyme activities of protease, lipase, amylase, alkaline phosphatase *etc*. in the herbivorous and carnivorous fishes with natural feeding habits and diet composition. Further, to establish the relationship between carnivorous and herbivorous fishes on the basis of digestive enzymes, if any, is the objective of the study. The main aim of the present study is to explore the diversity of the specific enzymatic activities in different species of fishes with different feeding habits. However, the evolutionary differences may be analysed based on biological databases with other computational techniques supplemented by the experimental data. The present study may also be helpful to explore the additional knowledge about species environmental changes. Till date, scanty information regarding it are available in the scientific literature.

MATERIAL AND METHODS

Sequence data

All the nucleotide sequence of digestive enzymes of 5 different species of fishes were collected from National Centre for Biotechnology Information (NCBI) in fasta format with their accession number as shown in Table 1.

Table 1.List of digestive enzymes with Genbank accession numbers of different fish species

Sl. No	Feeding Habits	Organism	Lipase		Amylase		Pepsin		Trypsin	
1.	Carnivorous	*Pangasianodon hypophthalmus* (Striped catfish)	Accession id	length	Accession id	length	Accession id	length	Accession id	length
			XM_026915805	1819 bp	XM_026911354	1,615 bp	XM_026921593	1,236 bp	XM_026921223	1,199 bp
		Tachysurus fulvidraco (Yellow catfish)	XM_027137105	2380 bp	AY940145	888 bp	XM_027167205	1448 bp	XM_027145167	1,002 bp
2.	Herbivorous	*Oreochromis niloticus* (Nile tilapia)	NM_001279753	2313 bp	GU433189	2313 bp	JQ043215	1,173 bp	AY510093	863 bp
		Cyprinus carpio	FJ716101	2,411 bp	XM_019115816	2,029 bp	XM_019063590	933 bp	XM_019113891	1,495 bp
3.	Omnivorous	*Danio rerio*	U57656	1,658 bp	NM_001014353	1,970 bp	NM_131056	468 bp	NM_001014353	1,970 bp

Sequence Alignment and Evolutionary Analysis

Phylogenetic analysis was performed on three sets of fish species on the basis of feeding habits, corresponding to different portions of the sequence gene representing digestive enzymes. The lengths expressed above refer to the final alignments. Nucleotide sequences were directly aligned using the program Clustal W[8]. Ambiguously aligned sequences were excluded from the final alignment. The nucleotide sequences were aligned as above, mentioned in Table 1. Aligned sequences were analysed under the criterion of maximum parsimony with the program PAUP, neighbor-joining (NJ) method, as implemented in the construction of phylogenetic tree[9-10]. Alternatively, we incorporated the search options, present in PAUP and NJ to test the stability of the obtained trees. Bootstrap resamplings were also performed to test the robustness of the trees[11]. Maximum parsimony analysis was performed only on informative characters. Program settings are detailed in the figure legends. The analyses were performed on Linux operating system.

RESULTS

During this work, we identified 70 different species of fish on their feeding habits. Out of them, 5 fish species were selected for the study of different digestive enzyme with distinct sequence-types (Table 1). Clustal W sequence alignment showed specific values, which represent the evolutionary distance shown in table 2. Then, we used the branch-site model to test for positive selection in individual codons for the lineage leading to divergence of digestive enzymes with carnivorous, herbivorous and omnivorous fishes. Further, the study based on cladistics analysis performed with a result of five equally parsimonious cladograms shown in figure 1(A-D). *Pangasianodon hypophthalmus* (Striped catfish), *Tachysurus fulvidraco* (Yellow catfish), *Oreochromis niloticus* (Nile tilapia), *Cyprinus carpio* and *Danio rerio* fishes associated evolutionary relationship consensus tree is presented, together with the NJ-tree. The topologies of trees are in good agreement, even though some discrepancies exist.

Table 2.Species used in the present study with GenBank accession numbers with clustal W value.

Organism	Lipase		Amylase		Pepsin		Trypsin	
Pangasianodon hypophthalmus (Striped catfish)	Accession id	Clustal w value	Accession id	Clustal w value	Accession id	Clustal w value	Accession id	Clustal w value
	XM_026915805	0.914075003	XM_026911354	0.0694275	XM_026921593	0.0871005	XM_026921223	0.414602
Tachysurus fulvidraco (Yellow catfish)	XM_027137105	0.69096	AY940145	0.0715371	XM_027167205	0.150067	XM_027145167	0.06461
Oreochromis niloticus (Nile tilapia)	NM_001279753	0.296712	GU433189	0.586395	JQ043215	0.308813	AY510093	0.282391
Cyprinus carpio	FJ716101	0.0902064	XM_019115816	0.857549	XM_019063590	0.760953	XM_019113891	0.857495
Danio rerio	U57656	0.0798773	NM_001014353	0.9056295	NM_131056	0.651466	NM_001014353	0.96099

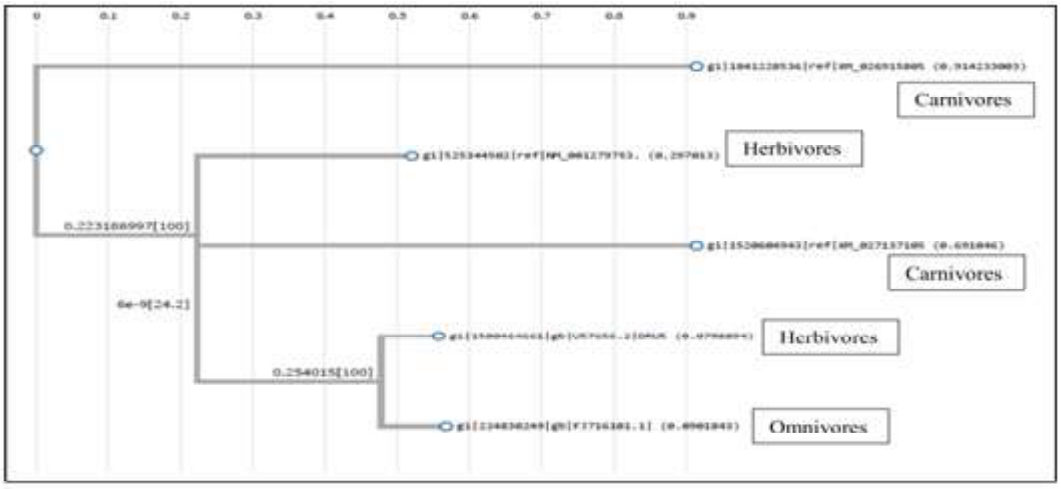


Fig. 1 (A)

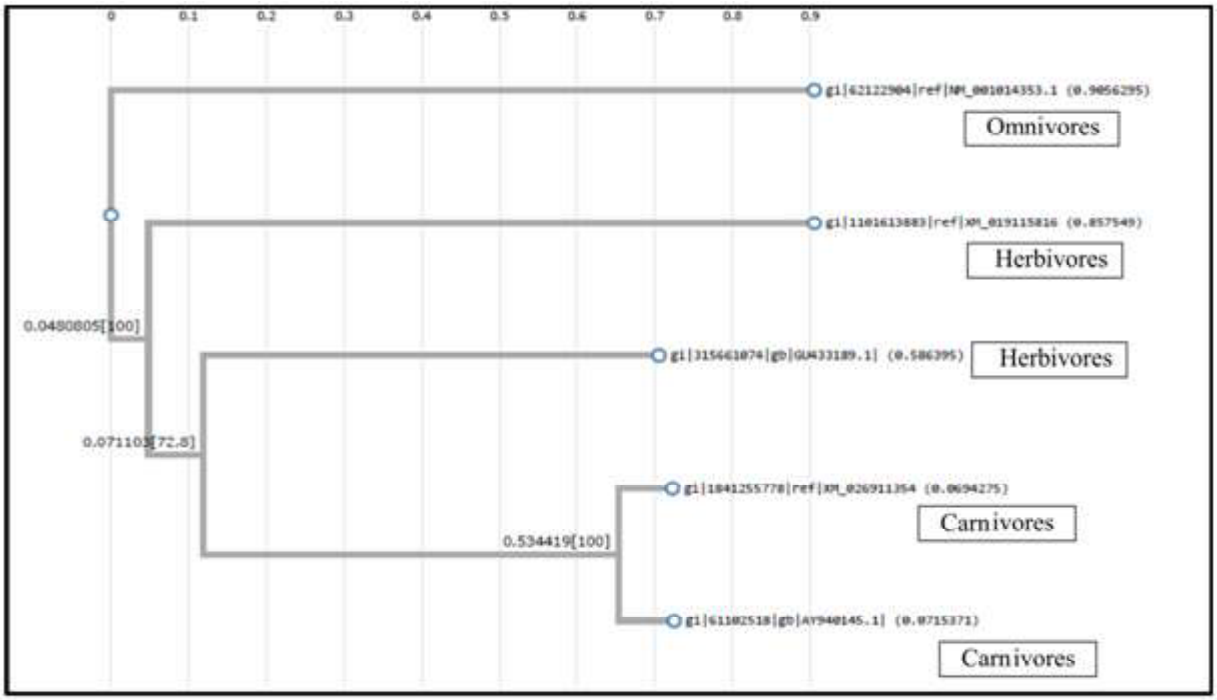

0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
gi|62122904|ref|NM_001014353.1 (0.9056295)
Omnivores
gi|1101613883|ref|XM_019115816 (0.857549)
Herbivores
0.0480805[100]
gi|315661074|gb|GU433189.1| (0.586395)
Herbivores
0.071103[72.8]
gi|1841255778|ref|XM_026911354 (0.0694275)
Carnivores
0.534419[100]
gi|61102518|gb|AY940145.1| (0.0715371)
Carnivores

Fig. 1(B)

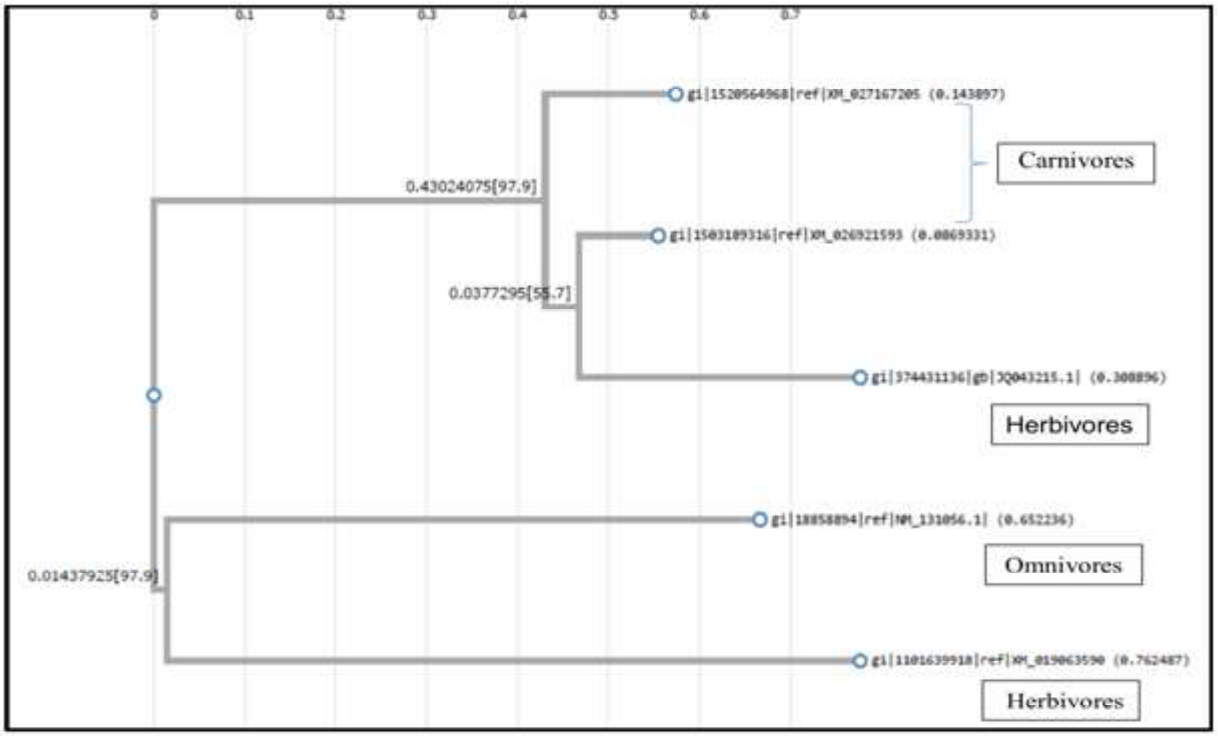

0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
gi|1520564968|ref|XM_027167205 (0.143897)
Carnivores
0.43024075[97.9]
gi|1503189316|ref|XM_026921593 (0.0869331)
0.0377295[55.7]
gi|374431136|gb|JQ043215.1| (0.308896)
Herbivores
gi|18858894|ref|NM_131056.1| (0.652236)
Omnivores
0.0143792[97.9]
gi|1101639918|ref|XM_015063590 (0.762487)
Herbivores

Fig. 1(C)

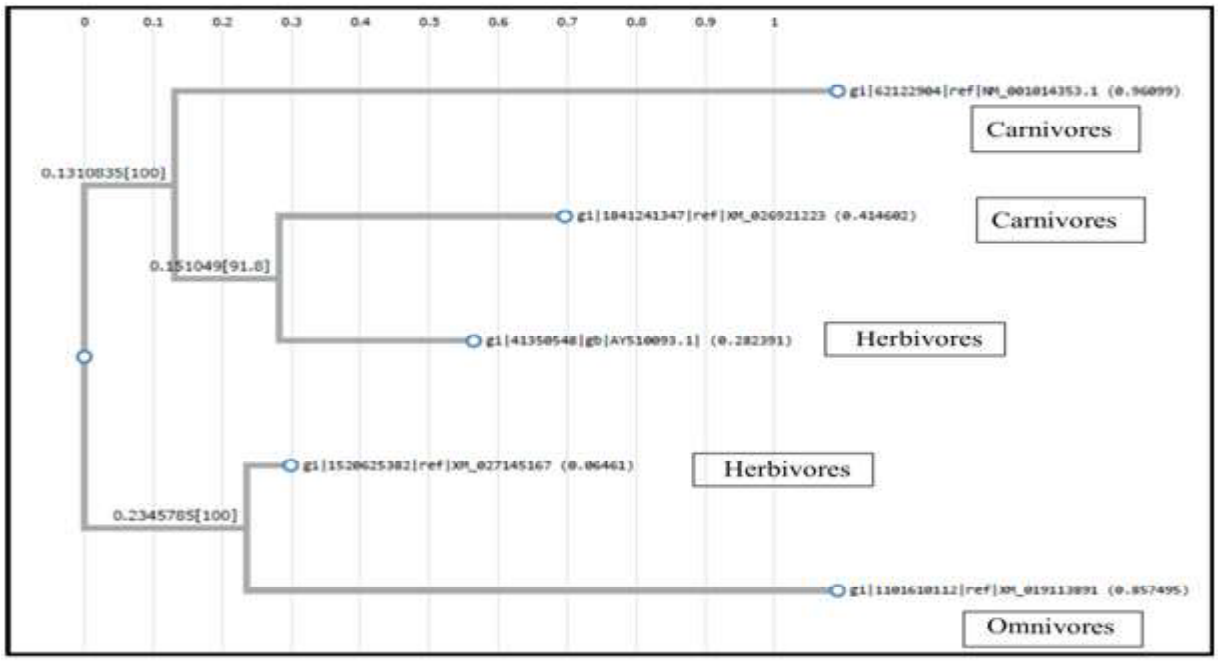


Fig. 1 (D)

Fig. 1- ***An N-J tree of the phylogenetic analyses were produced with PAUP according to the standard settings like uninformative characters ignored, Branch-and-bound search options and the distance scale is expressed and refer to the bootstrap values expressed in percentage.***

(A) Lipase enzyme rooted tree illustrating that carnivores are evolutionary distantly related with herbivores and omnivores with respect to lipase enzyme.

(B) In Amylase, Tree showing carnivore amylase enzyme less active which is responsible for carbohydrate digestion due to their feeding habit.

(C)Pepsin enzyme also showing their slightly less active in case of carnivores in compare to herbivores and omnivores.

(D)Trypsin enzyme more prominently active in carnivores with respect to others.

DISCUSSION

While the spectrum of digestive enzymes in bony fish is known to be the same as in other vertebrate animals, a very fewer digestive enzymes are studied in fish. The prevalent hypothesis is that digestive enzyme production in fish represents their feeding ecology and trophic niche under natural conditions, and thus is well associated with their diet[12]. Additionally, the proper conception of feeding protocols and diet formulations under conditions of fish larval culture requires an in-depth knowledge and understanding of the digestive processes that occur during early ontogeny to synchronize various types of feed with the output of various digestive enzymes. Thus, the assessment of the presence and digestive enzymes activity may be used as a comparative indicator of development of fish larvae, food intake habit and digestive capacity for survival and growth predictions. There are three main steps considered as maturation of digestive system functions in fish larvae namely- i) achievement of pancreatic secretion; ii) onset of the villi membrane enzymes in the intestine; and iii) development of the stomach and onset of acidic digestion[13].

In earlier studies, comparing digestive enzyme activities between different species was difficult due to

differences in methodology including different parameters in different environmental culture, having specific instruments used for analysis and species ontogenetic stages examined, as well as in the quantity and composition of diet[14-16]. Consequently, we decided to compare only the development of different digestive enzymes in catfish (carnivores), herbivores and omnivores, those reported from different freshwater and marine fish species. In this context, special emphasis was given on catfish species and other freshwater species with similar developmental patterns as done by earlier workers[17-18]. Enzymes involved in the digestion of protein, lipid and carbohydrate were present in butter catfish larvae at the time of hatching and before the onset of exogenous feeding. The developmental pattern of pancreatic, gastric and intestinal enzymes was closely related to the histological development of the digestive system, as described previously[19].

The study of the lipases during the early life stages of development in fish is an important issue because it permits the activity of the optimal utilization of lipids from food[20].In the present study, lipase activities in carnivores are more prominently active, while amylase enzymes are less active due to lack of less consumption of carbohydrate rich feed. Similarly Pepsin and trypsin enzymes are moderately active in contrast to the fish groups of other feeding habit like herbivores and omnivores as depicted in above figures. In case of herbivores amylase enzyme activities show more divergent with omnivores as well in carnivores. Thus, as the functional development of the digestive system help in developmental process i.e. dependent on the rearing conditions and the diet, a study with different feeding regimes and weaning points is needed for establishing it in future.

CONCLUSION

We reported here for the first time a broad spectrum of digestive enzyme of the fishes of different feeding habits including carnivores, herbivores and omnivores with their respective nucleotide sequences coding specific enzyme, representing the diversification. These data provide new horizon and useful information for further understanding of the phylogenetic classification and evolutionary significance of fishes with respect to structural and genetic changes. Moreover, this approach will very useful as basis for genetic conservation and selective breeding of fish species in fishery and aquaculture.

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Nutritional habit of *Channa striatus* (Bloch) from the paddy field of Madhepura District, Bihar

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Abstract:*Channa striatus* (Bloch) has been gaining importance as a common nutrition rich fish. Mean RLG value was found to be highest in the size group of 5.0-10.0cm (0.55) and lowest in the size group of 21.1-25.0 cm. (0.46). The results reveal that the species fall in the carnivorous category. Gut content analysis reveal that zooplankton was the highest percentage (12.50) occurrence in July, insects was found to be highest (23.53%) in June. The highest percentage of crustaceans was observed in February (22), annelids was highest in April (7.40), molluscs occurred in April (3.70) and occurrence of fish in the gut was in September (55.56). The highest percentage of plant matter noticed in January (15.09) andmucks and unidentified components was observed in December (23.26). Analysis of gut contentrevealed that in fry stages most dominant food item was zooplankton (68.52%), followed by insect (25.2%), annelids (5.01%),crustacean (1.02%) and plant matter (0.25%). In juvenile the most preferred food item being insects (31.16%), followed by trash fishes (30.82%), crustaceans (19.82%), annelids (12.18%), plant matter (4%) and molluscs (2.02%). In adult, a strong positive selection was observed for fish (46.37%), insects (12.45%), mucks and unidentified material (12.04%), plant matter (10.12%), crustaceans (9.16%), annelids (6.84%) and molluscs (3.02%). Gastro somatic index (GSI) was ranged from 1.15 ±0.36 (August) to 5.14 ±1.76 (April) for males and that of female from 2.24 ±1.10 (December) to 9.52±3.91 (September).

Key words: *Channa striatus*, RLG, GSI, Gut content analysis

INTRODUCTION

The striped snakehead, *Channa striatus* (Bloch), distributed throughout the South-East Asian countries and has been identified as a potential species for rearing in paddy fields, derelict and swampy water as it is a hardy and air-breathing fish. The fish breeds during south-west and northeast monsoons in flooded rivers and ponds and frequently found in flooded rice-fields[1]. Of freshwater food fishes, they are amongst the easiest to transport due to their air breathing abilities and hardness. Mortalities on transit are generally low. In local markets, they are kept in large bins or tubs in large numbers in little water without aeration. Several works have been reported on the feeding of *Channa* species from outside India. Studied onfood and feeding ratios of the Amur snakehead, *Channa argus* in water bodies in the lower reaches of the Amu Darya[2] while studied on the linear regression model to predict cannibalistic in juvenile *Channa striatus*[3]. Further, recorded feeding intensity and dynamics of *Channa argus* under different illumination[4]. Again studied on the food and feeding habits of *C. punctatus* in beel environment of Bangladesh[5].Several workers investigated food and feeding behaviour of *Channa* species in India. Among the mare: studied the feeding habits of the fingerlings of *Channa striatus*[6], worked on utilization ofnatural food by

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juvenile *C. gachua*[7] and studied the food preference of the juvenile of *Channa gachua*[8]. Meanwhile, investigation of food spectrum and intensity, food preferences and seasonal variation of *C. punctatus* were reported[9,10,11]. Again, Rao *et al.,* (1998) investigated was madeon the food and feeding habits of *Channa* spp. from east Godavari District[12]. Further, adaptation of alimentary tract to feeding habit in four species of *Channa* was also reported[13]. Although, over the last 10 years, its population has undergone a steady decline due to over exploitation, loss of habitat, pollution as well as destructive fishing. According to IUCN status[14], it has been listed among the 66 LR-nt fish species in India. The fish is well known for its taste, high nutritive value and medicinal qualities and is recommended as a diet during convalescence[15]. The paper deals with the food and feeding habit of *Channa striatus* from certain paddy field and wetland of Madhepura district, Bihar.

MATERIALS AND METHODS

The specimens of *C. striatus* were collected from paddy fields of Madhepura district of Bihar during August, 2018 to July, 2019. Food and feeding habits of *Channa striatus* were studied by examining a total of 645 digestive tracts. The guts were removed from the specimen after measuring and weighing each specimen to the nearest cm and gm respectively and were preserved in 5% formalin for subsequent analysis. The preserved guts were later uncoiled, cleaned off the attached fat and the length and weight were recorded.

Relative length of the gut (RLG): The ratio between the gut length and total length (RLG) was estimated by adopting the following formula[16]: Where, GL stands for total length of the gut and TL are the total length of the fish.

Feeding intensity (GSI): The feeding intensity or gastro-somatic index (GSI) was calculated using the following formula[17].

$$GSI = \frac{\text{Weight of the gut}}{\text{Total weight of the fish}} \times 100$$

Gut content analysis: Gut content analysis was done for examining seasonal variation in diet components. Further, the data were analysed for different size group to record basic changes in feeding habit if any and in various stages of growth. Both volumetric and occurrence methods were used for gut content analysis.

(i) Volumetric method: The content of each sample was taken as a unit and various items are expressed as % volume by eye inspection[18]. The content of each gut was vigorously shaken with distilled water and then a drop of the content was examined under microscope. The area occupied by each food item was estimated arbitrarily. At least ten such drops were examined and the average of each of the drops was recorded.

(ii) Occurrence method: In this method, the number of guts containing a particular item of food was expressed as a percentage of the total number of gut examined[19]. This method, apart from describing the qualitative analysis of the diet also gives the frequency of a particular food item occurred in the gut and thus helps to understand the preference of any particular food item.

RESULTS AND DISCUSSION

(a) Relative length of the gut (RLG): Mean RLG value was found to be highest (Table-1) in the size group 5.0-10.0cm (0.55) and lowest in the size group 21.1-25.0 cm. (0.46). This indicates a gradual decline in RLG values as the fish grows. The relative length of the gut in *C. striatus* was found to decrease with the increasing length of the fish indicating the change of feeding habit from omni-carnivorous diet in the fry stage to a highly carnivorous in the adult stage. Similar case was also noticed by[20] in *Ompok pabda*; and[21] in *Notopterus notopterus* from different habitats and considered these fish species as carnivorous. A change in the feeding habit of fingerlings of *Channa striatus was reported*[6]. Furthermore, a change in feeding habit of *Channa* in different life stages is reflected in the RLG value at different length groups[13]. Shortened alimentary canal in adult fishes of *C. barca was also observed by*[22]. Thus the low RLG value in adult and high percentage of animal diet in food composition of *C. striatus* confirms that it has a high carnivorous type of feeding habit.

Table-1: RLG value in various length groups of *C.striatus*

Size group (cm)	RLG Value
5 - 10	0.55 (±0.02)
10.1 - 15	0.53 (±0.06)
15.1 - 20	0.50 (±0.02)
21.1 - 25	0.46 (±0.01)

(b) Gut content analysis: The percentage composition of food items in the gut of *C. striatus* as observed in different months has been summarized in the Table-2. The gut content of *C. striatus* have been group into 8 broad categories i.e. zooplankton, insects, crustaceans, annelids, molluscs, fishes, plant matter, mucks and unidentified components. It was seen that there were considerable variations in the percentage of different food items during different months of the year. *Zooplankton* was the highest percentage (12.50) occurrence in July and that of lowest (3.92) in June. Insects percentage varies from 13.95 (December) to 23.53 (June). The highest percentage of Crustaceans was observed in February (22) and lowest (10.05) in May. Percentage occurrence of Annelids was highest in April (7.40) and lowest in June (1.96). The highest percentage of molluscs occurred in April (3.70) and lowest in May (1.61). The highest percentage of occurrence of fish in the gut was in September (55.56) and lowest in December (20.93). The highest percentage of plant matter noticed in January (15.09) and that of lowest percentage in April (1.85). The highest percentage of mucks and unidentified components was observed in December (23.26) and lowest in February (2.0). Seasonal variation showed a slight variation in feeding habit which might be due to fluctuation in the availability of different food items in different season. Percentage occurrence of annelids and molluscs was highest during pre-monsoon season while zooplankton and insects were also found to be the highest percentage during monsoon season. Again, the highest percentage of fish, crustaceans, plant matter and mucks and unidentified components were observed during post monsoon and winter season. The main food items of *C. striatus* were insect eggs, gastropods and aquatic vegetation during summer and maximum food intake was recorded during post-monsoon and pre-monsoon and minimum in winter was reported[9]. Feeding was found to be intensive during February-May and August-October and low during June-July and November-January. From the present study it can be inferred that *C. striatus* changes its food habit with the change in seasons. These findings agree with those of[23] who reported that *Tilapia nilotica* changed its food habit with the change in season.

(c) Food composition in various life stages: Variation of food composition in various life stages is recorded in Table- 3 and the same is summarized below:-

Fry: Analysis of gut content revealed that in fry stages most dominant food item was zooplankton (68.52%), followed by insect (25.2%), annelids (5.01%), crustacean (1.02%) and plant matter (0.25%).

Juvenile: The most preferred food item being insects (31.16%), followed by trash fishes (30.82%) and crustaceans (19.82%). Among other items annelids (12.18%) are notable. Plant matter constituted only 4% of the gut content molluscs constituted 2.02%.

Adult: Fish was found to be the most dominant (46.37%) item in the gut. The next food items in order of preference were - insects (12.45%), mucks and unidentified material (12.04%), plant matter (10.12%), crustacean (9.16%), annelids (6.84%) and molluscs (3.02%).

Table 2: Seasonal variation of food items in *C. striatus*

Month	No. of fish examined	Zooplankton		Insects	Crustaceans	Annelids	Molluscs	Fishes	Plant Matter	Mucks and others
		Protozoa	Rotifers							
Jan	53	-	5.66	16.78	18.86	-	-	28.30	15.09	15.09
Feb	50	-	8	16	22	-	-	32	20	2.0
Mar	45	4.44	6.67	20	17.78	-	-	35.56	11.11	4.44
Apr	54	-	-	20.37	18.52	7.40	3.70	48	1.85	-
May	62	6.45	-	19.35	10.05	4.84	1.61	48.02	9.68	-
Jun	51	-	3.92	23.53	17.65	1.96	-	43.14	5.58	3.92
Jul	56	3.57	8.93	17.86	14.19	-	-	46.43	5.36	3.57
Aug	58	-	-	20.69	17.24	-	3.45	51.73	3.45	3.45
Sep	63	-	-	19.05	20.63	-	-	55.56	4.76	-
Oct	60	5	6.67	20	13.33	-	-	50	-	5
Nov	50	-	-	18	14	-	-	24	20	24
Dec	43	-	9.30	13.95	18.60	-	-	20.93	13.95	23.26

The surface feeders feed on surface plants and animals, the mid or column feeders feed on subsurface food organisms and the bottom feeders feed on mud and decaying substances and also bottom flora and fauna was observed by[24]. The dominant food item of *C. punctatus* as fish fingerlings and insects as well as carnivorous feeding nature was analyzed by[9,10]. The juvenile of *C. striatus* like *C. punctatus* was mainly fed on the animal food viz. crustaceans, insects, molluscs and fishes[5].

Table-3: Variation in food items (%) in different life stages of *C. striatus*

Food Items	Fry (0-5 cm)	Juvenile (5-10 cm)	Adult (<10 cm)
Zooplankton	68.52	-	-
Insects	25.2	31.16	12.45
Crustaceans	1.02	19.82	9.16
Annelids	5.01	12.18	6.84
Molluscs	-	2.02	3.02
Fishes	-	30.82	46.37
Plant matters	0.25	4	10.12
Mucks & unidentified components	-	-	12.04

Table-4: Average Gastrosomatic index (GSI) in various months

Months	Male	Female
August	1.15 (±0.36)	7.95 (±4.31)
September	4.29 (±1.88)	9.52 (±3.91)
October	4.17 (±1.72)	6.22 (±1.89)
November	2.05 (±0.83)	2.26 (±1.21)
December	2.04 (±0.88)	2.24 (±1.10)
January	2.72 (±0.66)	2.52 (±0.49)
February	3.15 (±0.65)	3.37 (±0.81)
March	3.14 (±1.03)	3.57 (±1.04)
April	5.14 (±1.76)	3.84 (±1.53)
May	3.14 (±1.03)	3.57 (±1.04)
June	2.07 (±0.84)	3.84 (±1.53)
July	2.87 (±1.22)	3.57 (±1.18)

(d) Gastrosomatic index (GSI): The observation on feeding intensity was based on gastrosomatic index (GSI) taken on monthly basis and the results have been summarized in Table- 4. It ranged from 1.15 ±0.36 (August) to 5.14 ±1.76 (April) for males and that of female from 2.24 ±1.10 (December) to 9.52 ±3.91 (September). Sharp rise and fall in the feeding intensity was noticed in various months. The GSI showed a steady increase from February-May onwards and then an abrupt fall was noticed in June-July. However, it increased once again during post monsoon months. The variation of GSI in females was more pronounced and they had high feeding intensity in August- September. During winter (November-January) the GSI was found lower in both sexes. Feeding intensity in winter was low because the species hibernate underneath aquatic weeds and in mud. Another reason for low feeding intensity in winter because of their low metabolic rate as the fish is poikilo thermics animal. Further, the reduced feeding intensity in pre-monsoon month may be due to the development of gonads. The poor feeding during breeding season may be attributed to the development of gonad which occupies the major space of the abdominal cavity[17,25,26]. Cannibalistic nature was also recorded specially when food availability was very low. Feeding intensity, however, improved after spawning was over in post monsoon months.

CONCLUSION

In the present study, mean RLG value was found to be maximum in the higher length group and minimum in the lower size group. The results indicate the species fall in the carnivorous category. There is also variation in the percentage composition of different items of food in the gut in different months. It can be inferred that *C. striatus* changes its nutritional habit with the change in seasons. The gut content analysis of *C. striatus* revealed a distinct variation in food intake of the species at different life stages. It revealed that in fry stages most dominant food item was zooplankton followed by insect, annelids, crustacean and plant matter. Similarly, in *Juvenile* insect was the most preferred food item, followed by trash fishes, crustaceans, annelids, plant matter and molluscs. Again in adult, fish was found to be the most dominant item in the gut and the next food items in order of preference were - insects, mucks and unidentified material, plant matter, crustacean, annelids and molluscs. Monthly variation of GSI in *C. striatus* revealed that the maximum feeding intensity in females was from August through October and lowest value was recorded in winter (November-January).

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Analytical survey of contaminated samples from soil, water, vegetables and fruits in relation to human parasites

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Abstract: An attempt was made to study the possible sources of transmission of protozoan cysts and helminthe eggs to man in the Samastipur town of Bihar from media such as soil, water and raw edibles (vegetables & fruits). A total of 2091 samples comprising 402 soil samples, 526 water samples, 375 fruit samples and 788 vegetables samples were examined. The highest prevalence rate (30.6 percent) was observed in soil samples and the lowest (12.05 percent) in vegetable samples. Out of the contaminated samples, 5.93 percent, 4.02 percent, 9.90 percent were position for *Entamoeba histolytica, Giardia intestinalis* and *Ascaris lumbricoides* respectively.

Key words: Protozoan cysts, helminthe eggs, *Entamoeba histolytica, Giardia intestinalis* and *Ascaris lumbricoides*

INTRODUCTION

In course of studying the prevalence and pattern of intestinal parasitic infections in man in the Samastipur town of Bihar it was thought desirable to explore out the particular fact of transmission dynamics, i.e., the possible source of their transmitting medium or the vector such as soil, water, fruit, and vegetable.[1]

For this purpose, the most common protozoans viz., *Giardia intestinalis, Entamoeba histolytica and Ascaris lumbricoides* among the helminthe parasites were selected considering their very high prevalence rate in the Samastipur town of Bihar.[2]

MATERIALS AND METHODS

Soil, water and fruit samples were collected from different wards and it was endeavored to assess the amount of contamination with the free living cysts of *Giardia intestinalis, Entamoeba histolytics* and the embryonated eggs of *Ascaris lumbricoides* in them.

For this purpose, two slides of each sample were prepared by two different methods for their subsequent microscopic examination. The methods used were as follows:

1. Sedimentation by simple centrifugation:

Water used for washing the fruit samples was kept for an hour for sedimentation. In case of water samples, they were kept for sedimentation without adding water to it. But in the case of soil samples, this process was repeated thrice after every three hours. The collected sediments were mixed with water and centrifuged at 2000 r.p.m. for two minutes. After this the whole contents of centrifuged tubes was poured down leaving only a few drops. A drop was taken on a micro slide and after putting a coverslip it was subjected to microscopic examination.[3]

2. Floatation by saturated zinc- sulphate solution

The seconds slides was prepared by floatation method using saturated Zinc sulphate which gave better results.

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RESULT AND DISCUSSION

Out of a total of 2091 samples, 402 soil samples, 526 water samples, 375 fruit samples and 788 vegetable samples were collected from different localities of the town and were examined to find out the prevalence rate of the most common intestinal parasites:

Out of the total examined 2091 samples, 415 samples (19.85 percent) were found to be contaminated with one or the other parasite. Of these contaminated samples, 124 samples (5.93 percent) were infected with *Entamoeba histolytics*, 84 samples (4.02 percent) with *Giardia intestinalis* and the remaining 207 samples (9.90 percent) with *Ascaris lumbricoides* respectively. Cysts, trophozoites, eggs and larva of some other intestinal parasites were also observed but they were not taken into consideration in the present study.[4]

The following is the account of the various examined samples.

SOIL SAMPLES

402 Soil samples from different parts of the town were collected and examined. These samples were taken from the crop- field, ponds, ditches & river banks, road sides and fruit orchards.

Of these 141, 85, 74 and 102 samples were collected from crop-fields, ponds, ditches & river banks, road sides and fruit orchards respectively.

Out of these 402 examined soil samples, only 123 samples (30.60 percent) were contaminated. These comprised 37 samples (9.20 percent) infected with *E. histolytics* 24 samples (5.97 percent) with *G.intestinalis* 62 samples (15.42 percent) with *A.lumbricoides.*

WATER SAMPLES

Water samples were collected for examination from various sources such as hand- pump, well, irrigation canal or drain, municipal supply and from river & ponds of which 171, 166, 62 and 85 water samples were examined from the above sources respectively.

A total of 526 samples of water were collected from various sources, from which the town people drink water and were examined. Out of theses 134 water samples (25.47 percent) were found to be contaminated 45 samples (8.55 percent) with *Entamoeba histolytics,* 31 samples (5.89 percent) with *Giardia intestinalis* and 58 samples (11.03 percent) with *Ascaris lumbricoides.*

Highest prevalence of the parasites was observed from the water samples of river and ponds. Out of 85 such samples examined, 28 samples (32.94 percent) were infective.

FRUIT SAMPLES

375 fruits samples of locally available cheap seasonal fruits, purchased from the markets or collected from the fields were examined to assess the degree of contaminated- 18 samples (4.80 percent) were *Entamoeba histolytics*, 10 samples (2.67 percent) with *Giardia intestinalis* and 35 samples (9.33 percent) with the embryonated eggs of *Ascaris lumbricoides.*[5]

VEGETABLE SAMPLES

Vegetable samples were collected for examination from the vegetable fields or purchased from the local markets. Of these 140, 176, 165, 134, and 173 samples were of Radish, Carrot, Sag, Mint leaves and Coriander leaves respectively.

Out of these 788 examined vegetable samples, only 95 samples (12.05 percent) were contaminated. These comprised 29 samples. (3.68 percent) infected with *E.histolytica*, 15 samples (1.90 percent) with *G. intestinalis*, 51 samples (6.47 percent) with *A. lumbricoides.*[6]

A review of the literature, regarding the transmission of intestinal parasites through different median reveals that not many workers have carried out the analysis of these media for the free- living stages of humans parasites, i.e., the cysts of *Entamoeba histolytics, Giardia intestinalis* and embryonated eggs and larvae of *Ascaris lumbricoides.*[7]

The general prevalence rate of the intestinal parasites was recorded to be 19.85 percent. Out of 2091 examined samples were from soil, 134 samples from water, 63 samples from fruits and 95 samples from vegetables respectively.

28.37 percent prevalence was reported from soil samples of river banks, where water is easily available for ablution. It leads to heavy concentration of human's fecal matter. The results are in conformity with the finding of Haich H.C. (1965)[8]. Pranavkshari *et.al* (1978)[9] also observed that areas adjacent to rivers, ponds and open sewage canals have higher rate of contamination.

In the case of 526 water samples collected for examination from various sources, 134 samples (25.47 percent) were found to be infected for intestinal parasites. Chaurasia (1993)[10] recorded the prevalence of many

helminthic parasites in the well's water of India and attributed that ablution near the wells may be the main reason. Of the 375 cheap seasonal fruit samples from areas of the district, 63 samples (16.30 percent) were found to be contaminated with the cysts of protozoans and eggs of helminthic parasites.

Returning of human excreta directly to soil as a matter of habit and convenience, bring about contamination of food and water, which in turn infects the population of the Samastipur with intestinal protozoan and helminthic parasites.

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Studies on organic pollution level from different water bodies at Madhepura District (North Bihar)

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Abstract: The present investigation was carried out to evaluate organic pollution level from different water samples in vicinity of Madhepura district. A total of 16 water samples were collected from different source. The sources were Ganga (4), Kosi (4), Waste water (4) and tube well (4) from different localities. All the collected samples were subjected to use quality parameters viz pH, dissolved oxygen, biological oxygen demand, chemical oxygen demand and chlorine following standard methods of BIS. Dissolved oxygen value ranged between 4.05-9.3 mg/l. Biological oxygen demand ranged between 1.85-3.8 mg/l. Chemical oxygen demand value between 3.2-95.3 mg/l and chloride value ranged between 1.85-76.82 mg/l.

Key words: Madhepura district, dissolved oxygen, biological demand, chemical oxygen demand, chloride.

INTRODUCTION

The quality of life is closely associated with the quality of environment. With increasing industrialization and pollution growth, water source available for various purposes such as drinking, recreation, aquaculture, agriculture, have been adulterated with industrial as well as human wastes. Pollution water contains solids and dissolved organic compounds that impart an offensive odor and serve as an excellent medium for the growth and multiplication of microorganisms.[1]

A survey conducted by World Health Organization in 2001, at least 20,000 people die every day in developing countries of the world because of unsanitary water supplies. Sewage consists of approximately 99.9% water, 0.02-0.03% suspended solids and other soluble organic and inorganic substances. Dissolved oxygen is an important indicator for aquatic plants and animals need to survive. Respiration by aquatic animals, decomposition, and various chemical reactions consume oxygen.[2-3]

If more oxygen is consumed than is produced dissolved oxygen level decline and some sensitive animals may disappear.[4] In view of above points in consideration the physico chemical evaluation viz pH, dissolved oxygen biological oxygen demand, chemical oxygen demand and chloride of water samples in the vicinity of Madhepura district were carried out.

MATERIALS AND METHODS

Sixteen samples (4- Ganga water, 4-Kosi water, 4-waste water and 4-Tube well water) of 1000 ml each were collected from different areas and sources in Madhepura district. The collected samples were immediately transferred to laboratory and subjected for physico-chemical characteristics as per the standard methods (BIS

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1991, APHA 1992). Dissolved oxygen was calculated using Winker-Azide method.

For measuring biological oxygen demand, water samples were including at 27^0C for 3 days. Oxygen is also consumed during nitrification; therefore, 1 ml of 0.05% allylthio urea should be added to water sample to check over estimation of BOD. Chemical oxygen demand was estimated by titration method and the chloride ions were estimated by titration with silver nitrate solution.

RESULT AND DISCUSSION

The study reveals that the pH of samples from all sources was ranging in between 6.4-7.8 that is within prescribed limit of BIS. The sample from all sources tested was found to be in ranged between 4.05 to 9.3 mg/l. The amount of dissolved oxygen for different water bodies (Table 1) show that most highly oxygenated among them was the tube well water. If the oxygen level in water less than 4 mg/1, then it is highly contaminated, and that water is not fit for aquatic life.[5]

In the present study biological oxygen demand (BOD) value ranged between 1.8 to 3.8 mg/1. If there is a large quantity of organic waste in the water supply, there will also be a lot of bacteria present working to decompose in this waste. In this case the demand for oxygen will be high, so the BOD levels will begin to decline. When BOD level are high, DO levels decrease because the bacteria are consuming the oxygen that is available in the water? Since less DO is available in the water, fish and other aquatic organism may not survive.

Microorganisms such as bacteria are responsible for decomposing organic waste. As the waste is consumed through the water, BOD levels will begin to decline, and DO levels will to rise. The DO levels will fluctuate daily and seasonally. They are also varying with water temperature.

Cold water holds more oxygen than that of warm water. The most critical time for many aquatic animals are early morning on hot summer days, when river flows low, water temperature are high, and plants have not been producing oxygen since sunset. Maximum pollution occurs in Allahabad city due to presence of sewage water.

Chemical oxygen demand (COD) is one the major parameters per determination organic pollution in water bodies from (Table 1) it could be seen that, the highest value for COD was for waste water and lowest was for tube well water. High value of COD indicates various chemicals present in the water body, this dearly shows that the amount of organic matter is high and hence the water body is more polluted.[6]

The chloride constitute one of the major pollutants present in water, they are lethal if in excess. Chloride usually occurs as NaCl, $CaCl_2$ and $MgCl_2$. High chloride content may harm metallic pipes and structures, as well as growing plants. The presence of chloride also affects the conductivity of water. The table by loose for the chlorides in the water bodies showed that its concentration was more in waste water than that of any other water bodies. The high value of chlorides clearly indicates towards more pollution.

Table 1. Evaluation of physiological characteristics.

Sl. No.	BIS standard	pH	DO (mg/l)	BOD (mg/l)	COD (mg/l)	Chloride
	Source	6.5-8.5	10.0	30.0	250.0	250.0
1.	Ganga Water	7.0-7.5	6.6-8.3	3.3-3.8	10.1-10.6	45.1-51.65
2.	Kosi Water	7.0-7.3	6.2-7.8	3.5-4.05	4.13-4.51	51.3-51.4
3.	Waste Water	6.8-7.2	3.85-4.15	3.4-3.8	70.1-95.8	71.3-76.8
4.	Tube Well Water	7.0	9.1-9.6	1.35-1.85	3.1-3.6	1.55-1.85

Sallu *et al* (1996) studied water quality parameters of Ganga River. They found that pH exceed the permissible limit but the increase was not much harmful, the low dissolve oxygen (0.6-1.7) at Baruni though out the study, which is harm tull for aquatic iife d ittl other purpose. To some extent ont result are in agreement with the study as we also found low dissolved oxygencontent in a few sources (4.5-9.3 mg/I). Sigh et al 1999 observation of biochemical and chemical demands of certain pollution stretch of river Ganga. They found that pH, dissolved oxygen, BOD COD and chlorides content were much below the standard limits our result are in concurrence with this study. Winker, L.W. conrincted a study for determination of dissolved oxygen in water. He found that

the river showed higher concentration of oxygen. In our study the dissolved oxygen value was within the desirable limits. From the studies carried out on limited number of samples it dearly indicates that majority of sources tested for organic pollution levels in response to their potability; fulfills test parameters. Slightly of higher concentration undesirable quantities chloride.

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Studies on the religion wise prevalence rate of total and specific protozoan and helminthic intestinal parasites in rural area of Saran, Bihar, India

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Abstract: Present study on prevalence percentage of total and specific intestinal parasites of rural areas of Saran district of Bihar was studied and for which 15732 single stool samples from 15 C.D. blocks, comprising 9079 males and 6653 females, were collected on random sampling basis and microscopically examined. Religion wise, the highest prevalence rate was observed in the muslims (71.75%), followed by the Hindus (67.8%) and Sikh & Others (64.11%) respectively. while in all the specific protozoan and helminthic parasites in the Hindus, Sikhs & Others, the males showed their dominance over the females.

Key words: Prevalence, protozoan, helminthic, intestinal parasites, Saran, Bihar

INTRODUCTION

A parasite is very small life forms that can either an irritant or destroyer or both. The ones that cannot live in the presence of oxygen are commonly found in the intestinal tract, a place containing very little oxygen, where they nourished and propagate.

Two thirds of the world's population lives in the less developed countries that lack proper sanitary facilities and safe drinking water supply, which leads to transmission of enteric pathogens. Most people are not aware of the danger to their health posed by parasites.

Parasites have been known to man, since soon after the differentiation of life began in the world. One can find a lot of literature of epidemiological work on human parasites, carried out in the past in the U.S.A., U.K. other parts of the world and India as well, but little attention has been paid towards the plight of rural population. So far Bihar state is concerned there has been very little work, note worthy among them are the observation of Chaurasia (1993)[1], who studied the prevalence percentage of intestinal parasites in the rural areas of Vaishali. Saran district of Bihar is devoid of industries and very fertile land. Therefore, a large chunk of male population has immigrated to different parts of the country in search of their livelihood. This is why it was proposed to study the prevalence rate of Instestinal parasites in man in rural areas of this district. Comprising 15 C.D. Blocks, in this district, the rural population constitutes more than 90% of the total population of the district.

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MATERIAL & METHODS

Stool specimens of 15732 human subjects, obtained by random sampling basis from different rural areas of Saran district and belonging to different sex, food-habits, religion and economic strata, were collected and examined

to find out the prevalence rate of intestinal protozoan and helminth parasites. The study was made between January, 2002 to Dec. 2004 and on average 25 to 35 stools were collected and examined every day. The total collected specimens comprised about 0.67 percent of the total rural population of the district. Details of sex, age, religion, food-habit, income, gastro intestinal disorder (if any), source of drinking water, toilet facility etc. of the individuals were recorded on the questionaire.

The collected faecal samples were subjected to microscopic examination. With a view to identify the larvae, cysts, tapeworm proglottids, adult worms, mucus, blood-stains and consistency of sample. Three slides of each stool sample were made by three different methods for their microscopic examination. The methods employed are Direct Smear method in which saline preparation was done for protozoan trophozoites. Iodine stain preparation was first prepared as per recommendation of Dobell and O. Conner. The least expensive and most satisfactory results for helminth eggs and embryo was obtained by employing simple floatation technique of Maplestone for concentration of faecal cysts, ova and larvae, the formol ether concentration method was found to be the best of all the three methods employed. It is the modified version of Ritchie's formal Ether Technique (Ridley and Howgood (1956)[2]. Prasad, Sinha and Kapoor (1978)[3] suggested kerosene or turpentine as a cheaper substitute in lieu of relatively dearer ether in the formol-ether concentration technique.

RESULT & OBSERVATION

To study the religion wise prevalence rate of total intestinal parasites, the rural population of Saran district, was divided into three categories viz Hindu (Hn), Muslims (Mu) and Others (Oth) comparising the Sikhs, Christian, Buddhists, Jains etc. It was observed that there was some variation in the food-habit of the people due to their religion, and it had some impact on the prevalence rate of intestinal parasites.

***Hindu Population*: -** A total of 10243 stool samples of the Hindu population was examined, out of which 6945 samples (67.80%) were found infested with intestinal parasites, possessing one or more species of the protozoan or helminth or both parasites.

In this group of population, the prevalence rate of only protozoan, only helminth, mixed, total protozoan, total helminth and all the specific protozoan and helminth intestinal parasites was recorded as such:

Table 1. Prevalance percentage of infected samples collected from different localities

Name of group / Species of Parasites	No. of Infected Samples	Prevalence percentage
Only Protozoan	1439	14.05
Only Helminth	4478	43.72
Mixed	1019	9.76
Total Protozoan	2458	24.00
Total Helminth	5497	53.66
E. Histolytica	1308	12.77
G. Intesitinalis	1133	11.06
E. Coli	270	2.63
T. hominis	135	1.32
O(P)	11	0.11
Roundworm	3797	37.07
Hookworm	1124	10.97
T. trichiura	179	1.75
H. nana	192	1.87
E. Vermicularis	105	20.40
Taenia Sps.	81	0.79
S. Stercoracis	57	0.56
O (H)	21	0.20

Muslim Population: - About 27% of the studied population of Muslims altogether 4291 samples were examined and 3079 samples (71.75%) contained one or more species of intestinal parasites.

The prevalence rate of only protozoan, only helminth, mixed, total protozoan and total helmith intestinal parasites was observed as 638 samples (14.87%), 1973 samples (45.98%), 488 samples (11.37%), 1126 samples (26.24%) and 2461 samples (57.35%) respectively.

The frequency of specific protozoan and helminth intestinal parasites was seen as such:

Table 2- Prevalance percentage of infected samples collected with name of species

Name of Species	No. of Infected Samples	Prevalence percentage
E. Histolytica	596	13.89
G. Intesitinalis	522	12.16
E. Coli	146	3.40
T. hominis	54	1.26
O(P)	5	0.12
Roundworm	1675	39.04
Hookworm	513	11.95
T. trichiura	102	2.38
H. nana	93	2.17
E. Vermicularis	61	*28.40
Taenia Sps.	42	0.98
S. stercoracis	35	0.81
O (H)	8	0.19

Sikh and Others:- This category of population comprised of the Sikhs, Christians, Buddhists, Jains and people of other minority communities, excluding the Muslims.

Out of 1198 samples, 769 samples (64.19%) were found to be infected with one or more species of intestinal parasites.

Only protozoan, only helminth, mixed, total protozoan and total helminth parasites were observed in 158 samples (13.19%), 500 samples (41.74%), 100 samples (8.35%), 258 samples (21.53%) and 600 samples (50.08%) respectively.

DISCUSSION

Considering the poor living and sanitary conditions of the surroundings, illiteracy density of population, low per-capita income and seasonal variations, it was proposed to study the prevalence pattern of protozoan and helminth intestinal parasites in the rural population of Saran district of Bihar state. Prior to this work, Korke (1925)[4], Sharma in Jharkhand and Chaurasia (1993)[1] and Kumar, P & Inam, N. *et al* in Bihar, east of river Narayani has done such investigation. It is the first such epidemiological survey to be undertaken in western Bihar.

The perusal of relevant literature reveals the note worthy contributions of Lane (1917)[5], Mhaskar (1917)[6], Korke (1925)[4], Srivastava (1953)[7], Baghchi and Prasad (1961)[8], Amin *et al.*(1962)[9], Choudhary and Schiller (1968)[10], Sen (1968)[11],Rao *et al.* (1971)[12], Sanyal *et al.* (1972)[13], Rao and Rao (1975)[14], Veeranan (1977)[15], Deokinandan *et al* (1979)[16], Sinha (1980)[17], Prasad (1983)[18], Verma (1985)[19], Bariar (1987)[20], Bhatia and Juyal (1989)[21], Chaurasia (1993)[1], Mirdha and Samantray (2002)[22] and Singh (2004)[23].

There is very little information about the correlation of prevalence of intestinal parasites with that of the religion. The contributions of Korke (1925)[4], Sinha (1980)[17], Prasad (1983)[18], Verma (1985)[19], Bariar (1987)[20], Chaurasia (1993)[1], Kumar and Singh are note worthy.

Chermin (1954)[24] and Choudhary and Schiller (1968)[10] observed higher prevalence of Roundworm and *T. trichiura* in the muslims, than the Hindus, while Choudhary and Schiller (1968)[10] also observed the higher incidence of Hookworm in the Hindus.

In the present study, the highest prevalence rate of intestinal parasites (71.75%) was observed in the Muslims, followed by (67.80%) in the Hindus and the lowest (64.19%) in Sikh and other community. The same

condition was found with the respect to only protozoan, only helminth and mixed intestinal parasites. Our finding is in very close proximity with that of Chaurasia (1993)[1], who observed 71.87% prevalence rate in the Muslims, as against 71.75% of ours. It is also in confirmity with the findings of Sinha (1980)[17], Prasad (1983)[18], Verma (1985)[19] and Singh (2004)[23].

The highest prevalence rate in the muslims may be assigned to their big families, acute poverty, illiteracy, unhygienic living conditions, habits of living in clusters (Choudhary *et al*, 1968)[10] and stress on non-vegetarian food etc. It was observed that majority of the Muslims were very poor, lived in narrow dark lanes and were fond of keeping poultry with them. This factor may contribute to the unhygienic conditions, suitable for favouring the higher incidence of intestinal parasites in them.

The lowest prevalence rate in the Sikhs, Christians and other minority Communities may be attributed to their better socio-economic conditions and awareness towards sanitation.

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Role of chironomid larvae in the diet of catfish, *Clarias batrachus*

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Abstract: Six months-long experiment was carried out in a fish pond at Kahara Block in Saharsa (Bihar), India from September 2018 to February 2019 to evaluate the role of Chironomid larvae in the diet of catfish, *Clarias batrachus*. The composition of the benthic macro-invertebrates at the bottom indicated that Chironomidae was most dominant group in this pond. The body-weight percentage of the organisms showed that Chironomids and Oligochaetes were major two groups. The quantitative and qualitative studies of Chironomid larvae indicated that there was monthly variation in the abundance of Chironomids. The highest (3695.19 m^{-2}) and the lowest (517.15 m^{-2}) abundance of Chironomids in 3 samples were recorded in the month of January 2019 and October 2018, respectively. Gut content analysis suggested that Chironomids was dominant food item in the diet of *Clarias batrachus*. The maximum 768 and minimum 25 occurrences were recorded in the months of December and October 2018, respectively in 5 fishes sampled from the experimental pond. The electivity indices suggested a shifting to Chironomid larvae from negative selection to positive selection in different months.

Key words: Catfish, *Clarias batrachus*, diet, Chironomid larvae, electivity indices

INTRODUCTION

Bihar is endowed with rich and extensive fisheries resources. Fish and fisheries have been a part and parcel of life and culture of the people of Bihar, particularly northern Bihar from time immemorial. Selection of species is also an important factor of consideration for successful fish culture, because all fish species are not suitable for intensive aquaculture in the fish ponds. Catfish is an important group of fishes and its popularity is increasing day by day showing a promising future for commercial culture[1]. *Clarias batrachus* (Linn.), locally known as 'magur' is an important catfish in the local market for aquaculture, it is a delicious and widely accepted fish. However, this fish is rarely cultured in farmer's level in Saharsa (Bihar), although it can thrive in any type of freshwater habitat and can survive well in extremely oxygen depleted conditions[2]. One of the main setbacks of its culture is the non-availability of fry to stock as per their demand. However, some pioneer works on its breeding technology were done by Rahmatullah *et al.* and Mollah *et al.*[3,4]. Considerable amount of work has been done which included the larvae rearing aspects of this species[5]. However, no detailed study has been conducted on the role of Chironomid larvae in the diet of *C. batrachus* in a fish pond.

Chironomidae is one kind of benthic organisms found in large quantities in the mud. These are one of the most common and essential organisms in an aquatic medium. The bottom fauna forms a very important element in the food chain and a source of potential food for *Clarias*

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batrachus. Thus, the capacity of fish production of an aquatic environment is often evaluated on the basis of the kinds and abundance of Chironomid larvae. The Chironomidae is considered to be one of the most important group of fish food and is most widely distributed and frequently the most abundant group of insects in freshwater environments[6]. Larval Chironomidae have been reported to be important secondary producers in the benthic environment of lakes[7] and to be an important element of invertebrate communities of shallow lakes[8,9]. Despite the enormous literatures on various aspects of the ecology of Chironomidae there is a dearth of information on the biology and ecology of Chironomid populations in fish pond environments[10,11]. The principal reason for such deficiencies is probably due to the difficulty in identification of the various stages coupled with a large number of species frequently encountered within even a small waterbody[6]. Considering the importance of this unexplored area, the study was undertaken in pond, subjected to yearly drying up at the end of one fish culture cycle.

Considering the above facts, it was felt necessary to conduct a research work on the role of Chironomid larvae in the diet of *Clarias batrachus.*

MATERIALS & METHODS

The experiment was carried out from September, 2018 to February, 2019 in the pond water at KaharaBlock in Saharsa (Bihar), Indiafor a period of six months. *Clarias batrachus* at the average size of ±23.39 cm were collected from commercial fish trader and were stocked in the pond. The stocking density 1 fish m^{-2} was maintained for better growth of fish. To ensure better survival and production of fish'sproper feeding management measure were taken.

Collection of Chironomid larvae from pond bottom: Composition of the bottom mud was recorded with visual identification and subsequent quantification was accomplished in accordance with the weight of the types of benthos. Bottom mud was collected from three different locations in the experimental pond (middle and two corner) using an Ekman Grab. Chironomid larvae were collected by using a standard brass sieve (mesh size 0.92 mm). Each sample was washed on the sieve with water in order to washout the soil content from the sieve. Chironomid larvae remaining within the sieve were collected with care and poured in to small vials containing 5% formalin. These were carried to the laboratory for detailed quantitative and qualitative analysis:

Microscopic slide preparation and examination: Chironomid larvae were mounted in polyvinyl lacto phenol, each individual was mounted on a separate slide. The head capsule was separated from the body with the help of a needle. The body and head capsule of each individual was placed on the same slide under separate cover slips. Care was taken to remove excess water from the body by touching with tissue paper before placing on a slide because water along with ethanol makes the mounting difficult. Then, the specimens were ready for identification.

Sampling of fish: Fishes were sampled monthly by using a cast net and 5 fishes were taken for gut content analysis.

Gut content analysis: The gut contents of the collected fish samples were examined in thisexperiment. The composition of food was recorded and the components were quantified according to the body weight of food items present. The abdomen of the individual fish was cut by the help of scissors and gut content were taken out carefully and guts were collected and stocked in vials containing 5% formalin until the gut contents were examined. During analysis the guts were taken out from the vials and kept on petri dish. Then each gut was pieced and a small portion was washed with distilled water. Then, the washed water was sampled on a glass slide and observed under microscope. Thus, the qualitative and quantitative study of the Chironomid larvae was done.

Electivity Index: In order to gain an idea of the proportion of organisms in the diet relative to the proportion measured in the benthos, an index of selective feeding or electivity was calculated using the formula given by Ivlev[12] as cited by Brown and Oldham[13].

$$E = \frac{(Pg - Pm)}{Pg + Pm}$$

Where:

E = Electivity index value

Pg = Relative content of Chironomid larvae in the gut, expressed as percentage of total ration

Pm = Relative proportion of the same organism present in the bottom mud

The numerical assessment of gut contents and collected benthic faunal data provided the required information for measuring electivity index in this study. Electivity or selection for an organism is expressed by values from +1 to -1.

Enumeration of benthos: Bottom mud samples were taken with the help of an "Ekman dredge" which covered an area of 225 cm^2. The Chironomid larvae were enumerated for the study of abundance in the investigated area. The specimens were counted species wise, recorded and converted to their abundance (m^{-2}).

The abundance was expressed as density (individual m^{-2}) by following formula:

$$n = \frac{o}{(a\ x\ s)} x\ 10000$$

Where:

n = Number of individuals in one sq. meter
o = Number of individuals totally/actually counted
a = Area of samples in square centimetre
s = Number of samples taken at one sampling spot

Statistical analysis: The data obtained from monthly abundance of Chironomid fauna from the studied pond was tabulated for statistical analysis. Analysis of variance was done with the help of computer package SPSS version 10.0. The mean values were compared by Duncan's New Multiple Range Test at 5% level of significance.

RESULTS

Chironomid population: Quantitative and qualitative studies of Chironomid larvae were made in every month both from the bottom mud of the experimental pond and from the gut of the sampled fishes. A summary of the data on the population abundance of the Chironomid larvae recorded during the experimental period is provided in Table-1.

Table-1: Abundance of Chironomid larvae in different months from 3 locations in bottom mud of the experimental pond during September 2018 to February 2019.

Month	Average no. of Chironomid larvae
September	592.59
October	517.15
November	1866.67
December	3365.94
January	3695.19
February	2681.48

The monthly abundance of Chironomid larvae in the bottom mud of the experimental pond during the study period was found to be between 517.15 to 3695.19 individual m^{-2}. The second highest abundance was 3365.93 individual m^{-2}. The maximum abundance of Chironomid larvae was found in the month of January 2019. The second highest abundance was recorded in the month of December 2018. The minimum abundance was recorded in the month of October 2018.

Gut content analysis of Chironomid larvae: Monthly recorded number of Chironomid larvae in the gut of 5 sampled fishes during the study period was found to vary between 25 and 768 in number. The second highest occurrence was 410 larvae. The most Chironomids were found in the intestine of the sampled fishes. The maximum occurrence of Chironomid larva was found in the month of December 2018. The second highest occurrence was recorded in the month of November 2018. The minimum occurrence was recorded in the month of October 2018. (Table 2-4).

Table-2: Monthly observation of gut content of chironomid larvae in the experimental pond during September 2018 to February 2019

Month	No. of samples observed	Buccal cavity	Pharynx	Oesophagus	Stomach	Intestine	Total	Average
September	5	0	3	0	15	12	30	6.0
October	5	0	3	0	13	9	25	5.0
November	5	0	7	0	33	370	410	82.0
December	5	0	6	0	52	710	768	153.6
January	5	0	0	0	21	252	275	55.0
February	5	0	0	0	19	18	37	7.4

Table-3: Composition (%) of benthic organisms present in the bottom mud of the experimental pond during September 2018 to February 2019

Month	Chironomidae	Oligochaete	Mosquito & other insects	Mollusca
September	22	25	16	37
October	23	26	13	38
November	32	25	10	33
December	25	33	11	31
January	35	30	9	26
February	29	31	12	28

Table-4: Composition (%) of diet as revealed by the stomach content analysis of the sampled from the experimental pond during September 2018 to February 2019

Month	Chironomidae	Oligochaete	Mosquito & other insects	Organic matter & debris
September	11	5	46	38
October	15	5	50	30
November	25	22	35	18
December	65	27	6	2
January	76	19	5	0
February	59	25	16	0

Table-5: Monthly variation observed in the electivity for chironomid larvae during September 2018 to February 2019

Month	Proportion in gut (%)	Proportion in mud (%)	Electivity Index (E)
September	12	21	-0.272727
October	15	23	-0.210530
November	25	32	-0.122810
December	65	25	0.444444
January	76	35	0.369369
February	59	29	0.340909

Electivity index: Electivity indices were calculated for every month and it varied from -0.27 to +0.44. The high electivity was observed in December 2018 and the lowest value was calculated for the month of November 2018. For the first two months, the fishes showed negative selection in feeding the Chironomid larvae. For the last three months the fishes showed greater preferences to the Chironomids (Table-5).

DISCUSSION

A total of 837 Chironomid larvae were collected from the sampling stations in the pond during 6 months of study period (September, 2018 – February, 2019). Monthly observed total no. of Chironomid larvae in the bottom mud of the experimental pond during the study period was found to be between 517.15 and 3695.19 individual m^{-2}. A distinct seasonal variation in the population of Chironomids was observed in this study. The observed lower number in February 2019 might be due to heavy grazing by *Clarias batrachus*, which agrees with the observation of Brown and Oldham[13]. The ecological conditions of the ponds were probably favourable for the growth of these organisms. The soft bottom mud of the tropical fish ponds has been found to exert a significant influence upon the occurrence of the bottom organisms especially the Chironomids. There were qualitative and quantitative variations of the

Chironomid larvae. Higher abundance in the pond seems to be due to better feeding, environmental conditions and probably breeding. Seasonal changes in the population abundance of Chironomidae were investigated by many workers[13,14,15]. A distinct seasonal variation in total Chironomidae was observed in this study. Olfsson had similar observation where he found the highest number of individuals in June in the pond and lowest number was recorded in August[16].

Food composition of sampled fishes in the experimental pond suggested that Chironomids comprised a major component of the diet of *Clarias*. The food comprised mainly of insects, polychaetes, oligochaetes and debris. This result is similar to that observed by Dewan[17]. The electivity indices showed negative selection (E =-0.27) to Chironomids at first and thereafter it shifted to the greater preferences (E = +0.44) over the time elapsed as the fish grew older in months. Ali and Begum observed positive selection for insects[18]. The information on electivity of *Clarias* on Chironomids seems to be lacking as well. Since, Chironomids were found to dominate in the diet of *Clarias* as it grows, efforts should be made to enhance Chironomid production in the catfish ponds for better fish production.

CONCLUSION

The composition of the benthic macro-invertebrates in the bottom and indicated that Chironomidae was the dominant group. The body-weight percentage of the organisms showed that Chironomids and Oligochaetes were two major groups. The quantitative and qualitative studies of Chironomid larvae indicated that there was monthly variation in the abundance of Chironomids. The highest abundance of Chironomids 3695.19 individual m^{-2} in 3 samples was recorded in the month of January 2019. The lowest abundance 517.15 individual m^{-2} in 3 samples was observed in the month of October 2018.

The gut content analysis showed that Chironomids was the dominant food item in fish diet as they grow up. Maximum Chironomids were found in the intestine. The maximum occurrence of Chironomid larvae, 768 in number, was recorded for 5 fish samples in the month of December 2018. The minimum occurrence was recorded in October 2018.The electivity indices calculated for Chironomid larvae in each month showed that fish had negative selection for Chironomids for the first 2 months; a shift in the electivity to greater preferences to Chironomids was observed thereafter. It may be concluded that a greater abundance of Chironomid larvae may be required in the *Clarias* culture ponds for better fish production.

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Toxicity of ovary and kidney due to excessive intake of fluoride content in female albino rat

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Abstract: The main objective of this research study was to find out the toxic effect of fluoride in female rat during diet intake. When the doses of sodium intake were increased in rat it showed a great toxic effect on the reproductive as well as physical functions of the rat. Higher intake of sodium leads to significant increase in size of the ovary, uterus, kidney etc. Rats which had ingested NaF showed increases in the weights of the ovaries and also the relative weights of the uterus and kidney. The results indicate that exposure of female rats to NaF in drinking water has adverse fetotoxic effects.

Key words: fetotoxic, lysosomal, histopathological.

INTRODUCTION

Although fluoridation of water supplies is practiced in many places in the hope of reducing the incidence of dental caries, there is continuing concern that exposure to fluoride could cause toxic effects. An epidemiological study to assess whether fluoride could affect human birth rates using a U.S. database of drinking water systems showed an association of decreasing total fertility rate with increasing fluoride levels.[1] Fluoride is considered as one of the important constituent for prevention of dental carries tooth decay.[2] These studies have suggested that fluoride toxicity may cause adverse effects in the reproductive system of organisms living in fluorosis endemic areas. Additionally, it was also found that in vitro exposure of human sperm to fluoride (250 mm) lead to altered lysosomal activity, glutathione levels and caused morphological anomalies producing a significant decline in sperm motility[3]. Multiple animal models have also shown that fluoride toxicity decreases fertility in most species studied, but so far these studies have concentrated mainly on the reproductive toxic effects of fluoride on male animals. Thus endemic fluorosis has been seen many parts of the world that cases damage to different tissues such as skeleton, teeth along with other soft tissues such as kidney, spleen brain etc.[4,5]

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MATERIAL AND METHOD

Experimental study

15 female albino rats having the average weight of about 120-150 gm were bought from the local market and were kept in the laboratory in the university department for its further study. All the rats were kept in the cage under normal laboratory conditions with care and fed water and pellet.

15 rats were randomly divided in to three groups. Each of the group contained 5 rats each. Group 1 the rats were treated with distilled water that contained no fluoride concentration. Group 2 the rats were treated with 10 mg of sodium fluoride in water, and group 3 of the rats were treated with 20 mg of sodium fluoride for at least 30 days. Each of the animals was weighed before & after experiment was carried out. After 30 days of the protocol each of the rats were sacrificed and the tissues of the ovary were dissected for the further study of the histopathology as described earlier as well as different concentrations of the hormone found in ovary.[6,7]

Histopathological study of the ovary

The dissected tissues of the ovary were then collected and immediately fixed by immersion in Bouin's liquid for 24 hours. The samples were then processed according to different histological technique followed by dehydration, xylene was used for clearing and then it was fixed in paraffin wax. About 5 micrometers thick secretion of the ovarian tissues was stained in hematoxylin-eosin for the morphological study.[8] The sections were then mounted in DPX for the further study.

Collagen study of kidney

For the study of collagen in kidney the tissues of the kidney was dissected, cleared and weighed. The tissues of the kidney was then homogenized and then homogenate for the biochemical analysis.

RESULT & DISCUSSION

From the above analysis the result obtained showed that the structure of the ovary observed in group 1, group 2 and group 3 showed various differentiations in the histopathological structure. When observed in control group it was found that there was no change in the structure of the ovary while in group 2 & group 3 there was great differentiation. In group 2 the ooplasm showed degenerative changes without any nucleus & nucleolus. In group 3 rats fed with 20 mg of sodium fluoride also showed degenerative changes in the ovum. The ooplasm was seen shifted towards the one side and the nucleus & the nucleolus was not visible. The estrogen level of fluoride treated rats showed decline in its concentration. Finally the research of fluoride on female albino rat shows that fluoride has adverse effect on the different reproductive organelles of the rats as well as it reproductive behavior. The analysis of kidney showed great variation in total collagen count. NaF when fed in 10 mg caused a significant decrease of about 78% in total collagen count and when the rat was fed with 20 mg of NaF showed complete list of the collagen fibril packing with respect to the control group.

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Developmental changes in adult male albino rat due to manganese

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Abstract: Manganese is a compound which is essential for normal physiological functions of the body but however when taken in excessive amount may lead to physiological disorders which may even lead to abnormal development of the reproductive tissues & several structural changes with certain hormonal changes too. Food, water, tea etc. etc are the basic sources of Mn in our daily life, however pesticides and other occupational exposure are also considered as toxic potential source.

Key words: Manganese, Parkinson disease, myometrium, morphological changes

INTRODUCTION

Metals are the constituents that are found in nature in considerable amount all over the earth, crest, in plants, in animals, water bodies etc, such as lead , mercury, cobalt, zinc etc. Most of the these metals are very important for the physiological as well as biological functions of the body & when taken in inadequate amount may lead deficiency or some other disease in different organism at different levels[1]. Excessive Mn may also cause manganism, which is a type of neurodegenerative disorder which has the symptoms like those of Parkinson's disease[2,3]. Metal manufacturing industries, compounds containing metals such as pesticides, fertilizers are the important source of metal toxicity that reaches to human & animals through its diffusion & transportation to water bodies.[4] Mangaese is regarded as one of the very important dietary element. In the body of human approximately 12 mg of Mn is present and usually in bones. There are several studies which suggest that most of the neurological effects or dysfunction is usually related to dose [5]. The normal requirements of Mn are constituently fulfilled by diet, which is the basic source of Mn while in very minor amount contributed by air & water. There are several studies which shows that manganese in air & water may lead to intellectual impairment in school children. There are numbers of research done that shows the several biochemical changes due to manganese intake. There are various paper related to toxic effects of manganese on the mammalian reproductive cells leading to lower sperm count and quality of semen[6].

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MATERIAL & METHODS

Animals treated

For carrying out the research based on effect of manganese on development of mice 20 female pregnant mice were bought from the local market of Madhepura .

These mice were then kept in the cage in university department under the 12 hours day and night cycles at normal room temperature.

Experiment performed

The selected pregnant female mice were then divided into four groups group a, group b, group c and group d. In group a, mice of control group were kept which were fed with normal tap water and food. Group b mice were fed with 1.0mg/kg manganese mixed in diet, group c mice were fed with 10mg/kg of Mn^{2+} while group d mice were fed with 30mg/kg of Mn^{2+}.

Physical Evaluation

For the physical evaluation all the physical status including weight of the mice were noted down so that study on physical status can be observed easily.

Histological evaluation

For histological evaluation the treated mice were sacrificed & the different reproductive tissues including ovary, oviduct & uterus were dissected out for the further studies following different steps. The step includes the following:-

- Sacrificing of animals
- Taking out of reproductive tissue.
- Keeping the tissue in 4% paraformaldehyde
- Dehydrating of the tissue in xylene or ethanol
- Embedding of the tissue in paraffin
- Sectioning of the tissue
- Staining of the sectioned tissue by Eosin
- Observation under the microscope.

For the analysis of FSH & LH levels ELISA test was performed for its complete study.

RESULT & DISCUSSION

The result on the toxic effect of manganese on the female mice shows that there was great variation in the functional activity of both groups of the mice.

Weight of the tissue

When the weight of the ovary, uterus oviduct was done there was found great variation in the mice treated with manganese compared to control group. The weight of the ovary of group 4 was very less in contrast to other group of mice which has been shown below:-

Group a> Group b> Group c> Group d

Similar result was obtained with the weight of uterus & oviduct.

Histological observation

When the thin section of the reproductive tissues was observed under the electronic microscope there was seen certain morphologiocal changes in the structure, these findings has also been given in different investigations[7]. The myometrium layer became thick with exposure to Mn^{2+}. Thus there was increase in the thickness of mymetrium with increase in the magnesium dose. Several studies on the thickening of the myometrium shows that it can leads to reduction of proliferation activity bringing about irregularity in the fetus development.

There was also seen certain drawback in FSH & LH level with intake of higher doses of Mg. FSH level decreased with increase in the dose, thus mice kept in group d had less FSH formation which is one of the symptoms of infertility. One of the factors resulting in the hormonal changes may be due to structural changes of the ovary due to excessive Mn intake. While observation on LH showed that there was very minute change in its level due to Mn^{2+} intake.

Thus the studies shows that exposure of Mn^{2+} in the female mice lead to variation in the weight as well as histological morphology of the reproductive tissues which is one of the drawback for increasing the fertility rate of mammals and thus functioning as toxic to both the fetus & embryo[8].

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Analysis of water quality parameters of tributaries of the river Kosi Madhepura district, North Bihar

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Abstract: The present study was undertaken to analyze water quality parameters in tributaries of Kosi River in Madhepura district of North Bihar, India. Three different tributaries, named locally as Kirkhidhar, Gomtidhar and river Sursar were selected for investigations. These, over flow channels and old river bed of Kosi River, are one of the prime sources of water in the study area. Though, these water resources are continuously subjected to tremendous anthropogenic pressure leads to degradation of water quality and depletion of biota. The samples were collected monthly from January 2018 to December 2018. Physico-chemical parameters analyzed were temperature, transparency, conductivity, pH, dissolved oxygen, free carbon dioxide, total alkalinity, calcium, magnesium, chloride, nitrate, phosphate and BOD. Frequent flooding during monsoon season decreases some variables to a considerable extent. Correlation coefficient analysis showed high significant positive and negative correlation. The recorded parameters were within the prescribed limit of WHO and BIS except nitrate-nitrogen. With respect to water quality, tributaries studied was not highly polluted, water is suitable for irrigation and domestic purposes. But, there is an indication of increasing pollutant due to disposal of human and animal wastes and agricultural runoff, therefore, a proper monitoring of water quality is needed to avoid contamination.

Key words: Kosi River, Tributaries, Physico-chemical parameters, Correlation, Threats

INTRODUCTION

India has a vast network of rivers and its tributaries[1]. These are one of the important natural water resources uses for drinking, irrigation, transportation, electricity and provide livelihoods for thousand and thousand people all over country. Several studies have reported status of rivers after receiving a variety of pollutants altering physico-chemical and biological characteristics[2,3].

Kosi basin located in north of the Ganga River in Bihar state has a vast aquatic resource in terms of rivers and its tributaries. Kosi River is attributed as a major river of Kosi basin. It rises in Tibet, travels a distance of 724 Km southwards before merging with Ganga near Kursela. On its way to plain, Kosi River is joined by a good number of old river beds and over flow channels 'dhars'. These water bodies were reportedly yielding a good quantity of fish till recently[4].

Kosi River and its tributaries drain the plain of north Bihar is an important water resources for irrigation and domestic purposes. Tributaries of this river are currently facing tremendous pressure due to anthropogenic activities and agricultural runoff leading to degradation of water quality and depletion of biota. Most of the tributaries remain less examined aquatic ecosystems with regard to the baseline information. Therefore, the present study is

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an attempt to provide the basic data on physico-chemical characteristics of water in three tributaries of river Kosi located in Madhepura district of North Bihar.

MATERIALS AND METHODS

The present study has been carried out in Kosi basin (24º55' N to 27º31' N latitudes and 83º21'E to 88º17' E longitudes) to examine water quality status of three different tributaries of river Kosi at Madhepura district of North Bihar. The selected tributaries are termed locally as Bhirkhi dhar, Gomati dhar and river Sursar. Water current and size of these tributaries vary extensively during monsoon and non-monsoon season and rises many times during the flood periods. Samples were taken at monthly basis from January 2018 to December 2018.

Samples were collected by using polyethylene bottle of two liter capacity. Sampling bottles were first cleaned with water and then socked in 1:1 HNO_3 for 24 hours. Bottles were finely rinsed with distilled water. Again bottles were rinsed with sample water before filling. Sample was collected by immersing bottle directly from the depth of 10-15 cm below the surface water. The physico-chemical parameters analyzed on spot were water temperature, transparency, pH, DO and alkalinity. Samples were brought to the laboratory to analyze electrical conductivity, calcium, magnesium, chloride, nitrate-nitrogen, phosphate-phosphate and BOD. Analysis of the samples was done as per standard methods[5, 6].

RESULTS AND DISCUSSION

The results of the present study with respect to range (the maximum and the minimum), mean and standard deviation are presented in Table 1 and correlation data in Table 2 to 4.

Temperature regulates rate of chemical reactions and biological activities within water. High temperature decreases solubility of DO[7]. Water temperature ranged from 18.3-33.2 (26.9±4.5)ºC in Bhirkhi dhar, 19.2-34.9 (27.9±4.7)ºC in Gomti dhar and 19.5-35.8 (27.8±4.9) ºC in river Sursar (Table 1). The minimum temperature was recorded during winter and maximum observed in summer. Variation in temperature depends on the season, geographic location, atmospheric temperature and sampling time[8]. Water temperature was fairly high in all tributaries as located in subtropical region. Temperature showed a high negative correlation with DO and pH and a high positive correlation with magnesium and BOD (Table 2 to 4).

Secchi disc transparency measures the ability of water to transmit sunlight through the water column. The penetration of light is necessary for photosynthesis. Water transparency varied from 6.1-41.8 (25.1±11.7) cm in Bhirkhi dhar, 9.12-42.3 (26.7±10.3) cm in Gomti dhar and 9.6-43.2 (26.4±11.1) cm in Sursar river (Table1). The minimum transparency was recorded in monsoon and the maximum recorded during winter. The reduction in the light *penetration in water during monsoon* limits photosynthesis causes lowering of pH and DO consequently affects aquatic life. From monsoon onward water transparency increases and reaches higher in winter. Transparency of tributaries water more controlled by the suspended silt particles and less by the phytoplankton. Water transparency showed a high positive correlation with EC (r=0.805; p>0.001) in Bhirkhi dhar and a moderate negative correlation with free carbon dioxide (r=-0.781;p>0.001) in Gomti dhar (Table 2 and 3).

Water capability to transmit electric current is known as electrical conductivity and serves as tool to assess the purity of water. EC is affected by suspended impurities and depends on amount of ions present in water. EC ranged from 89.7-216.6(160.3±44.96) ìmohs/cm in Bhirkhi dhar, 115.5-396.9 (242.2±88.06) ìmohs/cm in Gomti dhar and 126.7-445.9 (290.8±96.16) ìmohs/cm in river Sursar (Table 1). EC was potable, soft and good to water quality. The maximum EC was observed during summer and the minimum recorded in monsoon.EC was affected by dilution factors, low EC during monsoon could be due to increase water level dilutes dissolved salts and ions, whereas, high EC in summer might be related to decrease water level concentrates content[9]. EC was much below the value recommended by WHO[10]. EC showed a high positive correlation with chloride, nitrate-nitrogen and phosphate-phosphorus in all tributaries, however, revealed a high positive correlation with calcium and alkalinity and a negative correlation with free carbon dioxide in Gomati dhar and river Sursar (Table 2 to 4).

pH measures relative amounts of free hydrogen and hydroxyl ions present in water. Water with more free hydrogen ions is acidic and with more free hydroxyl ions is basic[11]. pH varied within the range of 7.1-7.8 (7.4±0.2) in Bhirkhi dhar, 7.3-8.1 (7.7±0.2) in Gomti dhar and 7.2

to 8.0 (7.7±0.2) in river Sursar. pH range was alkaline in nature ranging within the permissible limit (6.5-8.5) recommended by WHO[10].The pH is good for promoting high primary productivity. The minimum pH was recorded during summer and the maximum observed in winter. Low pH observed during summer might be due to high temperature and decrease water level, whereas, high pH in winter could be due to greater phytoplankton density consequently removal of free carbon dioxide by photosynthetic uptake increasingpH[12]. With respect to the pH, water of all tributaries is suitable for drinking, bathing, washing, and irrigation. The pH exhibited a moderate positive correlation with DO (r=0.759; p>0.001) in Kirkhi dhar and a strong positive correlation (r=0.845; p>0.001) in Gomti dhar and (r=0.888;p>0.001) in river Sursar (Table 2 to 4).

Dissolved oxygen (DO) affects the growth, survival, distribution, behavior and physiology of aquatic organisms. The principal source of oxygen in water is atmospheric air and photosynthetic plankton and plant. DO range within the range of 4.7-6.7(5.3±0.5) mg/l in Bhirkhi dhar, 5.0-7.9 (5.9±0.8) mg/l in Gomti dhar and 5.0-7.5 (6.1±0.7) mg/l in rivers Sursar (Table 1). The maximum DO was recorded during winter and the minimum was observed in summer. High DO coincided with low temperature probably due to oxygen solubility increases with decreasing temperature[13].

The highest DO was recorded in summer due to low solubility of oxygen at high temperature and elevated microbial activities for degrading organic matter consume more oxygen.

Free carbon dioxide exists in water in the form of dissolved gas. The carbon dioxide released in water during respiration by biota and during decomposition of organic matter by microbes. Free carbon dioxide varied from 1.2-3.5 (2.5±1.1) mg/l in Bhirkhi dhar, 1.7-6.5(4.2±1.7) mg/l in Gomti dhar and 1.2-4.7(2.9±1.3) mg/l in river Sursar (Table 1). Tributaries studied were contain less amounts of free carbon dioxide which is below the permissible limit (22 mg/l) recommended by WHO[10]. The maximum free carbon dioxide was recorded during monsoon and the minimum observed in winter, though, it was not recorded in some months. The highest free carbon dioxide in monsoon could be due to addition of agricultural runoff contains biodegradable organic wastes consequently elevated microbial activities releases more carbon dioxide. The free carbon dioxide showed a strong negative correlation with alkalinity (r=-0.863; p>0.001) and calcium (r=-0.819; p>0.001) in Gomti dhar (Table-3).

Total alkalinity (TA) is important for aquatic life because of its buffering capacity against the rapid pH change[14, 15].The alkalinity varied from 95.5-137.9 (118.5±11.4) mg/l in Bhirkhi dhar, 102.4-150.9 (129.1±12.9) mg/l in Gomti dhar and 101.0-154.4 (130.6±29.1) mg/l in river Sursar (Table 1). The alkalinity was within the permissible range (50-200 mg/l) recommended by BIS[16]. The maximum alkalinity was recorded during summer and the minimum observed in monsoon. Water alkalinity was primarily due to bicarbonate concentration. Total alkalinity showed a strong positive correlation with calcium, chloride, nitrate-nitrogen and phosphate-phosphorus in Gomati dhar and in river Sursar (Table 3 and 4).

The calcium dissolved in natural water is one of the most common mineral that contribute to water hardness. Calcium plays various structural and functional roles in living organism[17].The calcium content ranged from 10.7-23.3 (15.7±3.6)mg/l in Bhirkhi dhar, 8.2-25.2 (18.3±4.3)mg/l in Gomti dhar and 9.6-26.3 (17.9±5.1)mg/l in river Sursar (Table 1). Calcium was much below the guideline value (75mg/l) recommended by BIS[16]. The minimum calcium was recorded in monsoon and the maximum observed during summer. Variation in calcium content might be due to dilution effects, decrease in calcium during monsoon could be attributed to increase water level dilutes content, whereas increase in calcium in summer could be related to decrease water level concentrates content. Calcium exhibited a strong positive correlation with chloride, nitrate-nitrogen and phosphate-phosphorus (Table 2 to 4).

Magnesium is often associated with calcium in all kinds of natural waters. Magnesium requires for *chlorophyll* synthesis and acts as a limiting factor for the growth of phytoplankton[17].The magnesium ranged from 4.1-14.1(9.0±2.6) mg/l in Bhirkhi dhar, 5.2-16.1(10.7±3.5) mg/l in Gomti dhar and 6.1-15.6 (10.1±2.9) mg/l in river Sursar (Table 1). Magnesium was below the guideline value (35 mg/l) recommended by BIS[16]. In terms of the magnesium, water is good for irrigation. The maximum magnesium was recorded in summer and the minimum

during winter. The highest amount of magnesium was recorded during summer might be due to decrease water level concentrates content. Magnesium showed a high positive correlation with BOD (r=0.868; p>0.001) in Sursar river (Table 4).

Chloride is widely distributed in natural water as salts of sodium chloride (NaCl), potassium chloride (KCl) and calcium chloride ($CaCl_2$). Higher concentration of chloride indicates organic pollution mainly of animal origin[18]. Chloride ranged within the range of 6.9-21.9(12.9±4.3) mg/l in Bhirkhi dhar, 8.8-28.5(16.8±6.1) mg/l in Gomti dhar and 8.7-28.1 (17.9±6.3)mg/l in Sursar river. The chloride was much below the maximum permissible limit (250 mg/l) recommended by WHO[10]. This could be attributed to absence of industrial and domestic wastewater discharge in the studied tributaries. In terms of chloride, all tributaries are unpolluted. The maximum chloride was recorded during summer and the minimum observed was in monsoon. Variation in chloride was due to dilution factors, the increase water level during monsoon dilutes content and decrease water level in summer concentrates content. Concentration of chloride was high in river Sursar could be attributed to human and animals waste contamination. The chloride ions cannot remove from water body biologically, as it is highly soluble in water. Chlorides how a strong positive correlation with nitrate-nitrogen and phosphate-phosphorus in all tributaries (Table 2 to 4).

The nitrate ion is the common form of nitrogen found in all kinds of natural waters, though, only little *amount of nitrate occur in water*[19]. Nitrate-nitrogen varied in the range of 0.007-0.177 (0.101±0.1)mg/l in Bhirkhi dhar, 0.014-0.273 (0.123±0.1)mg/l in Gomti dhar and 0.008-0.198 (0.094±0.1)mg/l in river Sursar (Table 1). Nitrate-nitrogen is much below the permissible limit (45 mg/l) recommended by WHO[10]. The maximum nitrate-nitrogen was recorded during summer and the minimum observed in monsoon. The lowest nitrate-nitrogen was recorded in monsoon might be due to addition of rain and floodwater dilutes content, however, highest during summer probably due to decaying nitrogenous organic matter adding nitrogen and decrease water level concentrates content. The concentration of nitrate-nitrogen was relatively higher in Gomti dhar due to added agricultural runoff and anthropogenic contamination. The nitrate-nitrogen showed a high positive correlation with phosphate-phosphorus(r=0.908; p>0.001) in Bhirkhi dhar, (r=0.925; p>0.001) in Gomti dhar and (r=0.888; p>0.001) in Sursar river (Table 2 to 4).

Phosphorus is the first limiting nutrient supports *growth* of algae and *plants* which offer food and *habitat* for fish and smaller organism living in *water*. The most common form of phosphorus used by biological organisms is phosphate. Phosphate-phosphorus concentration ranged from 0.039-0.394(0.173±0.1) mg/l in Bhirkhi dhar, 0.051-0.427 (0.191±0.1)mg/l in Gomti dhar, and 0.052-0.408 (0.194±0.1) mg/l in river Sursar. Phosphate-phosphorus was above the guideline value (0.1 mg/l) prescribed by WHO[10]. The maximum phosphate-phosphorus was recorded in summer and the minimum observed during monsoon. Agricultural runoff from adjacent paddy fields contain high phosphorus is main source of phosphate in the studied tributaries. Variation in phosphate-phosphorus was due to dilution factors, decrease water volume concentrates content might be the reason of high phosphate during summer, however frequent flooding in monsoon dilutes content. At the recorded level, phosphate-phosphorus not possesses any threat to human health. Phosphate-phosphorus showed a positive correlation with BOD in Gomati dhar (0.725; p>0.001) and in Sursar river (0.793; p>0.001).

Biochemical Oxygen Demand (BOD) is amounts of dissolved oxygen required by microbes to break down organic material in the water body. BOD value indicates degree of water pollution could be attributed to the maximum and the minimum biological activities. BOD of unpolluted waters should be less than 2mg/l while those receiving waste waters may have value up to 10 mg/l[20]. BODin the tributaries water ranged from 1.5-3.2 (2.2±0.5) mg/l in Bhirkhi dhar, 1.5-3.6 (2.5 ±0.6) mg/l in Gomti dhar and 1.7-3.9 (2.6±0.7) mg/l in river Sursar. BOD was much below the permissible limit (6 mg/l) recommended by WHO[10]. The maximum BOD was recorded during summer and the minimum recorded was in winter. BOD directly affects the amounts of dissolved oxygen in tributaries water. The BOD was highest during summer might be due to the high temperature and high decomposition rate of organic matter by microbes consume more oxygen[21, 22]. However the BOD was lowest in winter due to low temperature reducing microbial activity that consumes less oxygen.

Table-1: Range (min. and max.), mean and standard deviation of physico-chemical parameters.

	Bhirkhi dhar			Gomati dhar			Sursar river			WHO (2004) limits
Parameters	**Min.**	**Max.**	**Mean ± SD**	**Min.**	**Max.**	**Mean ± SD**	**Min.**	**Max.**	**Mean SD**	
Water Temperature	18.3	33.2	26.9 ± 4.5	19.2	34.9	27.9 ± 4.7	19.5	35.8	27.8 ± 4.9	30-35
Transparency	6.1	41.8	25.1 ± 11.7	9.12	42.3	26.7 ± 10.5	9.6	43.2	26.4 ± 11.1	N/A
Electrical Conductivity	89.7	216.6	160.3 ± 44.9	115.5	396.9	242.2 ± 88.0	126.7	445.6	290.8 ± 96.1	750
pH	7.1	7.8	7.4 ± 0.2	7.3	8.1	7.7 ± 0.2	7.2	8.0	7.7 ± 0.2	6.5-8.5
Dissolved Oxygen	4.6	6.7	5.3 ± 0.5	5.0	7.2	5.9 ± 0.8	5.1	7.5	6.1 ± 0.7	5.0-7.0
Free Carbon Dioxide	1.2	3.5	2.5 ± 1.1	1.7	6.5	4.2 ± 1.7	1.4	4.7	2.9 ± 1.3	22
Total Alkalinity	95.5	137.9	118.5 ± 11.4	102.4	150.9	129.1 ± 12.9	101.0	154.4	130.6 ± 29.1	50-200 (BIS, 2004)
Calcium	10.7	23.3	15.7 ± 3.6	8.2	25.2	18.3 ± 4.3	9.6	26.3	17.9 ± 5.1	75 (BIS, 2004)
Magnesium	4.1	14.1	9.0 ± 2.6	5.2	16.1	10.7 ± 3.5	6.1	15.6	10.1 ± 2.9	35 (BIS, 2004)
Chloride	6.9	21.9	12.9 ± 4.3	8.8	28.5	16.8 ± 6.1	8.7	28.1	17.9 ± 6.3	250
Nitrate-Nitrogen	0.007	0.177	0.101 ± 0.1	0.014	0.273	0.123 ± 0.1	0.008	0.198	0.094 ± 0.1	45
Phosphate-Phosphorus	0.039	0.394	0.173 ± 0.1	0.051	0.427	0.191 ± 0.1	0.052	0.408	0.194 ± 0.1	0.1
BOD	1.5	3.2	2.2 ± 0.5	1.7	3.6	2.5 ± 0.6	1.7	3.9	2.6 ± 0.7	6.0

*Values in mg/l,except Temp. (°C), Transp. (cm), EC (µmohs/cm) and pH.

Table-2: Correlation coefficient (*r*) matrix of physico-chemical parameters of Bhirkhi dhar.

Parameter	Temp.	Trans.	E.C.	pH	DO	Free CO_2	TA	Ca^{2+}	Mg^{2+}	Cl^{1-}	NO_3^--N	PO_4^{2-}-P	BOD
W. Temp.	_	-0.515	-0.149	-0.732	-0.892*	0.265	-0.210	0.127	0.878*	0.290	0.216	0.300	0.738
W. Trans.		_	0.805*	0.482	0.650	-0.740	0.527	0.504	-0.179	0.538	0.561	0.564	-0.221
EC			_	0.088	0.367	-0.863*	0.731	0.788	0.150	0.832*	0.861*	0.838*	0.225
pH				_	0.759*	-0.187	0.223	-0.069	-0.674	-0.127	-0.246	-0.169	-0.534
DO					_	-0.446	0.374	0.147	-0.743	0.002	0.013	-0.021	-0.587
Free CO_2							-0.714	-0.719	0.022	-0.662	-0.681	-0.659	-0.115
TA							_	0.736	-0.059	0.652	0.674	0.543	0.248
Ca^{2+}								_	0.374	0.892*	0.877*	0.844*	0.519
Mg^{2+}									_	0.511	0.501	0.569	0.732
Cl^{1-}										_	0.916*	0.951*	0.559
NO_3^--N											_	0.908*	0.515
PO_4^{2-}-P												_	0.485
BOD													_

*Significant at p>0.001

Table-3: Correlation coefficient (*r*) matrix of physico-chemical parameters of Gomti dhar.

Parameter	Temp.	Trans.	E.C.	pH	DO	Free CO_2	TA	Ca^{2+}	Mg^{2+}	Cl^{1-}	NO_3^--N	PO_4^{2-}-P	BOD
W. Temp.	_	-0.627	0.154	-0.786	-0.957*	0.314	0.046	0.077	0.754	0.289	0.245	0.349	0.728
W. Trans.		_	0.613	0.711	0.736	-0.781	0.604	0.598	0.118	0.518	0.521	0.462	-0.100
E.C.			_	0.157	0.003	-0.744	0.886*	0.879*	0.528	0.954*	0.951*	0.947*	0.538
pH				_	0.845*	-0.399	0.166	0.122	-0.481	0.068	-0.008	0.001	-0.479
DO					_	-0.436	0.091	0.067	-0.639	-0.123	-0.112	-0.191	-0.604
Free CO_2						_	-0.863*	-0.819*	-0.155	-0.642	-0.693	-0.619	-0.102
TA							_	0.917*	0.468	0.837*	0.841*	0.852*	0.438
Ca^{2+}								_	0.561	0.844*	0.889*	0.825*	0.424
Mg^{2+}									_	0.615	0.588	0.642	0.716
Cl^{1-}										_	0.944*	0.976*	0.661
NO_3^--N											_	0.925*	0.596
PO_4^{2-}-P												_	0.725
BOD													_

*Significant at p>0.001

Table-4: Correlation coefficient (*r*) matrix of physico-chemical parameters of river Sursar.

Parameter	Temp.	Trans.	E.C.	pH	DO	Free CO_2	TA	Ca^{2+}	Mg^{2+}	Cl^{1-}	NO_3^--N	PO_4^{2-}-P	BOD
W. Temp.	_	-0.565	0.127	-0.809*	-0.913*	0.204	0.027	0.017	0.846*	0.299	0.230	0.409	0.817*
W. Trans.		_	0.711	0.539	0.675	-0.692	0.693	0.693	-0.202	0.555	0.583	0.477	-0.066
E.C.			_	-0.010	0.090	-0.711	0.881*	0.874*	0.464	0.918*	0.878*	0.926*	0.626
pH				_	0.888*	-0.117	0.121	0.078	-0.596	-0.149	-0.092	-0.211	-0.614
DO					_	-0.250	0.189	0.135	-0.690	-0.137	-0.070	-0.183	-0.638
Free CO_2						_	-0.658	-0.717	-0.060	-0.689	-0.609	-0.574	-0.231
TA							_	0.894*	0.350	0.889*	0.895*	0.846*	0.487
Ca^{2+}								_	0.304	0.887*	0.900*	0.785*	0.496
Mg^{2+}									_	0.576	0.520	0.647	0.868*
Cl^{1-}										_	0.948*	0.938*	0.716
NO_3^--N											_	0.888*	0.633
PO_4^{2-}-P												_	0.793
BOD													_

*Significant at p>0.001

CONCLUSION

While assessing the results of various parameters, it was observed that all tributaries (dhars and river) have a good water quality suitable for irrigation, fisheries, cattle watering and other domestic uses. Physico-chemical parameters recorded were within the permissible limit, except phosphate. The studied tributaries are not polluted till date but there is an indication of increasing pollutant mainly due to the disposal of anthropogenic wastes and agricultural runoff, therefore, to prevent further contamination a proper monitoring of water quality of tributaries is needed.

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Profitable management practices of fish culture ponds in Koshi region of Araria District in Bihar

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Abstract: Koshi region of Araria district has large number of ponds which receives flood and rain water. The pond of this region is very fertile and well known for fish culture. A survey has been conducted to study the present package of practices of fish culture in the area through a pre tested questionnaire. The preliminary information of the survey has been depicted in the table. It is interesting to note that only 15-20% farmers are adopting the recommended package of practices and 25% adopting partially where as large number of farmers have not adopted it due to unknown reason. The annual production of the fish in this region ranges from 1-8 tone/ha showing the ample scope of improving the annual yeild if the farmers adopt the recommended practices properly. The extension missionary has to play an important role to educate the farmers by demonstrating the technology to make the fish culture a source of livelihood and strong source of income.

Key words: Koshi region, Araria, fish culture, recommended package, profitable management, improved production

INTRODUCTION

Aquaculture has acquired a special significance not only because of its contribution to food resources but also for contribution to quality of our diet. India is getting good amount of foreign exchange by exporting the aqua products by adopting the scientific management techniques. Bihar is located in the eastern part of India where Araria district situated at latitude 26.135' and longitude 87.464' respectively posesess a large cultivable stagnant fresh water area in the form of pond and tanks, out of which only small part is utilized for fish culture. Most of the farmers involved in this region for fish culture are not using scientific method and practices due to lack of proper knowledge so they are getting minimum fish production (1000kg/Ha), while few farmers using the recomended scientific techniques and are getting maximum production up to 8000-9000 kg/Hain the same area. If we can provide the proper knowledge regarding culture technique, site selection, pond construction, stocking of quality seeds, fed quality feed etc then only they will achieve the desire level of production and a day will come when our state stop to import the fish and start to export the fish to the other states. Although aquaculture development in the region is taking place at a rapid rate however integrated efforts are necessary to increase the present level of production through both horizontal and vertical expansion.

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MATERIALS & METHOD

A survey was conducted in Araria district where randomly farmers were selected for collection of data. Practices of fish culture in the area through a pre tested questionnaire.

RESULT & DISCUSSION

Present investigation reveal that most of the farmer involved in culturing of fish in this region are either not using or using little available scientific techniques and thus unable to attain the targeted level of production, either it may be due to lack of awareness or facility. Intention of the present paper is to give the proper knowledge regarding the fish culture so that people of this region get maximum benefit.

MANAGEMENT OF CULTURE POND

(A) DRYING

The phase refers to pond preparation to ensure maximum survival and proper growth of culture fish and involve repair of embankment, ploughing of the pond in summer month.

i.Oxidation of organic deposited it by exposing to air.

ii. Killing the pathogenic agent, unwanted fish etc.

iii. Removal of heavy organic deposits which releases obnoxious gases like Hydrogen sulphide (H_2S).

(B) PLOUGHING OF THE POND IN SUMMER MONTH:

Ploughing the pond bottom helps in removal of the obnoxious gases and exposing the organic deposited to the air. As per necessity either shallow or deep ploughing is carried out in the direction of perpendicular or parallel to the length of pond.[1]

15% farmers adopted in full. 25% adopted partially and 60% not adopted.

(C) APPLICATION OF PLANT –BASED PISCICIDES IN PERINEAL POND

Four methods manual, mechanical, chemical and biological are used to eradicate weed and unwanted fish from ponds, among which Biological method using fish and duck is probably the best method for controlling weed.[2]

17% farmers adopted in full, 28% adopted partially and 55% not adopted.

(D) APPLICATION OF MANURE

Primary production of the pond is not enough to fulfil the requirement of culture animals as organic as well as inorganic fertilizer is applied to increase the primary production. Among organic fertilizer cow dung is most preferable as it supplies all the nutrients need by the primary production[3].

It consider better to apply cow dung in two doses instead of one good quality of zooplankton can be maintained for a longer period if a dose of 10000kg/ha/yr.

13% farmers adopted in full, 26% partially adopted and 61% not adopted.

(E) APPLICATION OF THE LIME

The first step in fertilization of pond is the application of quick lime (Cao) which raises the water pH and act as antiparasite, killing bacteria and parasite and their larval stages. Liming also helps to decompose the organic matter, thus increase the productivity and possibly it counteracts the harmful effects of excess magnesium, sodium, potassium ions etc. Quick lime @ 500kg /ha is generally spread on the bottom.18% farmers adopted in full, 25% adopted partially and 57% not adopted.[3]

(G) STOCKING OF FINGERLING

Stocking phase consist of evaluation of carrying capacity of the pond, relative density and combination of selected species. High production of fish from a pond is possible if a Culture considering the type of pond management. The stocking rate varies from 6,000-8,000 no/ha fingerling depending upon the carrying capacity of the pond.

18% farmers adopted in full, 30% adopted partially and 52% not adopted.

(H) SIX VARIETIES AS PER RECOMMENDATION

Choice of a suitable species is an important factor in delivering high yield, where as proper -combination of 3-5 species of carp gives best result.

Generally Indian major carp such as *Catla, Rohu, Mrigal* and exotic carp like Silver carp, Grass carp and Common carp are grown together in a pond and gave a high yield. 20% farmers adopted in full, 26% adopted partially and 54% not adopted.

(I) APPLICATION OF STOCKING RATIO:

16% farmers adopted in full recommended stocking ratio 4:3:3: 31% adopted partially and 52% not adopted.

(J) APPLICATION OF FEED

Feed should be done @ 2-3% of the biomass is to ensure proper growth of cultured of fish artificial feeding enhances fish production Supplementary feeding is necessary so that high density of fish can be maintained. Natural food production in the pond is limited but it

provides all the essential nutrients to the fish, Hence, a mixture of natural and artificial food gives best results. The artificial food should be such that it is acceptable to fish and is easily digestible; it should have good nutritive value and high conversion rate etc.

Finely powdered rice bran, mustered oil cake or groundnut oil cake are use as food for fish, fry, fingerling of IMC and exotic carp in mixture (1:1) and is considered to the best artificial food, which is given daily @2-3% of the total fish biomass. It is either sprayed over the water surface or kept as thick paste in shallow, wide open mouth earthen pots in the pond. Feeding initially done in the morning and evening and should not change suddenly.[4] Artificial feeding enhances fish production and attributes 50 to 80% of total fish production in ponds in Germanya to artificial feeding. 18% farmers adopted in full, 18% adopted partially and 67% not adopted.

(K) REGULAR NETTING OF HEALTH MANAGEMENT

The practices vary often convert into causative agent of serious epizootic to the cultured host and closed water bodies; otherwise it leads to growth retardation and mortality 17% farmers adopted in full, 25% adopted partially and 58% not adopted.[5]

(L) FISH PRODUTION (achieved)

Fish production adopted in full fish culture practices, partially or not adopted fish farmers realise additional profit by 8000kg/ha, 4000kg/ha, 1000kg/ha. Both management and material involved the economics of three input levels of fish production under Araria district.

(M) SALE PROCEEDS (Rs/ha)

8 lakh adopted in full, 4 lakh partially and 1 lakh not adopted.

Table 1 - Survey of farmers fish culture

S.No	Recommended practices	Adopted in full	Adopted partially	Not adopted
1	Ploughing in the pond in summer month	15%	25%	60%
2	Application of plant based piscicides in perrineal pond	17%	28%	55%
3	Application of manure @10 tons/ha/year	13%	26%	61%
4	Application of lime @500kg/ha	18%	25%	57%
5	Stoking of fingerling	18%	30%	52%
6	Six varieties as per recommendation	20%	26%	54%
7	Application of stoking ratio	16%	31%	52%
8	Application of feed @2-3% of the biomass	18%	18%	67%
9	Regular netting for health management	17%	25%	58%
10	Fish production	8000 kg/ha	4000 kg/ha	1000 kg/ha
11	Income/ha	8 lakh	4lakh	1lakh

Table 2- Production cost, income and returns in composite fish culture CFc&SP (Randhir 1984)[6]

	High level inputs RS	**Intermediate RS**	**Low level inputs RS**
1 Pond development	2,000	2,000	2,000
Maintenance of embankment	2,000	2,000	2,000
Fingerling	12,000	12,000	12,000
Feed	30,000	20,000	
Manure	1,000	600	
Lime	2,000	800	
Health management, Netting, Misc.	2,000	2,000	1,500
Total	51,000	39,000	16,500
Cost of production	8,00,000	4,00,000	1,00,000
Net Income	7,49,000	3,61,500	83,500

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Studies on the toxic effect of arsenic in the reproduction of female mice (*Mus musculus* Linn. 1758)

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Abstract: Arsenic is regarded as one of the most toxic element that functions as toxicant in humans as well as the laboratory animals. Exposure of high doses of arsenic not only leads to malfunctioning of the reproductive organs but also leads to change in its structure. In mice induction of methylated or inorganic arsenic repeatedly leads to reproductive toxicity as well as developmental toxicity. Arsenic given to the pregnant mice in repeated oral state also lead toxic effect in its development respectively.

Key words: Toxic, *Mus muscullus,* arsenite, histomorphological, diestrous cycle.

INTRODUCTION

Arsenic is among the most measurable quantities of element found in water, soil & air also in earth crest.[1,2] China, India, America etc are the countries where arsenic concentration has been reported very high.[3,4,5] Arsenite, arsenate are the different compounds of arsenic that causes chronic as well as acute damage to human beings and other organisms. Organs that usually get affected due to the impact of arsenic are those that are involved in excretion, absorption etc.[6] Toxicity of arsenic does not only occurs due to diet containing high amount of arsenic content but also through inhaling arsenic in industries & also when we come across to herbicides & pesticides.[7] Arsenic for most of the bacteria function as respiratory metabolites trace quantity are also regarded as very essential in few animals such as rats, goat, chicken, and other species. There are also certain evidence that shows that arsenic has adverse effect on human reproductive health[8] which may cause certain health hazards such as low birth weight, spontaneous abortions, infant mortality etc.[9,10,11,12] According to united states environment protection agency arsenic are also regarded as risky for human health.[13] The toxic effect of arsenic is also seen in reproductive system of rodents.[14]

As we know there is great impact of high arsenic intake in the organism although there are very few evidences which tell us about the toxicity of high arsenic intake.

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MATERIAL AND METHODS

Experimental animals

Female mice belonging to the species *Mus musculus* was selected for the study toxic effect of arsenic in reproduction. 40 mice were bought which were about 45 days from the local market and was kept in laboratory for

its further study. All the mice were kept in normal laboratory condition and fed with standard diet and normal tap water. These mice were then allowed to mate. 18 female pregnant mice having average body weight were then selected & divided into three groups each having 6 animals. Each of the groups was dived into control group, group 1 & group 2. Control group of the mice were fed with normal diet, group 1 of the mice were fed with diet having 100 ppm arsenic & the other group was fed with 200 ppm ie high doses of arsenic for about three weeks. The animals were then sacrificed for the study of different reproductive organs.

Collection of the blood & tissue sampling

The sacrificed species were then kept in the laboratory for its histomrphological study. Blood was collected from each of the mouse and was kept separately for its reproductive hormonal study. The pregnant mice were then dissected to study the different reproductive organs. The tissues of the reproductive organs were the dissected out. The sectioned tissues were then stained with hematoxylene & eosine to study the histomorphology.

RESULT & DISCUSSION

- **Histopathological study**
  After the completion of experiment it was found that there was great changes in morphology of the group 2 mice tissue compared to the group 1 & control group. In the pregnant mice of control group there was no changes observed in the morphology of uterine structure while in group 1 & group 2 the uterine horn (the point where the uterus and the fallopian tubes meet) appeared very thin.
- **Weight of the ovary & uterus**
  There was great variation found in the weight of the ovary & uterus. The weight of the group 2 mice was very much decreased with respect to the group 1 species. When the LH, FSH & Estradiol level found in plasma was estimated, its quantity was also decreased. Thus decreased FSH, LH, estradiol lead to decrease in the ovarian activities. Since all the above are very important for the reproduction to take place successfully. Thus the diet containing high arsenic quantity may lead to decreased reproduction. Thus decreased plasma level may lead to diestrous cycle.
- **Degeneration of ovarian & uterine cell**
  High arsenic level in rat also lead to degeneration of different cells found in ovary and uterine which acts as a inhibition factor for the reproduction in mice (Chattopadhyay *et al*. 2003).[14]

CONCLUSION

From the above observations it is found that increase in the level of arsenic lead to decreased production of healthy follicles & increased apoptotic follicles, thus inhibiting the process of reproduction in mice. Great difference was observed in mice treated with 200 & 100 ppm of arsenic in diet in contrast to control group where the process of reproduction was found to be very normal.

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A comparative analysis of estimation of free amino acids in larval haemolymph of the mutant strains of *Antheraea mylitta* Drury

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Abstract: This research article is an effort to study the relative biochemical estimation of free amino acids in the larval bodies of three mutant strains viz: Daba-blue, Daba-yellow and Daba-almond of tasar silk worm, *Antheraea mylitta*. Results obtained are indicative of the fact that the Am-blue mutant strains of *Antheraea mylitta* carry greater number of free amino acid as compared to Am-yellow and Am-almond mutant strains of *A. mylitta.* The concentration of free amino acids during commercial crop season in all the mutant strains have been found relatively greater than the seed crop season. The number and concentration of free amino acids have been found to present variation in relation to genetic makeup, dietary and ecological conditions.

Key words:- Ecotype, Free Amino Acid, Haemolymph, Mutant Strains

INTRODUCTION

Antheraea mylitta is an indigenous traditional tasar producing insect, which exists in the forms of nearly 20 ecotypes distributed all over the ecological zones of tropical tasar belts in our Country. The different ecotypes of *Antheraea mylitta* in spite of having the same chromosomal number differ among themselves in their quantitative and qualitative characters. These ecotypes are uni, bi and trivoltine. Different local names have been assigned to each group by the tribals of this area eg. Daba, Raily, Model, Naila, Bogari etc. Among these, the Daba ecotypes of *Antheraea mylitta* Drury are distributed to the different localities of tasar belts particularly in the states of Bihar, Jharkhand and Madhya Pradesh. The tribals usually reared this ecotype by preparing the eggs in captivity and rear the larvae outdoor on foliages of *Terminalia arjuna, Terminalia tomentosa* and *Shorea robusta* during Seed Crop (July-August) and Commercial Crop Seasons (September- October).

Agarwal and Jolly (1974)[1] while working on biochemistry of tasar silkworm have mentioned that free amino acids in the larval haemolymph vary with the different concentrations of diet. Drilhon, *et al.* (1951)[2] have reported that the concentrations of free amino acids in the larval haemolymph has a tendency to show increase or decrease according to physiological conditions at different stages of their life cycle. Sharma and Pandey (1990)[3] while working on the variations of free amino acids contents have found that there exists a great variation in the number of amino acids among different ecotypes of tasar silkworm. Florkin (1959)[4] has reported qualitative and quantitative changes in the free amino acids contents at different stages

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in the life cycle of insects. Chen (1962)[5] has reported similar pattern of amino acid variations in Calliphora.

Chen (1960)[6] in *Drosophilla* shows maximum increase of amino acids at 72 hours after oviposition and steady dropping after 96 hours, corresponding to the period of most intensive growth as determined by wet weight if the values were calculated percent body weight. The pattern of free amino acids in *Drosophilla* was quite similar to the larvae of mosquito *Culex pipiens*.[7]

MATERIALS AND METHODS

The larval haemolymph of three different mutant strains of *Antheraea mylitta* Drury, in relation to their free amino acid was considered for the relative biochemical analysis. The fifth larval and final instars of almost similar age were selected for the analysis of biochemical content of free amino acids. The larval culture was carried out on *Terminalia arjuna* host plant of tasar worm as per the method suggested by Jolly (1973)[8]. During larval collection cautions were taken for obtaining the disease free larvae. The pathological analysis in respect of disease free larvae was carried out before taking out the larval haemolymph under aseptic condition.

The estimations of free amino acids of larval haemolymph of three different mutant strains of *Antheraea mylitta* was carried out by paper chromatography method. In this method the larval haemolymph of three different mutant strains were collected separately and deproteinised with ethanol (70%v/v) and there after centrifuged for ten minutes at 300 rpm. The protein free clear supernatant was evaporated to dryness on a water bath. Fats and lipids were removed by extracting the residue with ether (1ml) and residue was further dissolved in iso-propanol (1.5ml, 10% v/v) and all the extract were prepared in the same manner and used for two dimensional paper partition chromatography for the comparative biochemical analysis of larval haemolymph of three different mutant strains. 1ml haemolymph was used at every stage of analysis. Analar chemicals were used for the experiment.

Paper Chromatography Method: 10 ml of the extract was applied in each case on whatmann no. 1 filter paper (20x20 cm). The chromatography were first run butanol: acetic acid: water (4:1:5, v/v) in ascending manner and after overnight thorough drying the second run was made with phenol water (4:4, v/v). The chromatograms were dried at room temperature.

Development of spots: The chromatograms were sprayed with ninhydrin solution in acetone (0.25% w/v) sprayed chromatograms were heated in an air oven at 100°C for 15 minutes. The identification of amino acids was made as follows:

a) Comparing R.F values with known standards, running under the same experimental conditions

b) Using various specific spray reagents as given below:

i. Sakaguchi- reagents for arginine and other monosubstituted guanine derivatives (Sakaguchi, 1925)[9]
ii. Isatin for proline and hydroxyl proline as described by Hackman and Lazarus (1956)[10]
iii. Ehrlich's reagent for tryptophan, cytruline and urea (Smith, 1953)[11]
iv. Folin's reagent for amino butyric acid and their isomers (Pant and Agarwal, 1963)[12]

The results of the experiment in relation to variations of different amino acids contents were tabulated and presented in tables in the forms of traces (+ -), present (+), absent (-) and higher concentration (++) for both the seasons. Data obtained were recorded in the tables.

RESULTS

The relative biochemical estimations concerning free amino acids contents in the larval bodies of three mutant strains of *Antheraea mylitta,* namely Am- blue, Am- yellow and Am- almond along with control have been carried out during the seed crop and commercial crop seasons and results so obtained are recorded in tables 1 to 5.

Table 1 reveals the number and concentration of free amino acids in the larval haemolymph of Am - blue mutant strain of *Antheraea mylitta* during the seed crop and commercial crop season. As per table the total number of free amino acid (F.A.A.) is 17 in seed crop seasons and 19 in commercial crop seasons. In seed crop season, the concentration and argine, asparagin, cystic acid and glutamine have been found in traces. The leucin and glycine are altogether absent apart from this other amino acids have shown normal concentration. During the commercial crop season, the concentration of aspartic acid, beta alanine, glutamic acid, lysine, methionine-sulphoxide and serine have been recorded in abundance showing greater concentration. Glycine and histidine are in traces.

Apart from this all other amino acids have shown normal concentration in the larval haemolymph of the Am-blue mutant strain of *Antheraea mylitta*. It is worthwhile to mention that Am-blue mutant strain carry all the 19 free amino acids in its biochemical concentration during the commercial crop season.

Table 2 further accounts for the number and concentration of free amino acids (F.A.A.) in the larval haemolymph of Am-yellow mutant strain of *Antheraea mylitta* during the seed crop and commercial crop seasons.As per the table the total number of free amino acids is recorded 16 during the seed crop season in which glutamine, methionine-sulphoxide and valine are totally absent. The cystic acid, glycine, lysine and theronine have been found in traces and apart from this remaining free amino acids have shown normal concentration. During the commercial crop season the total number of free amino acids is recorded 17 in which leucin and methionine-sulphoxide are altogether absent and asparagines, lysine and serine have been recorded in traces.

Table 3 indicates the number and concentration of free amino acids (F.A.A) in the larval haemolymph of Am-almond mutant strain of *Antheraea mylitta* during both the seasons of rearing. As per table during the seed crop season the total number of free amino acids in the larval haemolymph of Am-almond mutant strain is recorded 15 out of which 19 in which cystine, glutamine, lysine and valin are totally absent and cystic acid, alpha9alanine, arginine, leucin and proline have been recorded in traces. During the commercial crop season the total number of free amino acids in the larval haemolymph of Am-almond mutant strain has been recorded 16 and the free amino acids namely cystine, serine and valine are altogether absent. Aspartic, cystic acid and threonine are recorded in traces. Apart from this remaining free amino acids have shown normal concentration.

Table 4 accounts for the number and concentration of free amino acids in Am-green (control) in its larval haemolymph during the seed crop and commercial crop seasons. Table clearly shows that during the seed crop season the total number of free amino acids (F.A.A.) is only 15 out of 19in which Beta-alanine, glutamic acid, leucin and tyrosine have been found missing. The concentration of cystic acid, glutamic acid, methionine-sulphoxide, proline, threonine and valine are recorded in traces. The concentration of gelycine and histidine is recorded in abundance showing greater concentration of free amino acids. During the commercial crop season the total number of free amino acids in the larval haemolymph is 16 out of 19 in which Beta-alanine, glycine and lysine are totally missing. The cystic acid, histidine, lysine, threonine and tyrosine have shown their concentration in traces. Arganine, asparagines and methionine-sulphoxide have been found in abundance in Am-green control of *Antheraea mylitta*. The rearing amino acids have shown their normal presence in the larval haemolymph of control.

A comparative picture in respect of number and concentration of free amino acids in the larval haemolymph of three mutant strains along with control during the seed crop and the commercial crop seasons has been summed up in table 5 and Figure 1.

Table 1: Table showing free amino acid content in the larval haemolymph of Am-blue mutant strain of *A. mylitta* during seed crop season and Commercial crop season

Sl. No.	Free amino acid	Seed Crop Season	Commercial Crop season
1	Alpha-alanine	+	++
2	Arginine	+-	+
3	Asparagine	+-	+
4	Aspartic acid	++	++
5	Beta-alanine	+	++
6	Cystic acid	+-	+
7	Cystine	+	+
8	Glutamic acid	++	++
9	Glutamine	+-	+
10	Glycine	-	+-
11	Histidine	+	+-
12	Leucin	-	+
13	Lysine	+-	++
14	Methoinine-sulphate	+	++
15	Proline	+	++
16	Serine	+	+
17	Threonine	++	+
18	Tyrosine	+	+
19	Valine		

++Abundant, +Present, +-Trace, -Absent, Am-*Anthearea mylitta*

Table 2: Table showing free amino acid content in the larval haemolymph of Am-yellow mutant strain of *A. mylitta* during seed crop season and Commercial crop season

Sl. No.	Free amino acid	Seed Crop Season	Commercial Crop season
1	Alpha-alanine	+	++
2	Arginine	+	+
3	Asparagine	+	+-
4	Aspartic acid	++	++
5	Beta-alanine	+	++
6	Cystic acid	+-	++
7	Cystine	+	+
8	Glutamic acid	++	++
9	Glutamine	-	+
10	Glycine	+-	+
11	Histidine	+	+
12	Leucin	+	-
13	Lysine	+-	+-
14	Methoinine-sulphate	-	-
15	Proline	+	++
16	Serine	+	+-
17	Threonine	+-	+
18	Tyrosine	++	++
19	Valine	-	+

Table 3: Table showing free amino acid content in the larval haemolymph of Am-almond mutant strain of *A. mylitta* during seed crop season and Commercial crop season

Sl. No.	Free amino acid	Seed Crop Season	Commercial Crop season
1	Alpha-alanine	+-	+
2	Arginine	+-	+
3	Asparagine	+	++
4	Aspartic acid	+	+-
5	Beta-alanine	+	+
6	Cystic acid	+-	+-
7	Cystine	-	-
8	Glutamic acid	+-	+
9	Glutamine	-	+-
10	Glycine	+	+
11	Histidine	+	+
12	Leucin	+-	+
13	Lysine	-	+
14	Methoinine-sulphate	+	+
15	Proline	+-	+
16	Serine	+	-
17	Threonine	+	+-
18	Tyrosine	++	+
19	Valine	-	-

Table 4: Table showing free amino acid content in the larval haemolymph of Am-green (Control) mutant strain of *A. mylitta* during seed crop season and Commercial crop season

Sl. No.	Free amino acid	Seed Crop Season	Commercial Crop season
1	Alpha-alanine	+	+
2	Arginine	+	+
3	Asparagine	+	++
4	Aspartic acid	+	+
5	Beta-alanine	-	-
6	Cystic acid	+-	+-
7	Cystine	+	+
8	Glutamic acid	+-	+
9	Glutamine	-	+
10	Glycine	++	-
11	Histidine	++	+-
12	Leucin	-	+
13	Lysine	+	-
14	Methoinine-sulphate	+-	++
15	Proline	+-	+
16	Serine	+	+
17	Threonine	+-	+-
18	Tyrosine	-	+-
19	Valine	+-	+

Table 5: Table showing a comparative picture concerning the total number of free amino acid in the larval haemolymph of three mutant strains and its control of *A. mylitta* during seed crop season and Commercial Crop season

Sl. No.	Strains of *Anthearea mylitta*	Seed Crop Season	Commercial Crop season
1	Am-Blue	17	19
2	Am-yellow	16	17
3	Am-almond	15	16
4	Am-green (Control)	15	16

++Abundant, +Present, +-Trace, -Absent, Am-*Anthearea mylitta*

DISCUSSION

A comparative analysis of estimation of free amino acids in various mutant strains of *Antheraea mylitta* reveals the under given results

- All the three mutant strains of *Antheraea mylitta*, namely Am-blue, Am-yellow and Am-almond on account of their genetic diversities present evident variations in respect of number and concentration

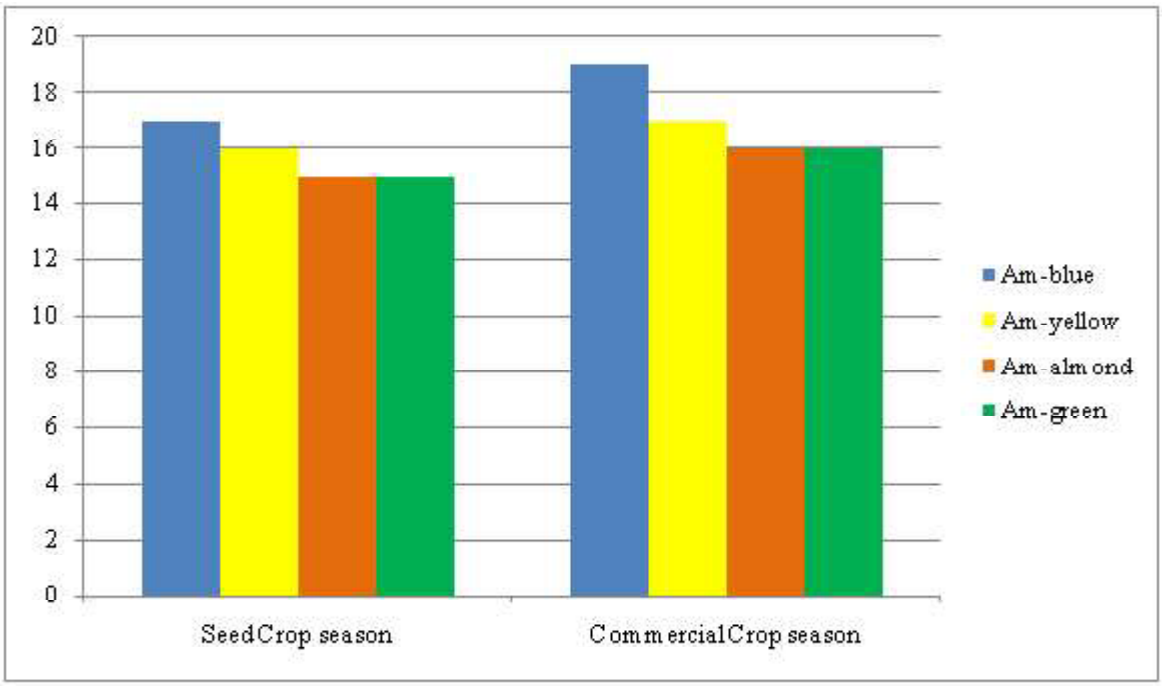


Figure 1: Bar Diagram showing a comparative picture concerning the total number of free amino acid in the larval haemolymph of three mutant strains and its control of *A. mylitta* during seed crop season and Commercial Crop season

of free amino acids in their larval haemolymph during the seed crop and commercial crop seasons.

- Among the three mutant strains the number and concentration of free amino acids are relatively greater in the larval haemolymph of Am-blue followed by Am-yellow and Am-almond respectively.
- The number and concentration of free amino acids among all the three mutant strains of *Antheraea mylitta* in spite of their relative differences are evidently and significantly greater and higher than the control during both the seasons of estimations.
- On account of seasonal differences the number and concentration of free amino acids in the larval haemolymph of all the three mutant strains and control have been found relatively greater during commercial crop season than the seed crop season.

The number and concentration of free amino acids have been found to present variation in relation to genetic makeup, dietary and ecological conditions.[13] Agarwal *et al* (1975)[14] reported that the healthy tasar larvae of *Antheraea mylitta* on account of genetic robustness carry greater number of free amino acids than the unhealthy larvae with poor genetic vigour. Auclar while working on *Galleria melonella* has found evident variation in the free amino acid concentration in relation to ecological factors and dietary variation. Pandey (1989)[15] while working on different ecotypes of *Antheraea mylitta* has mentioned that the ecotypic differentiation on account of genetic divergence is the most effective factors for the variation in free amino acid content at the different stages of life cycle of *Antheraea mylitta.* Sinha has reported that the tasar larvae reared on mature leaves carry greater number of free amino acids as compared to tender and coarse leaves. Verma (1992)[16] while working on different ecotypes of *Antheraea mylitta* maintains that the physiogeneic factors are very much responsible for evident variation in the free amino acid content among the different ecotypes of *Antheraea mylitta.*

CONCLUSION

In the light of aforesaid information, it is logical to assume that the relative differences in amino acids among the three mutant strains owing to their different

physiogeneic constitution is a natural outcome and very much inconformity with the works of earlier investigators. The greater concentration of free amino acids during commercial crop season at the larval, pupal and adult stages of life cycle among three mutant strains are certainly associated with diverse nutritional and ecological conditions. Since the nutritional and ecological condition during Commercial crop season are more favourable to the mutant strains than the seed crop season, thus, the differences in the free amino acids content in relation to two different seasons appear to be fully justified.

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Study on foraging behavioral pattern in *Solenopsis geminata*

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Received : 20th December, 2019 ; Revised : 4th February, 2020

Abstract: Ants are one of the primitive species following wasp like in habit forming small and large colonies. *Solenopsis* is among the several ant species which comes under the household ants or harvester ants. In the present paper attempt has been made to study the foraging pattern or behavior among the ant's species *S.geminata*. Since foraging behavior has been studied in most of the species but here research has been done to study about the fire ant *S.geminata*. Although, this study was a complicated process. Foraging in ants does not involves only study about the behavioral pattern but also the different activities opted by the ants to avail food.

Key words: *Solenopsis geminata*, foraging, Madhepura, colonies

INTRODUCTION

The order hymenoptera consist of more than 100000 species of insects and some of the species are still to be discovered. They may be free living, phytophagous, predatory, entomophagous parasitic or social insects. Certain wasps, bees and ants are polymorphic and social insects. They have very diversified behavior, habits, nest building, caste system, chemical communication system etc. In almost all types of habitats insects are found due to their behavioural adaptability or plasticity[1]. Although such types of behaviour pattern are usually associated with foraging genes[2].

Study on Ants (Family: Formicidae) are an exceptionally diverse and widespread group of arthropods, containing over 13,000 extant species estimated to comprise up to 20% of animal biomass[3,4]. Foraging process is quite complicated, by obligate & eusocial found among the ant colonies since foraging occurs for the benefit of entire colony rather than the individual[5]. For the survival ants need to leave their nest & forage for food. Among the ant colony survival of the fittest usually depends on optimal path between their nest & the food source[6,7]. Thus foraging decision may include where to search, how long to search at the site given, whether or not to return to a site where search was previously conducted. Nest and food source indeed play important roles in ants' foraging behavior.

The tropical fire ant species *Solenopsis geminata* is among one of these species having the family hymenoptera.

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MATERIAL AND METHODS

Selection of the sites:-

Different colonies of the *S.geminata* were selected at one of the government school in Madhepura district. The sites selected were present at only one place so that study can become easy. All the colonies were then marked with colony A, Colony B & Colony C. Each of the colonies contained 50 ants each. Each of the ant group was then separated with silver wire net with pin holes so that the

ants could not escape outside from it. And each of these ants nest were located merely 50-70 cm in each.

Colony A: - Colony A was fed with the grinded wheat flour (granule type) and was kept in plastic petridish. A hole was made at one side so that the study can be more clear following the method mentioned by Hooper & Rust(1997)[8].

Colony B: - Colony B was fed with anchovy which was soaked in water to soften and then dried in oven under 50° C to make sure that it can be turned into small granules (<2mm) & can be easy for the workers ant to carry it to their nest before carrying out the research.

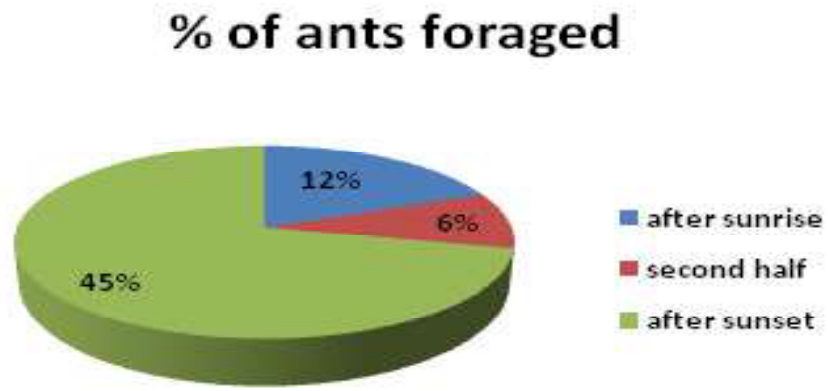
Colony C: - Colony C of the ant was fed with sugar in plastic petridish & further study was made.

Field trip:-During the study of foraging pattern field trip of the selected area were made three times in a day. Once in the morning between 8-10 am, second trip was between 12-2 pm in afternoon and the last trip was made during evening hour between 6-8 pm. The visit of the selected sites was made at least for 2 weeks and the result obtained was as follows:-

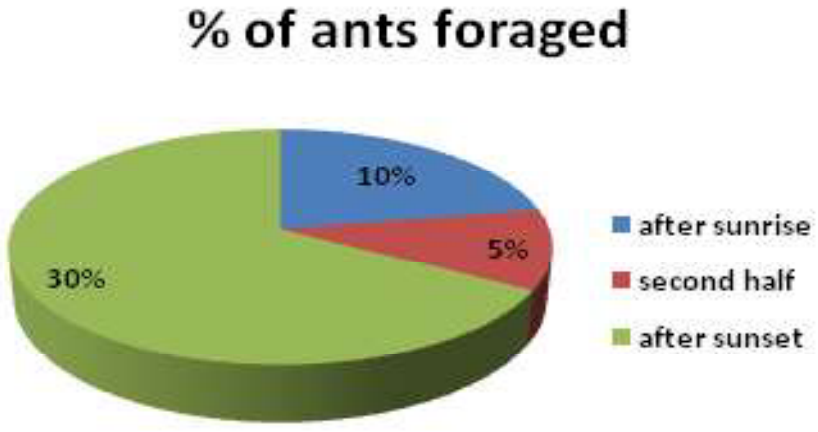
RESULT & DISCUSSION

The investigation on *S.geminata* shows that the all the three colonies were marked for the foraging study did not showed any differences, the only difference was the amount of removal of bait by the workers ant. The foraging behavior was higher after the sunrise since it made easy by the workers ants to search food. While it slowed down after sunset.

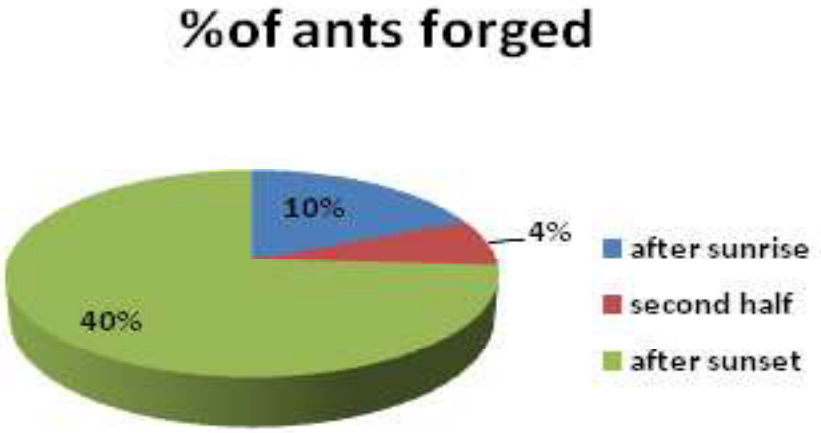
In colony A it was seen that in 50 of the ants 12% of the ants started foraging and return back to their nest carrying different bait of food size during the morning hour. However during the second half when the study was carried out it was found that only 6% of the ants foraged. And during the evening the foraging frequency was at the peak i.e 45%.



In colony B it was found that 10% of the ant species started foraging and return back to their nest very safely whereas during second half only 5% of the ants started to forage. In the evening time 30% of the ants were seen searching for food but they also returned back to their nest very safely which was the highest value.



And in colony C 19 % of the ants were seen searching for food in which population of ants searching for food was not very high. After sunset, 40% of the ant started foraging which was the peak value. While during second half only 4% of the ant could be seen searching for food having the least value.



However it was found that foraging was mostly affected by temperature, humidity, and other environmental factors are the base for foraging behavior in ants. Mostly the ants avoided foraging during day time due to the high environment temperature[9].

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The taxonomic status of *Lytocestus bharatae* Patil, 2018 – A critical study

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Received : 27th December, 2019 ; Revised : 24th February, 2020

Abstract: Patil (2018)[1] reported a new species *Caryophyllaeid* Cestode belonging to the genus *Lytocestus* (*Caryophyllidea: Lytocestidae*) which he recovered from *Clarias batrachus* at Gulbarga (Karnataka) India. The description of the said species *L. bharatae* appeared in *Review Research* **vol.7(6)** on page 1-4. His paper suffers from a number of lacunae. The present authors have pointed out the mistakes, after critically going through the description & observing the sketches provided, and suggested some remedial measures to delineate the existence of the species till then the species has been kept under uncertain status (incertae sedis)

Key words: *Lytocestus bharatae*, incertae sedis, review.

INTRODUCTION

Caryophyllidean cestodes are interesting group, with a unique morphology, evolutionary status, and genetic stability showing high degree of endemism. Only *Archigetes sieboldi* Leuckart (1878)[2] and *Glaridacris catostomi* Cooper (1920)[3] are represented from more than one geographical area.

These show monozoism and low fecundity yet are quite successful, the relative abundance in fishes is on account of "Vitelline cells" which can synthesize and store glycogen in the nuclei as a normal function-"a phenomena unique to animals kingdom"-Agarwal (1985)[4]. The increase in glycogen without concomitant increase in the number of vitelline cells or in egg size could lead to prolonged period of infectivity & utilisation of glycogen and not lipid as energy source. Glycogen being heavier than lipid, it can be utilised under anaerobic condition are consistent with life cycle that has benthic intermediate hosts-Mackiewicz (1981,a)[5].

Higher level of glycogen in *Caryophyllaeids* is indicative of lower calorific value than lipid which are the result of lower oxygen concentration in the enteric environment but low fecundity and relative abundance is due to K-selection strategies as an environmental condition Sahay *et al.* (2007)[6].

The order *Caryophyllidea* van Beneden (Carus, 1863)[7] include a group of Cestodes (one segmented) which has four families:

1) Caryophyllaeidae Leuckart (in Luhe,1970)[8]
2) Lytocestidae Wardle & McLeod (1952)[9]
3) Capingentidae Wardle & McLeod (1952)[9]
4) Balanotaenidae Mackiewicz & Blair (1978)[10]

These families have been erected giving cognisance to the relationship between inner longitudinal muscle, vitellarial follicles & position of testicular follicles.

Under the family *Lytocestidae* Wardle & McLeod (1952)[9], a number of genera have been reported. One of

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the genus is *Lytocestus* Cohn (1908)[11]which is characterised by "Body elongate tapering anteriorly, Scolex undifferentiated, inner longitudinal muscles in a ring around testes. Testes in broad median field of pre-uterine medulla, vas deferens convoluted, leading into compact parenchymatous mass not sharply demarcated from surrounding and containing numerous dorso-ventral muscle fibres, in which the thin walled ejaculatory duct is winding, cirrus with strongly muscular wall opening into deep narrow mid ventral pit. Ovary bilobed, with lateral lobe outside the inner longitudinal muscle sheath. Vitellaria surrounding inner longitudinal muscle sheath in testicular zone, no post ovarian follicles, uterus looped behind shell gland and then coiled between ovary and male terminalia where it is surrounded by a layer of tall radiating accompanying cells. Vagina also provided with a layer of accompanying cells opening mid ventrally directly behind cirrus." Parasites of mormyrid fishes.

More than fifty species have been reported in the genus *Lytocestus* Cohn (1908)[11], surprisingly majority of them are from the state of Maharashtra. Out of these quite a good number of *Lytocestus* species have been synonimed/ kept under incertae sedis / transferred to another genera, for example: Ash (2012)[12] without having a consistent opinion took following decisions:-

1) Synonimised *L.lativitellarium* Furtado & Kim Low (1973)[13] with *Lucknowia microcephala* Bovien (1926)[14].
2) *L.heteropneusti* Tandon *et.al*. (2005)[15], *L.jagtai* Tripathi *et. al*. (2007)[16] with *Lucknowia fossilisi* Gupta (1961)[17].
3) *L.attenuatus* Tandon *et. al* (2005)[15], *L.clariae* Tandon *et al*. (2005)[15] with *Bovienia indica* (*Lucknowia indica*) Niyogi *et.al.* (1982) [18].
4) *L. marthawadensis* Shinde *et. al* (1988)[19] was synonymised with *Pseudocaryophyllaeus ritai* Gupta and Sinha (1984)[20].
5) *L.parvulus* Furtado *et. al*. (1963)[21], *L.moghei* Murhar *et. al*. (1963)[22], *L. longicollis* Ramadevi (1973)[23] *L.majumdari* Poonam (2007)[24], *L.bokaroensis* Poonam (2007)[24] were synonymised with *Pseudocaryophyllaeus teniuicollis* Bovien (1926)[14].
6) Follicular vitellaria bearing forms like *L.alii* Jadhav *et. al* (1991)[25], *L.clariasae* Jadhav *et. al* (1991)[25], *L.naldurgensis* Kadam *et. al* (1998)[26], *L.kopardaensis* Shinde & Borde (1999)[27], *L.teranaensis* Kolpuke *et. al* (1999)[28], *L.batrachusae* Pawar & Shinde (2002)[29], *L.clariasae* (minor) Pawar *et.al* (2002)[29], *L.subhapradhi* Jawlikar *et.al* (2008)[30], *L.murhari* Kaul *et.al* (2010)[31], were synonimised with *L.indicus* Moghe (1925)[32].

It seems that Patil (2018)[1] did not have access to the literature of 50 species of *Lytocestus* else he would not have compared his specimens with only 11 species:

1) (viz.; 1) *L.filiformis* Woodland (1923)[33],
2) *L.indicus* Moghe (1925)[32],
3) *L.alestesi* Lynsdale (1950)[34],
4) *L.birmanicus* Lynsdale (1956)[34],
5) *L.javanicus* Bovein (1926)[14], Mackiewicz (1962)[35].
6) *L.parvulus* Furtado (1963)[21],
7) *L.longicollis* Ramadevi (1973)[23],
8) *L.fossilis* Singh (1975)[36],
9) *L.marathawadensis* Shinde & Phad (1990)[19],
10) *L.alii* Jadhav and Gavhane (1991)[25],
11) *L.clariasae* Jadhav and Gavhane (1991)[25].

Knowing fully well that out of the above serial 6, 9, 10, 11 were not existing as these had undergone synonymy with *L.indicus* Moghe (1925)[32] vide Ash (2012)[37]

At least Patil (2018)[1] should have compared his specimen with granular vitellaria bearing forms viz.; *L.*(*adhaerens* Cohn 1908[11]; *chalisgaonensis* Khalse *et.al* 1999[38]; *govindae* Patil *et.al* 2002[39]; *shindei* Khadap *et. al* 2004[40]; *nagapurensis* Lakhe *et. al* 2004[41]; *paithanensis* Shelke 2007[42]; *punensis* Jadhav *et. al* (2008)[43]; *majaraensis* Solunke *et.al* 2012[44] and *khami* Jawle *et.al* 2011[45].

Following species of *Lytocestus* have been kept under INCERTAE SEDIS

1) *L.indica* Deshmukh *et. al* (2015)[46] by Sahay, Singh and Saxena 2018[47].
2) *L.bishnupurensis* Shomendra *et. al* (2003)[48] was synonimised with *L.indicus* by Singh, Sahay & Sadaf (2018)[48].
3) *L. jagtai* Tripathi *et. al.* (2007)[16] and *L.fossilis* Singh (1975)[36] were considered synonym of *Lucknowia fossilisi* Gupta (1961)[17] by Sahay & Ekka (2019)[49].
4) Sahay, Mandal, Saxena and Singh (2017)[50] held *L.heteropneusti* Tandon *et.al*. (2005)[15] valid and

not a synonym of *Lucknowia fossilisi* Gupta (1961)[17].

5) *L. mastacembelusi* Pardeshi (2016)[51] was kept by Sahay, Khalkho, Singh and Mandal (2019)[52] under incertae sedis.
6) *L. paithanensis* Kale (2017)[53] was held invalid species by Sahay, Khalkho, Ekka and Mandal (2019)[54].

The main aim of the study is to assess critically the status of *Lytocestus bharatae* Patil (2018)[1] as the description of the species suffers from a number of lacunae inadvertently committed by the author.

MATERIALS AND METHODS

Several research papers have been consulted &few slides observed.

DISCUSSION

The main characteristics of *Lytocestus bharatae* Patil (2018)[1] are as follows:

"Scolex (Head) elongated narrow anteriorly and broad posteriorly, neck short squarish slightly convex lateral margin, bears 1470 to 1480 (1475) number of testicular follicles, pre-ovarian median extending between base of neck to ovary, cirrus pouch median elongated cylindrical curved anteriorly, pre-ovarian transversely or obliquely placed, ovary dumbbell shaped distinctly bilobed near posterior end follicles being 28-32, (cortical or median not mentioned), isthmus present, Vagina starts from genital pore runs medially posteriorly opens in ootype, ootype large, irregular in shape post ovarian situated in the concavity of the ovarian lobe, uterus coiled, commence from ootype, runs posteriorly beyond isthmus taking a turn anteriorly pre-ovarian with several coils & open through uterine pore behind genital pore. Vitellaria are granular strips of medium width on each lateral side from base of neck to the posterior end of the worm."

Lacunae:

1) A number of spelling mistake in the said paper.
2) Ovarian lobes are cortical or not, has not been mentioned.
3) No mention has been made about the presence of oviduct, receptaculum seminis & utero-vaginal duct. Normally vagina and uterus joins anteriorly (some distance before the male genital opening) and forms utero-vaginal duct which opens through female genital opening behind the male gonopore.

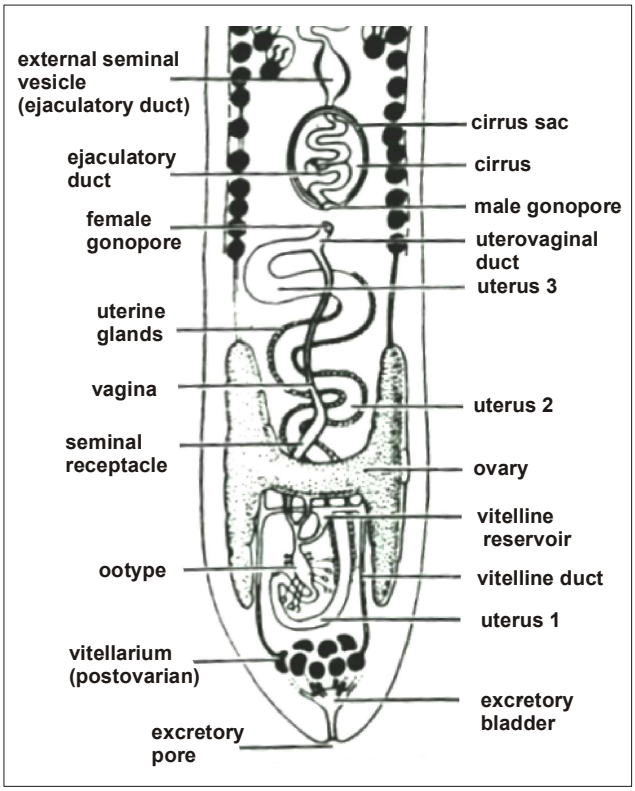


Fig- 1 Posterior end of a Caryophyllaeid (Taken from Mackiewicz, 1972*)*

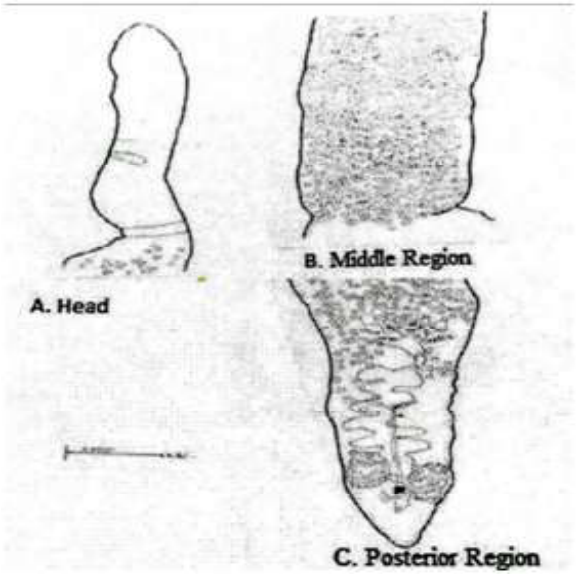


Fig- 2 Camera lucida drawing of *Lytocestus bharatae* Patil (2018) (from his research paper)

Besides, in some cases the posterior part of vagina becomes swollen to form receptaculum seminis, which continues behind the isthmus and receives a common duct arising from the isthmus & also receives common vitelline duct, continues further behind to end in the ootype as depicted in the figure below.(Fig -1)

1) The author of *Lytocestus bharatae* (?)[1] forgot to give the measurement of eggs.This indicates that worms studied were not fully grown.
2) Vitellaria have been mentioned to be granular. If that be so, it is a sign of immaturity.
3) *Lytocestus bharatae* (?)[1] has been separated from aforesaid 11 species on the basis of the number of testicular follicles (being 1470-1480 mean 1495). Species separation on the basis of number of testicular follicles has already been negated by Sahay, Singh and Saxena (2018)[47] because the range depicts that the worms studies were not of the same age and are in different stages of growth & maturity and that number of testicular follicles should normally be constant provided the worms are of same age. The range creates problem for example the range of testicular follicles 230-270 in *L.indicus* Moghe (1925)[32] fits very well in the range of testicular follicles 190-400 of *L.attenuatus* Tandon *et.al* (2005) [14], also with range 230-340 in *L.heteropneusti* Tandon *et.al* (2005) [15].

The range of testicular follicles 460-480 of *L.alii* fits very well in the range 400-500 for *L.follicularae* Bhure *et.al* (2010)[55].

Similar is the situation in the *L. Bharatae* (?) Patil (2018)[1] where the number of testicular follicles have been shown to be (1470-1480 mean 1475). This range nearly falls in the range of testicular follicles given for *L. govindae* Patil andYadav (2002)[39] where range is 1425-1475; in case of *L. punensis* Jadhav *et.al* (2008)[43] to the range of testicular follicles is 1400-1500, the testicular range of *L. Bharatae* fits very well in the range for *L. punensis*.

The range 1425-1475 of *L. bharatae* (?) fits well in the range 1200-1500 testicular follicles given for *L. teranaensis* Kolpuke (1999)[28].

If at all Patil (2018)[1] gives cognisance to the number of testicular follicles as criteria for species separation/ recognition, he ought to have compared his specimens with above two species (*L.govindae* & *L. punensis*) which he miserably failed.

Does it mean that Patil (2018)[1] believed in the synonymy of *L.* (*govindae, punensis*) with *L.indicus* Moghe (1925)[32], this he failed to mention.

Sahay, Singh, Kamal & Jha (2018)[57] negated the synonymy of *L.* (*chalisgaonensis*, *govindae, shindei, nagapurensis, paithanensis* and *punensis* with *L.indicus*) as proposed by Ash (2012)[37] on the ground that the above species do not possess 230-270 testicular follicles met in *L.indicus* neither the vitelline follicles are granular (Refer –*J.Exp.Zool.India* vol **21** pp.271-276).

Another mistake to be pointed out here is that Patil (2018)[1] has mentioned 11 species of *Lytocestus* in text but their authors name does not appear in the Reference except that of Shinde *et.al* (1990)[19] and of Singh (1975)[36].

The camera lucida drawings provided by Patil (2018)[1] &the structural similarity between the species in question & *L.indicus* is so strong that one is inclined to consider Patil's species a synonym of *L.indicus* Moghe (1925)[31].

CONCLUSION

On the above stated grounds the taxonomic status of *L. bharatae* (?) Patil (2018)[1] becomes doubtful unless the errors are corrected. Further it is suggested that Patil (2018)[1] should undertake electrophoretic study of the said species or go for molecular characterisation & karyotypic studies, provide photomicrographs of the said species in question, cross sections etc., in order to substantiate the validity of the species till then authors keep the said species under 'incertae sedis'. Provision of a good camera lucida is also needed.

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Effect of hypoxia on fresh water fishes of Madhepura

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Abstract: Whenever an organism's body or part of the body is deprived of adequate amount of oxygen supply at tissue level, on condition generally called as hypoxia develops. Fishes generally depend on dissolved oxygen in water for their respiratory needs. And this dissolved oxygen usually depends on various factors such as surface area of water, temperature, etc. so for survival of fishes, a constant concentration of dissolved oxygen is a must. This paper deals with the causes and effects of fishes.

Key words: Gills, hypoxia, fishes, dissolved oxygen.

INTRODUCTION

Most fishes depend on surface dissolved oxygen for their respiratory requirements. They usually possess a pair of gills from which exchange of gases take place. Gills are composed of short thread like filaments, which contain capillary network increasing the surface area. Fishes take O_2 rich water & brush it over their gills & oxygen crosses the thin filament via diffusion. The oxygen is then transferred over the entire body via blood. Water then exits the fish body through operculum, a flap like structure found on the sides of throat of bony fishes.[1,2]

Hypoxia, usually develops when the content of oxygen in the water body falls considerably. Hypoxia is subdivided into two categories.[3]

Local- when only a part of body is affected.

Generalized- when the entire body is affected.

Hypoxia is generally with two other terms

i. Hypoxemia- low arterial O_2 supply.

ii. Anoxemia- zero arterial O_2 supply.

The general cause for hypoxia seems to be decreased exposed surface area of water body. As oxygen is sparingly soluble in water, any form of vegetation on the water body surface, hampers the solubility rate adversely.[4]

For inland fisheries in Bihar, the growth of water hyaunth, on the surface of river water posses a great threat. The spread of this plant is usually uncontrollable sometimes even the entire pond is covered by it. In such a scenario, the fishes are deprived of both sunlight as well as dissolved oxygen.[5] The submerged plants, (which are add on sources of oxygen) on not receiving sunlight, as unable to perform photosynthesis, which releases oxygen. This causes the O_2 level to further dwindle.[6]

At peak deprivation level, the fishes suffer hypoxia. Hypoxia affects not only the respiratory system but all the other important system such as reproduction development as well.

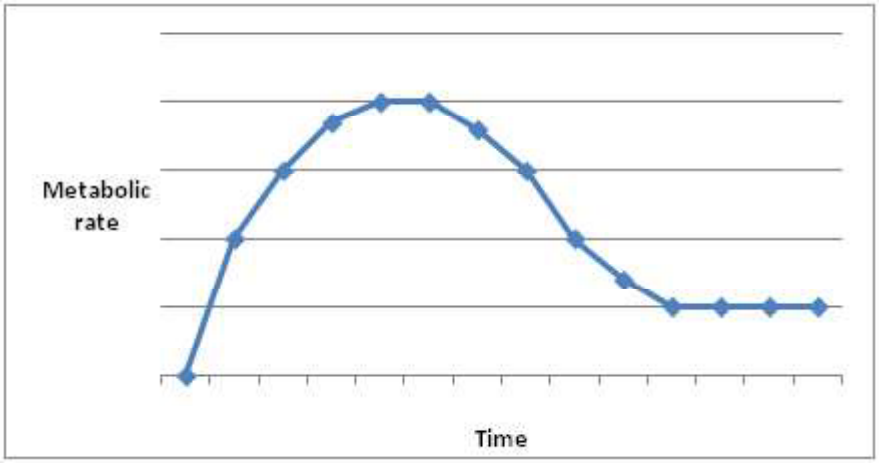
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MATERIALS & METHODS

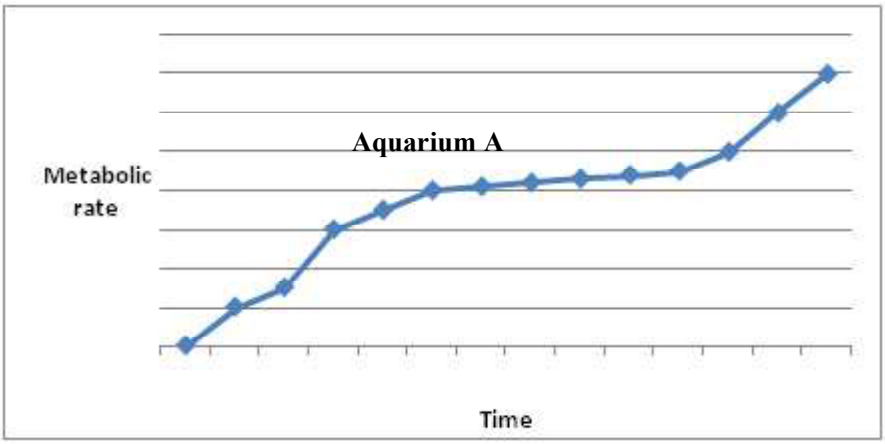
2 aquariums, 10 fishes (especially ones that are pregnant or producing eggs), Hydrilla, covers. An experiment was designed. The 2 aquariums were placed at 2 different spots. Water submerged plants such as hydrilla was introduced into the aquariums. Top of aquarium A was five fishes each were kept covered with air tight opaque material & negligible inlet of air (O_2) was allowed & it was kept in a room with no sunlight.

Aquarium B was kept in well lit room, with perforated lid, to allow inlet of O_2. The metabolic rates were monitored from time to time.

The following results were observed. Fishes in aquarium A & B were very active at first, feeding & moving actively. After two- three days, the activity of fishes in aquarium A was negligible, while that of aquarium B was constant.



Aquarium A



CONCLUSION

From the above experiment it can be said that lowering the amount of oxygen, lowered the activity of fishes, showing a sharp decrease in metabolic rates. Also it was seen that in fishes of aquarium A the reproduction of eggs was lowered almost to zero while in the fishes of aquarium B, the eggs count was good.

Hypoxia can always differ in terms of severity. In cases where hypoxia is sudden, it can kill thousands of fishes at a time (oil spills).

If hypoxia is gradual the fish body can adapt/adjust itself to low O_2 levels, usually by lowering their metabolic rates, & survive for some time.

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Studies on physico-chemical properties of ground water of Saharsa district (Bihar)

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Abstract: Findings of research on groundwater samples play a very crucial role to study the aquatic ecosystem and its effect on human beings since water is basic need for all living things to exist. Here in this paper different ground water samples derawn from various sites of Saharsa region was systematically studied to know more about the chemical and physical nature of water quality. The water samples were collected from 4 different sites of this region.

Key words: Groundwater, aquatic ecosystem, physico-chemical properties.

INTRODUCTION

Our planet earth despite surrounded by two third parts of it by water still there is a lack of reliable freshwater having standard physico-chemical quality. And the major problem has risen due to unpredicted human population growth, urbanization; overuse of land is increasingly becoming unsuitable & even dangerous for human consumption.[1,2] However freshwater as well as groundwater can be used for potable water. Groundwater is the basic need of all living things directly or indirectly and mostly in many agricultural areas it functions as a backbone for agricultural as well as domestic purposes. All the living creatures in the earth depend on the free available reserve water for their growth. In many parts of the world people still have no access to clean drinking water. Once the groundwater gets polluted the quality of water not only gets affected but the existence of life also gets threatened. The quality of water usually depends upon its physical, chemical and biological parameters or qualities. Groundwater does not only contain the purified water but it is occupied with several types of pollutants along with other type of substances which is hazardous to both health as well as life of the living organisms. The quality of groundwater is highly influenced by both natural & anthropogenic activities[3]. In rural areas mostly the agricultural areas excessive use of fertilizers has directly affected the ground water quality[4]. But beyond the qualitative aspect it is also advised to remain in vigilant on the level of the quality consumed by the people[5]. The report which has been recently given by Bihar government stated that the drinking water in rural areas is not safe due to which health of rural people are in risk[6]. In addition to all the problems mentioned above water logging is also a serious problem in many regions having impact on livelihood security[7]. Here in this research paper attempt was made to study the groundwater quality in Madhepura with respect to dissolved oxygen, sodium, potassium, calcium, magnesium & pH with several analysis of groundwater from different sources.

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MATERIALS & METHODS

Study area

The study was conducted in different villages of Saharsa district. For studying the water quality of wells following villages were selected Sardiha, Barahi, Dumrel, Baiyanathpur, Hansa, Nandanpur, Baruwari, etc. While for studying the water quality of hand pumps the hand pumps present in the following regions were selected Kandanlura, Baryahi, Maheshi, Mohanpur, Patarghat, Raghunathpur, Mirzapur, etc, Maheshi, Patwaha, Mohanpur, Kadanlura are the villages were selected for tube well water analysis. Rivers such as Bakra nadi, Sursa nadi, Simri nadi situated under the Saharsa district were chosen for analysis.

For the present study of quality assessment of drinking water of Saharsa district,10 open well, 10 hand pumps, 5 tube wells, 2 pond and 4 river ghats have been selected for sampling covering the entire area of the district. The sampling was done during rainy season of 2018-2019 by grab sampling method using 2.5 litre polythelene bottle marked with date, time and sampling point. In the case of well water, samples were taken from at least two meters deep from the surface of water in wells. In the case of hand pumps, it was run at first for two or three minutes and samples were collected. Water from the pond was simply drawn in plastic bottles and was filtered for removal of beverages. Similar treatment was done with the sampling or river water.

pH analysis:-

pH is considered as one of the most important operational water quality parameters, with the optimum pH required often being in the range of 6.5–9.5[8]. For the determining the pH of collected water samples electronic pH meter was used. pH is the term universally used to express the intensity of the acid ort alkaline condition of the solution.

Fluoride determination:-

For normal development of healthy teeth & bones fluoride is very important. However, higher concentration of F^- causes dental and skeletal fluorosis such as mottling of teeth, deformation of ligaments and bending of spinal cord[9]. Two calorimetric methods are highly used for the estimation of fluoride the SPADNS method & the Alizarian method. But here in this research work the first type of method i.e. the SPADNS was used for fluoride analysis since it requires little time for showing result.

Dissolved carbon dioxide determination:-

Since carbon dioxide is normal component of all natural water & it dissolves in water in varying amount. Therefore here in this research paper importance has been given in consideration of corrosive properties of CO_2 by titration with a standard alkali to pH 8.3.[10]

Chemical oxygen demand (COD)

COD is the oxygen requirement of a sample for oxidation of organic and inorganic matter. Determination of COD was done by standard method by simple oxidation & reduction method.[10]

Biological oxygen demand (BOD)

BOD is the requirement of oxygen due to biodegradable organic matter only is a biological phenomenon carried out at more or less natural condition. BOD was also determined by standard method.

RESULT AND DISCUSSION

This paper deals with the different results of the project including various aspects such as pH, fluoride determination, BOD, COD & dissolved carbon dioxide determination which has been discussed below in the table.

Table 1:- Different pollution parameters of well water (Ww) samples

Water sample	pH	Fluoride mgm/L	Dissolved CO_2	COD	BOD
Ww_1	8.00	1.52	39	3.12	0.9
Ww_2	7.98	1.00	38	3.18	0.8
Ww_3	7.90	0.23	34	3.52	1.41
Ww_4	8.60	0.26	37	3.00	1.61
Ww_5	6.96	0.15	29	3.26	1.00
Ww_6	7.85	0.89	35	3.01	0.86
Ww_7	7.69	0.24	36	2.68	1.21
Ww_8	7.89	0.97	29	3.60	1.32
Ww_9	8.60	0.98	28	3.64	1.13
Ww_{10}	7.92	1.08	36	2.48	1.10
Average	7.939	0.732	34.1	3.149	1.134

Table 2:- Different pollution parameters of hand pump water (HPw) samples

Water sample	pH	Fluoride mgm/L	Dissolved CO_2	COD	BOD
HPw_1	7.98	1.00	22.56	1.72	0.60
HPw_2	7.97	0.92	23.10	1.39	0.63
HPw_3	8.00	0.86	22.8	1.49	0.58
HPw_4	7.89	1.00	22.7	1.58	0.61
HPw_5	7.91	0.91	22.5	1.56	0.62
HPw_6	7.88	0.95	23.00	1.32	0.59
HPw_7	7.96	0.96	23.11	1.43	0.68
HPw_8	7.98	0.96	23.50	1.44	0.58
HPw_9	7.81	0.89	22.9	1.31	0.58
HPw_{10}	7.86	0.92	22.6	1.41	0.63
Average	7.924	0.937	22.8	1.465	0.61

Table 3:- Different pollution parameters of tube well water (TWw) samples

Water sample	pH	Fluoride mgm/L	Dissolved CO_2	COD	BOD
TWw_1	7.60	0.86	20.34	0.86	0.53
TWw_2	7.56	0.80	20.12	0.83	0.48
TWw_3	7.32	0.78	19.34	0.83	0.42
TWw_4	7.50	0.52	20.00	0.84	0.42
TWw_5	7.50	1.64	20.18	0.70	0.45
Average	6.096	0.92	19.99	0.812	0.46

Table 4:- Different pollution parameters of pond water (Pw) samples

Water sample	pH	Fluoride mgm/L	Dissolved CO_2	COD	BOD
Pw_1	8.32	0.92	46	4.30	4.11
Pw_2	8.41	0.83	49	4.29	4.32
Average	8.36	0.875	47.5	4.295	4.215

Table 5:- Different pollution parameters of river water (Rw) samples

Water sample	pH	Fluoride mgm/L	Dissolved CO_2	COD	BOD
Rw_1	8.40	0.53	48	5.33	3.26
Rw_2	8.32	0.69	39	5.41	3.18
Rw_3	7.92	0.58	46	5.39	3.44
Rw_4	8.24	0.51	43	5.37	3.38
Average	8.22	0.57	44	5.375	3.315

The above data given in the table reveals that the average pH values of different sources is pond water > river water > well water > hand pump water > tube well water. Since the tube well water is most protected from surface contamination, it is least polluted from surface contamination thus it is least polluted while open well ponds and rivers are most exposed to anthropogenic activities, they are most polluted in regard to pH.

When concerning about the fluoride concentration it has been found that hand pump water > tube well water > pond water > well water > river water. The average value of hand pump water is just above the desirable limit while that of tube well water is just below the desirable level. However the average values of fluoride concentration of pond water, well water & river water samples are much less than the desirable limit. Thus hand pump water is safe for drinking.

Study of Dissolved carbon dioxide helps to ascertain the life system of aquatic animals and plants and at the same time we can be sure of the quality of water regarding its utility purpose. According to the data mentioned above in the table the pond water sample records the highest

concentration of dissolved CO_2 while tube well water samples record the lowest. The order is given as-

Pond water > river water > well water > hand pump water > tube well water.

The chemical oxygen demand (COD) is an important parameter which assesses the pollution load in water. This parameter is determined by refluxing water samples with a known amount of potassium dichromate in sulphuric acid. In well water sample, COD runs from 2.48 to 3.68 ppm while for land pump water samples, the minimum and maximum values are 1.31 &1.72 ppm respectively. The tube well water samples record COD in between 0.68ppm to 0.86ppm. In pond water it is 4.29 & 4.30ppm while river water it runs from 5.33 to 5.41ppm. Since the river water is highly exposed to domestic waste therefore carries the higher average COD.

The order of the different water resources in respect of BOD content may be given as below:-

Pond water > river water > well water > hand pump water > tube well water. For open well water samples, the BOD value runs from 0.8 to 1.61 ppm. In the case of hand pump water samples, BOD runs from 0.58 to 0.68 ppm while for tube well water sample it is found in the range of 0.42 to 0.53 ppm. in the pond water it runs from 4.11 to 4.32 ppm and in river water samples it runs from 3.18-3.38. The high content of BOD in pond and river water may be attributed to the free inflow of runoff of agricultural field which carry the fertilizers and several insecticides, pesticides etc to increase the BOD load of pond water and river water.

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Physico-chemical analysis of drinking water in some rural areas of Madhepura, Bihar

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Abstract: Drinking water has great effects on human health and its quality matters a lot. This research is an effort to know the physico-chemical analysis of drinking water in rural areas of Madhepura district, Bihar. The parameters such as pH, dissolved oxygen, total dissolved solid, water temperature, alkalinity, chemical oxygen demand, biological oxygen demand, were analyzed from during the research period. Our results demonstrated that drinking water collected from different places of Madhepura such as Barahi, Chaura, Garhia, Manikpur and Pithahi was not found to be suitable for human health.

Keywords: Physico-chemical analysis, Water samples, pH, temperature

INTRODUCTION

Safe drinking water percentages have fallen drastically all over the world in the last few decades. The hardest hit parts are the developing countries & within such countries the worst are rural areas. Lack of awareness, resources and proper sanitization has tremendously lowered the potable water quality.[1]

70 % of the available water in India is polluted by industrial discharge in form of effluents, sludge etc. a large variety of water borne diseases are killing thousands of people every year.[2] Bihar faces over a hundred of water related problems every year. The quality of potable water is far less than the needs of the state.[3] Bihar is mainly drained by rivers in which river Kosi is one of the major rivers.[4]

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Though this forms the major source of water, people are mostly dependent on ground water sources like wells, tube wells etc.[5]

Groundwater is highly affected by seepage of chemicals in form of fertilizers, pesticides, weedicides or heavy metals like Mercury and radioactive elements.[6] To declare a water source as potable water resource, the water must undergo a few physicochemical tests. The present study aims at water quantity assessment in 5 villages of Madhepura. Various physical and chemical tests were done.

MATERIALS AND METHODS

Study of the area

Madhepura district forms an agricultural district of Bihar. It enjoys a warm and moderate climate with average temperature of about 25^0 C. It also receives a fair share of annual rainfall that helps recharge its aquifers naturally. It

lies between 25.92' latitudes and 86.79'longitudes. It is drained by river Kosi also known as Sorrow of Bihar.

Collection of raw data

Five villages namely Barahi, Chaura, Garhia, Manikpur and Pithahi were selected randomly. Water from different water bodies of these villages were collected in a sterilized glass bottles and were marked with date, time (10 am to 2pm) and the place and source of collection. The temperatures of the samples were noted. The samples were carefully brought to the laboratory for analysis.

Physico-chemical analysis

Parameters like temperature, pH, salinity, alkalinity, dissolved oxygen (DO), biological oxygen demands (BOD), chemical oxygen demand (COD), were measured.

Table 1- Data of bore well water analysis

Place	DO	COD	BOD	AK	TDS
Barahi	**2.7**	**440**	**410**	**45.2**	**192**
Chaura	**2.1**	**320**	**320**	**39.4**	**433**
Garhia	**1.8**	**360**	**360**	**48.1**	**426**
Manikpur	**1.9**	**220**	**280**	**39.4**	**315**
Pithahi	**2.3**	**422**	**400**	**44.2**	**401**

Table 2- Data of well water analysis

Place	DO	COD	BOD	AK	TDS
Barahi	**2.1**	**460**	**350**	**40.2**	**182**
Chaura	**2.6**	**220**	**370**	**32.5**	**333**
Garhia	**2.2**	**240**	**310**	**34.9**	**356**
Manikpur	**1.7**	**320**	**380**	**41.5**	**425**
Pithahi	**1.9**	**320**	**290**	**34.1**	**291**

RESULTS

1. Temperature- the temperature ranged from 18^0 C minimum to 30^0 C maximum.
2. pH – the pH ranged from slightly acidic 6.93 to slightly basic 7.55.
3. Salinity- the variation of salinity was negligible. It ranged between 5-6 ppt.
4. Dissolved oxygen- it is essential for aquatic fauna and flora (during night). It ranged from 1.7 Mg/L to 2.7 Mg/L.
5. Alkalinity- it ranged from 32.5 Mg/L to a maximum of 48.1 Mg/L.
6. BOD- higher BOD indicates pollution and presence of excessive nitrates and phosphates. It ranged from 280Mg/L to 410 Mg/L.
7. COD- it ranged from 220Mg/L to 460Mg/L.
8. Total Dissolved Solid- it ranged from 182 Mg/L to a maximum of 433 Mg/L.

CONCLUSION

The study analyzed the water quality in these villages. These parameters play an important role in maintaining the well being of the locals. Village water quality was most compromised than that of the other four villages which were studied. Though the level of pollution was not lethal but if proper awareness is not spread the water shall cease to remain potable in the next few years.

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Assessment of Nitric oxide scavenging activity and Total antioxidant activity of Chitosan extracted from carapace of freshwater edible crab *Sartoriana spinigera*

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Abstract: Antioxidants are produced by the living body to scavenge the free radicals. But due to increased exposure to undesirable factors such as pollutants, smoking, unbalanced diet, the body needs additional supplementary antioxidants. The study was conducted to observe antioxidative property in one such natural supplement that is chitosan. Chitosan is a cationic polysaccharide produced by N-deacetylation of chitin found in exoskeleton of arthropoda, especially crustacea. In this experiment, Chitosan was extracted from *Sartoriana spinigera* which is a freshwater edible crab of Jharkhand and its Nitric oxide scavenging activity and total antioxidant activity was studied. Percent of scavenging activity of Chitosan against Nitric oxide anion was found to be 21.95 %, 35.49%, 46.90%, and 66.04% at 0.5 mg/ml, mg/ml, 5 mg/ml and 10 mg/ml concentration respectively. Percent of total antioxidant activity of Chitosan was found to be 29.73 %, 33.78%, 50%, and 62.16 % at 50 µg/ml, 100 µg/ml, 200 µg/ml and 400 µg/ml concentration respectively.

Key words: ***Sartoriana spinigera,*** **exoskeleton, chitosan, antioxidants.**

INTRODUCTION

As metabolic processes take place in the human body for the purpose of oxidation, there is simultaneous formation of free radicals. Apart from *in vivo* production, free radicals also accumulate in the body as a result of anthropogenic activities, cigarette smoking, automobile gas exhaust and x-rays exposure. Free radicals are harmful to the body. The instability and reactivity of free radicals due to the lone electron in the outer shell can cause them to attack specific biomolecules in the body such as proteins and lipids[1]. Presence of abnormal amount of free radicals in the body leads to oxidative stress. To maintain normalcy, there is a balance between the quantity of free radicals and antioxidants that are produced by the body to scavenge free radicals and inhibit their deleterious effects in the body.

Nitric oxide and reactive nitrogen species (RNS) are free radicals that are derived from the interaction of NO with oxygen or reactive oxygen species[2]. Chronic exposure to nitric oxide radical is associated with various carcinomas and inflammatory conditions including juvenile diabetes, multiple sclerosis, arthritis and ulcerative colitis[3]. For estimation of total antioxidant potential, various scientists use different assay using DPPH, ABTS, Phosphomolyb denum complex and reducing power[4]. Due to increased concentration of free radicals in the body, additional supplements of antioxidants are needed to be consumed. Various medicinal plants and natural extracts have proved to have antioxidant properties against nitric oxide.

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However, natural antioxidants of animal origin are not acknowledged as much. Chitosan is one such natural source of antioxidants that is of animal origin. Chitosan is a linear polysaccharide formed by deacetylation of chitin. Chitosan is composed of two units: 80% of deacetylated unit-ß 1-4 linked D glucosamine and 20% acetylated unit-ß 1-4, N-acetyl-D-glucosamine. In this research, chitosan has been extracted from a freshwater crab *Sartoriana spinigera* abundantly found in Jharkhand, India. It is not only a delicacy, but is also of great ethnobiological significance and is used by the tribal people of Jharkhand against ailments such as bacterial infection, gastrointestinal problems.

The aim of this study was to extract Chitosan from carapace of *Sartoriana spinigera* and assess its Nitric oxide scavenging activity and Total antioxidant activity by phosphomolybdenum method.

MATERIALS AND METHODS

1. *Characterization of raw materials:*

Fresh specimen of freshwater crab *Sartoriana spinigera* were purchased from local market of Ranchi, Jharkhand. The specimens were sent to Zoological survey of India (ZSI), Kolkata for identification of species. The species were acclimatized for 7 days in laboratory of P.G. Department of Zoology, Ranchi University. The carapace was obtained.

2. *Extraction of chitosan:*

Extraction of Chitosan from the carapace was done following Takiguchi[5,6].

2.1 Demineralization:

20 g of carapace powder was soaked in 300 ml of 1N HCl for 24 hours. The residue obtained was washed with distilled water to obtain a neutral pH. It was dried and weighed.

2.2 Deproteinization:

Demineralized powder was added in 300 ml of 4% NaOH at 80°C in hot water bath for 24 hours with 3 intervals. The residue obtained was washed with distilled water to obtain a neutral pH. It was again dried and weighed. Chitin was obtained after deproteinization.

2.3 Deacetylation:

Chitin was then added in 300ml of 50% NaOH and heated at 100°C in hot water bath for 18 hours with 3 intervals. After every interval, NaOH was replaced by fresh quantity and residue washed with distilled water. The final residue obtained was brought into neutral pH. It was dried and weighed. The powder obtained was chitosan.

2.4 Characterization of chitosan:

Average degree of deacetylation of chitosan extracted from carapace of *Sartoriana spinigera* was found to be 78.53±2.03% following Brugnerotto *et al.*[7] FT-IR study for showed absorbance bands of chitosan extracted from *Sartoriana spinigera* showed seven peaks at 891.11cm^{-1},1029.99 cm^{-1}, 1427.32 cm^{-1}, 1577.77 cm^{-1},1662.64 cm^{-1}, 2885.51 cm^{-1} & 3259.70 cm^{-1}, that were similar to the standard chitosan for comparison[8].

3. *Antioxidative activity:*

3.1 Nitric oxide scavenging activity

Nitric oxide scavenging activity was estimated following Kumar *et al*[9]. The experiment of superoxide scavenging activity was conducted in Biogenics Laboratory, Hubli, Karnataka. Various concentrations (0.5mg/ml, 1 mg/ml, 5 mg/ml and 10 mg/ml) of samples and Butylated Hydroxy Anisole (BHA) were taken in different test tubes and made up to 3ml with 0.1M phosphate buffer (pH 7.2). Sodium Nitroprusside (5mM) prepared in buffered saline (pH 7.2) was added (1 ml) to each tube. The reaction mixture was incubated for 30 min at RT. A control without the test compound, but with an equivalent amount of methanol was maintained. After 30 min, 1.5 ml of above solution was mixed with 1.5 ml of Griess reagent (1% Sulphanilamide, 2% phosphoric acid and 0.1% N-1-Naphthyl Ethylenediamine Dihydro chloride). The absorbance of the samples was measured at 546 nm. Nitric oxide radical scavenging activity was calculated using the following formula:

$$\% \text{ NO radical scavenging activity} = \frac{(\text{control OD} - \text{sample OD})}{\text{control OD}} \times 100$$

3.2 Total antioxidant activity

Total antioxidant activity by phosphomolybdenum method was estimated following Prieto *et al*[10]. Various concentrations of chitosan (50 µg/ml, 100 µg/ml, 200 µg/ml and 400µg/ml) were mixed with 3 mL of reagent solution (0.6 M sulphuric acid, 28mM sodium phosphate and 4mM Ammonium molybdate)**.** The tubes containing reaction solution were incubated at 95°C for 90 minutes**.** Absorbance of the solution was measured at 695 nm using a UV-VIS SPECTROPHOTOMETER against blank after

cooling at room temperature. Methanol was used as blank. Ascorbic acid was used as standard.

STATISTICAL ANALYSIS

Correlation coefficient was calculated and regression graph were prepared to study the dependence of variables on independent variables. Student's t test between % scavenging activity of chitosan and standard was calculated.

RESULT AND DISCUSSION

Table 1: Scavenging activity of Standard (BHA) and Test (Chitosan) in different concentrations against Nitric oxide free radical

Sl. No.	Concentration (mg)	Standard – BHA (%)	Test - Chitosan (%)
1.	0.5	100	21.95
2.	1	100	35.49
3.	5	100	46.90
4.	10	100	66.04

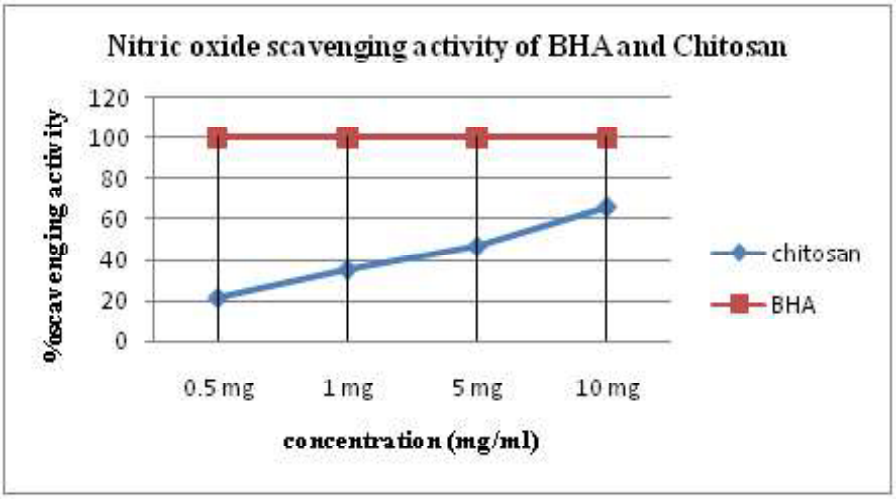


Fig 1: Scavenging activity of different concentration of Chitosan and BHA on Nitric oxide free radical

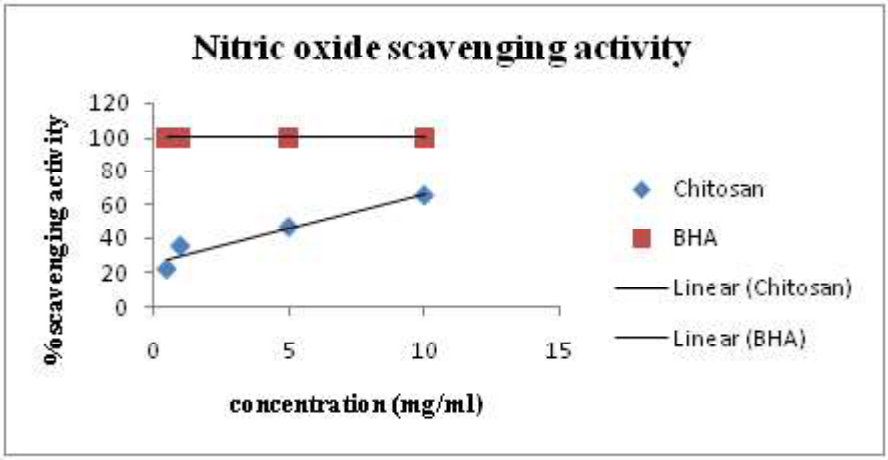


Fig 2: Graph representing correlation between concentration of BHA and chitosan and percent of nitric oxide scavenging activity

In the present study (Table:- 1) showed the percent scavenging activity of BHA against Nitric oxide anion was 100 %, 100%, 100% and 100% at concentration of 0.5 mg/ml, 1 mg/ml, 5 mg/ml and 10 mg/ml respectively. Percent of scavenging activity of Chitosan against Nitric oxide anion was found to be 21.95 %, 35.49%, 46.90%, and 66.04% at 0.5 mg/ml, 1 mg/ml, 5 mg/ml and 10 mg/ml concentration respectively.

Chitosan concentration showed positive correlation with its percent scavenging activity, indicating that as concentration of chitosan increased, its scavenging activity against Nitric oxide free radical also increased with the slope value b=4.09 (R^2= 0.96) (Fig:- 2).

The result of present study showed that chitosan extracted from carapace of *Sartoriana spinigera* exhibits antioxidant activity against Nitric oxide free radical.

No work on Nitric oxide scavenging activity of chitosan has been reported till date. However, studies on antioxidant activity of Balofloxacin and Prulifloxacin was conducted by Ranka[11] which showed that percent inhibition of NO with balofloxacin at 10mg/ml, 20 mg/ml, 30 mg/ml and 40 mg/ml was 34.49%, 49.09%, 60.42% and 73.14% respectively, whereas, inhibition of NO with Prulifloxacin at 10mg/ml, 20 mg/ml, 30 mg/ml and 40 mg/ml was 40.94%, 53.09%, 66.25% and 88.06% respectively. In vitro nitric oxide scavenging activity of three Bangladeshi medicinal plants were studied by Parul[12], and it was observed that *Phyllunthus freternus* showed percent scavenging at 50 µg/mL, 100 µg/mL and 200 µg/mL as 51.00±0.011 %, 60.16 ± 0.015% and 60.807± 0.005% respectively against nitric oxide. Leaves of *Triumpetta rhomboidae* at 50 µg/mL, 100 µg/mL and 200 µg/mL showed percent scavenging activity of 30.21± 0.008 %, 50.92± 0.013% and 53.942±0.005% respectively. Barks of *Casuarina littorea* showed nitric oxide scavenging activity as 29.008± 0.013%, 41.418±0.010% and 54.017± 0.006% at 50 µg/mL, 100 µg/mL and 200 µg/mL respectively.

Table 2: Total antioxidant activity of Standard (Ascorbic acid) and Test (Chitosan) in different concentrations

Sl.No.	Concentration (µg/mL)	Standard Ascorbic acid (%)	Test Chitosan (%)
1.	50	30.86	29.73
2.	100	41.97	33.78
3.	200	60.49	50
4.	400	79.01	62.16

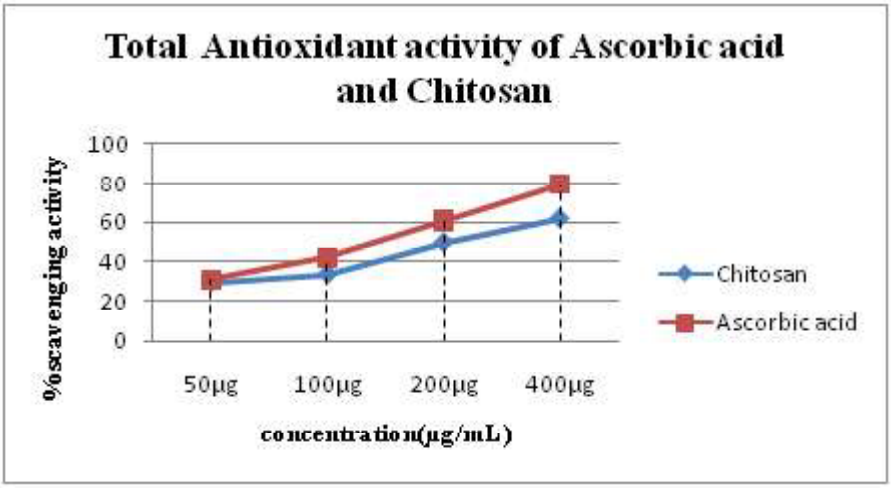


Fig 3: Total antioxidant capacity of different concentration of Chitosan and Ascorbic acid

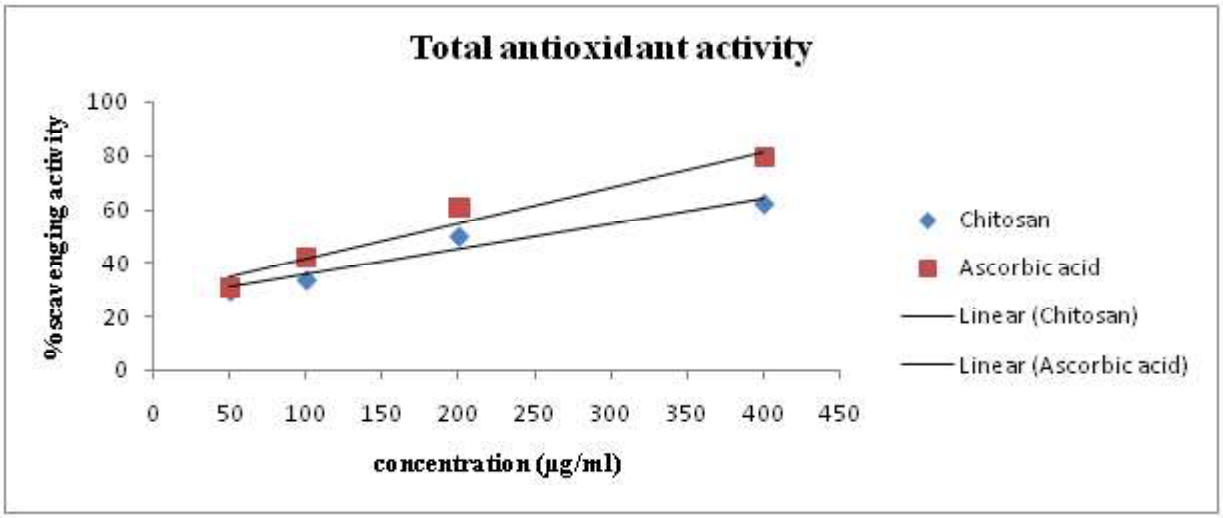


Fig 4: Graph representing Concentration of Ascorbic acid and chitosan used and their percent scavenging activity.

In the present study Table 2 showed the percent of total scavenging activity of Ascorbic acid was 30.86%, 41.97%, 60.49% and 79.01% at concentration of 50 µg/ml, 100 µg/ml , 200 µg/ml and 400 µg/ml respectively. Percent of total antioxidant activity of Chitosan was found to be 29.73 %, 33.78%, 50%, and 62.16 % at 50 µg/ml, 100 µg/ml, 200 µg/ml and 400 µg/ml concentration respectively.

Chitosan concentration showed positive correlation with its percent scavenging activity, indicating that as concentration of chitosan increased, its Total antioxidant activity also increased with the slope value b=0.094 (R^2= 0.97) (Fig:-4).

Aliyu[13] reported that the total antioxidant capacity of chitosan (TAC) was based on reduction of Mo(VI) to Mo(V) and subsequent formation of green phosphate/ Mo(V) complex at acidic pH.

Most of the studies on calculating Total antioxidant capacity have been done on plant extracts. Aliyu[13] observed that n-butanyl root extract of *Anchomanes difformis* possessed significant total antioxidant capacity equivalent to 90mg/g ascorbic acid at 500 and 1000 µg/mL concentration. Elkhamlichi[14] reported that ethyl acetate extract of the seeds of *Calycotoma villosa* also possessed significant total antioxidant capacity equivalent to 157.6 mg/g ascorbic acid at higher concentration (100 µg/mL).

The antioxidant capacity of chitosan extracted from carapace of *Sartoriana spinigera* against four free radicals showed that chitosan is highly efficient in scavenging the free radicals. Thus, chitosan can be encouraged to be used as a natural antioxidant in pharmaceutical industry.

CONCLUSION

The demand of supplement antioxidants has now bent towards an approach to nature and natural sources. In this study, Chitosan extracted from carapace of *Sartoriana spinigera* proved to have Total antioxidant activity as well as Nitric oxide scavenging activity. This study encourages the use of natural sources of antioxidants such as Chitosan to potentially replace synthetic drugs in the pharmaceutical market.

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Studies on erythrocytes & leukocytes count in normal & different types of leukemia cells

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Abstract: Red blood cells or the erythrocytes & WBC or the leucocytes are the cells that circulate in the blood & carries oxygen throughout the body. The counting of RBC indicates total number of red blood cells present in the given blood sample while the WBC represents the capacity to fight with diseases. It is regarded as one of the test which involves complete counting of blood & is very useful for the evaluation of the health & to have knowledge about certain disease.

Key words: erythrocytes, leucocytes, leukemia, hemocytometer

INTRODUCTION

RBC is the total number of red blood cells present in given volume of the blood. This test is performed to counts the number of red blood cells in a single drop (a microlitre) of blood & counted through either impedance or optical technology. By using microscopic evaluation of well-prepared blood smears qualitative estimation of RBC & WBC can be done efficiently. Decrease in RBC level also leads to anemia.[1] It is estimated that the normal range of RBC in blood varies according to the gender & age. Normal range of RBC in men varies from 4.5 to 6.2 million while in women it ranges from 4.0 to 5.2 million.

White blood cells also known as leucocytes play a major role in causing leukemia. The total number of WBC count also varies depending upon the age, gender & types of leukemia. A normal range of WBC is usually 5,000-10,000/ml in adults. White blood cells generally functions as an effectors cells of immune system or it is one of the important components of the body's defenses against any diseases. Decrease WBC count is often termed as leucopenia while decrease in WBC (blasts) causes medical emergency called leukostasis.[2-3] It circulates throughout the blood stream and the lymphatic system. However most of the studies show that high WBC count is generally related to coronary heart disease.[4-6] This paper concerns about the different RBC & WBC count in the patients having different types of leukemia.

Since in this paper, four different types of leukemia (blood cancer) patients have been studied about. Therefore attempt was made to find out different concentrations of RBC & WBC found in their blood cells. The major factor for the occurrence of blood cancer is abnormal increase in the concentration WBC that destroys the normal cells from its growth. There are various methods for finding out different concentrations of WBC & RBC count such as through hemocytometer using standard microscope system.

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MATERIALS & METHODS

Experimental trip

Before the analysis of erythrocytes & leucocytes in different leukemia cells, one of the cancer hospitals was selected for the research work in Bihar. The patients were selected suffering from different types leukemia having average age group. After the selection of the patient each of the patients depending on type of cancer were divided into five groups. In group 'a' normal patient were kept who did not suffered from any type of cancer so that study on comparison on erythrocytes & leucocyte count between the normal & diseased patient can be effective at great extend. In group 'b' patients suffering from acute Myeloid Leukemia (AML), in group 'c' patient having Chronic myelogenous leukemia (CML) were kept, in group 'd' patient suffering from Acute lymphocytic leukemia (ALL) were included and finally in group 'e' patient with Chronic lymphocytic leukemia (CLL) were kept. Number of patients per group with different types of leukemia has been given in table no 1.

Table 1-Different types of leukemia patients in five groups

Group name	Sample	No. of cases
Group a	Control	25
Group b	AML	14
Group c	CML	23
Group d	ALL	11
Group e	CLL	23

Procedure

Each of the patient were kept with great care provided by all the medical facilities so that none of the paitent get infected with other types of disease and were kept under the doctors supervision. Blood was drawn from each of the patient according to their division of groups to stop it from mixing with other patient's blood. Puncture of the vein was done to draw the blood evacuated tubes that contained anticoagulant ethylene diamine tetra acetic acid (EDTA)[7] to stop its natural clotting. The RBC & WBC concentration was further observed by using haemocytometer. The observation made has been explained using the table no 2 & 3 given below.

OBSERVATION

The finding on RBC count (in millions/mm^3) & WBC count (in thousand/ mm^3) in normal and different types of leukemia are shown in table no. 2 & 3.

Table no. 2:- RBC count (in millions/mm^3) in normal and different types of leukemia Control & normal mean value = 4.9 & SD = 0.356

Types of leukemia	Mean value	Combined SD(σ)
AML	1.8	0.395
CML	2.5	0.467
ALL	1.7	0.332
CLL	2.6	0.352

Table no. 3:- WBC counts (in thousand/mm^3) in normal and different types of leukemia Control & normal mean value = 7.0 & SD = 0.935

Types of leukemia	Mean value	Combined SD(σ)
AML	18.7	5.920
CML	241.5	25.958
ALL	49.3	6.583
CLL	154.0	19.387

RESULTS & DISCUSSION

According to the above observation it is found that the patient suffering from acute lymphocytic leukemia (ALL) had the most lower RBC count and the patient having Chronic lymphocytic leukemia (CLL) had higher mean value of RBC count. From the above explanation it is clear that the patient suffering from the different types of leukemia had lower RBC count. On the other hand since RBC is the most important component of the blood required for the proper functioning of the cells for growth & division.

When talking about WBC it is seen that the patient suffering from different types leukemia also had great differences in their WBC count. There was great inflammation of WBC count than the normal mean value. Patient suffering from Chronic Myelogenous Leukemia (CML) & Chronic Lymphocytic Leukemia (CLL) had higher value of WBC count with respect to normal WBC count. Since leukemia is a type of cancer mostly found in blood and bone marrow and its major cause is rapid production of abnormal white blood cells. As a result abnormal production of white blood cells is not able to fight infection and impair the ability of the bone marrow to produce red blood cells and platelets thus leading to leukemia.

Thus the overall result done on RBC and WBC count in different patient shows that change in the haemogram pattern is one of the major causes of leukemia also leading to anemia.

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Bacteriological analysis of groundwater samples in Saharsa district, Bihar

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Abstract:- Ground water samples in general & those drawn from different pockets of Saharsa districts need proper investigation with respect to bacteriological composition which determine the biological property of water, in addition to their investigating physiochemical profile. Samples from 30 different sites of Saharsa urban areas, collected from the ground water resource were analyzed for bacteriological population present in them by MPN test method. MPN stands for most probable number analysis is a statistical method based on the random dispersion of microorganism per volume in given sample. In the present research paper 6 different types of bacterias have been analised.

Key words: Bacteriological, physicochemical, *E. coli*, Coliforms, tetrameric, *S. typhimurium*

INTRODUCTION

According to an important quotation of living pharmacologist 'water is elixir of life' the fortifying importance of drinking water drawn from ground water resources can be easily understood. Since liquid water, used in drinking purpose as unseen physicochemical & bacteriological properties one can't be sure to what are the bacteriological population present in the sample in time & space. All the characteristics, of water nevertheless reflect ample dynamicism because the parameters governing these characters are highly changeable. The present investigation may seem to be repetition of the previous efforts. Yet targets has been made to investigate & standarise the bacteriological properties for one tetrameric seasonal package in a year.

Since water is regarded as one of the basic component of living organism & the increasing population density has increased its demand[1-4]. Rapid industrialization, urbanization, & other activities has led the water resources get contaminated are several of reasons which resulted in the degradation of aquatic population. There are about 1.2 billion people worldwide who are not having access to safe water[5–7]. Due to above reasons water resources have become more susceptible to contaminations with various microorganisms and other pollutants[8-10]. Contamination of ground water usually varies depending on seasonal changes, soil, rock type etc[11,12]. Besides the various organic & inorganic compound found in water most of these resources are also subjected to harmful aquatic organisms which may also lead to several disease and infections. Availability & supply is not only the major problem of water but its quality is the topmost reason for the utility of water. Major regions with contaminated water where the population is very high, not having the awareness about sanitations, hygiene practices, and industrial areas.

Bacteriological analysis is the method that is usually used to analyse the number of bacteria present in water &

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sometimes to study about the type of bacteria. It is one of the method which is generally conducted to get information about the quality of water. It is generally the micro biological analytical procedure where sample of water is used for the bacterial study.

There are various different types of micro-organisms that are found in water resources such as Colifoms which are the group of bacteria that are usually rod in shape. *E. coli* is an example of coliforms that are considered as the major microbial indicators for monitoring the quality of water samples. Coliforms are the bacteria that are present in environment and also found in the feces of warm blooded animals. Their presence in drinking water causes several diseases. These are also found in the intestine and gut of some animals. Most of the types of *E. coli* are harmless to humans but some of their strains are also harmful when present in water causing several diseases. In contrast to coliforms several other bacteria are also found such as *salmoneela*, *Legionella* etc, that causes several types of waterborne diseases which can be prove very hazardous to the livestock.

MATERIALS & METHODS

Study setting

The study for bacteriological analysis was conducted in different urban areas of Saharsa district in Bihar. The sample required was collected from 30 different water resources found in this area including the well water sample, hand pump water sample, tube well water, pond water, river water during the four different seasons from 2019-2020. The water samples were collected from different water resources during the morning hours in a clean plastic bottles & the bottle was immediately closed so that the water does not get contaminated with other types of bacteria. The study was performed during the morning hour when the water remained still in rivers and ponds.

Determination of bacteriological analysis

For the bacteriological analysis water samples from different water resources were collected and brought in the university laboratory to observe the presence of different types of micro-organisms by using the MPN test method. MPN test is considered as one of the specific test to determine the presence of bacteria count in the given samples. MPN stands for most probable number which statistically used for the estimation the numbers of bacteria in the given sample by inoculating broth in 10-fold dilutions. This method is basically used for estimation of bacterial cells in water and food.

For the bacteriological analysis MacConkey was dissolved in 100ml of distilled water and then autoclaved at 121°C for 15 minutes and then cooled at room temperature. About 10 ml of the prepared solution was then filled in test tube in which Durham's tube was added keeping it in inverted position for the determination of fermentation gas. The whole process is followed by series of tubes. Thus the method is usually completed in given three steps:-

- Presumptive test
- Confirmed test
- Completed test

RESULT & DISCUSSION

The major purpose for the bacteriological analysis water from the different water resources of Saharsa district was to determine the presence of bacteria in the given sample of drinking water used by the people in those areas. The research done on the bacteriological analysis showed the following result.

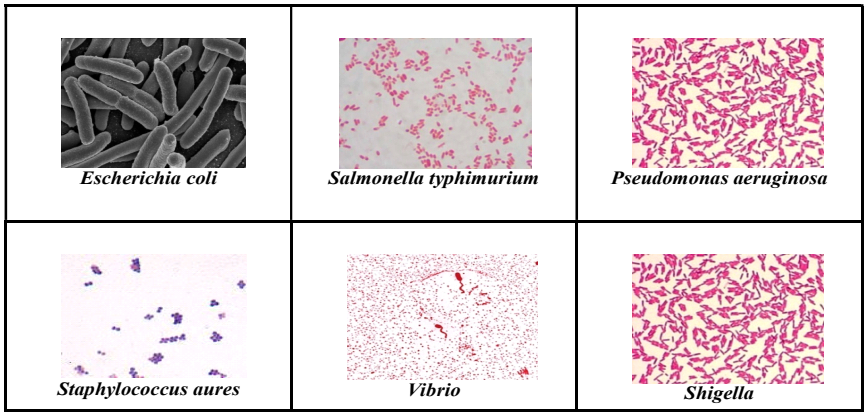


Fig. 1- Different types of bacteria observed

Table 1 :- Occurrence of bacteria in different seasons during the year 2019-2020

Rainy season (August- October)

Sl No	Microorganisms	Total count
1	*Escherichia coli*	25
2	*Salmonella typhimurium*	7
3	*Pseudomonas aeruginosa*	8
4	*Staphylococcus aures*	9
5	*Vibrio*	5
6	*Shigella*	4

Winter season (January-December)

Sl. No	Microorganisms	Total count
1	*Escherichia coli*	12
2	*Salmonella typhimurium*	11
3	*Pseudomonas aeruginosa*	5
4	*Staphylococcus aures*	7
5	*Vibrio*	6
6	*Shigella*	4

Autumn season (February- April)

Sl. No	Microorganisms	Total count
1	*Escherichia coli*	8
2	*Salmonella typhimurium*	5
3	*Pseudomonas aeruginosa*	7
4	*Staphylococcus aures*	9
5	*Vibrio*	7
6	*Shigella*	8

Summer season (May-July)

Sl. No	Microorganisms	Total count
1	*Escherichia coli*	24
2	*Salmonella typhimurium*	8
3	*Pseudomonas aeruginosa*	5
4	*Staphylococcus aures*	9
5	*Vibrio*	7
6	*Shigella*	6

The observation on the presence of bacteria in per 100 ml of water shows that *E. coli* & *Salmonella typhimurium* were found to be dominant in number having higher densities during the rainy seasons this generally implies that in areas where the sanitation facility is not properly developed has led to contamination of water resources. Since most of the water sample collected showed excessive growth of microorganisms, it can be considered as unfit for the intake by the people since it may also lead to various diseases.

CONCLUSION

All the water samples collected from the Saharsa region during the year 2019-2020 reflects the presence of *E. coli* & *S. typhimurium* in maximum in number obtained from all the water resources. The study reveals the prevalence of pathogen such as *E. coli* & *S. typhimurium* more during the rainy season than the other seasons for groundwater resources since the sewage & other organic waste present in those areas get washed and enter the water resources. Thus bacteriological counts from well water, tubewells water, pond water samples make the water more unfit for the drinking purpose. Thus monitoring of water resources is required for improving the quality of water.

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Different nutrients requirements of a laboratory rat

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Abstract: The laboratory rat is favoured for experiment during to its moderate size, extravagant reproduction & adaptation that is traceable. Change in diets has taught us about the nutritional requirements of rats & the diseases or effects which are caused by deficiency of these nutrition. Investigators monitor the weight gain & reproductive relation. Among the different strains of rat used reproductive characteristics such as onset of reproductive maturity, fertility, litter size, preweaning mortality vary considerably.

Key words: nutritional requirements, deficiency, extravagant reproduction

INTRODUCTION

The nutritional requirements change according to the reproductive activity, age & developmental state and hence are not static. There is considerable difference between male & female rats.[1] The nutrients which are being discussed here is as follows fat, proteins, calcium, magnesium, phosphorous, potassium, sodium, copper, iron & zinc.[2] Through there are more nutrients, emphasis on these nutrients is given here.

These nutrients help in maintaining most of the physiological activities inside the rat body.

- Fats are usually related with energy intake & body weight gain. Rats usually requires 112 k cal of energy per body weight. The daily intake of fat should be less than 70gm.[3]
- Proteins are building blocks of any organisms. High amount of protein is required especially during growth & reproduction.
- Calcium- It is used to maintain bone density as well as used in secondary signaling pathway. Increase in calcium ingestion is good for bone health & prevention of osteoporosis.[4]
- Magnesium- A long term dietary Mg deficiency increases blood pressure & induces sustained hypertension.[5]
- Phosphorous- It is an important nutrient, which is needed to make protein for growth maintenance, repair of cells & tissues. A high intake of phosphorous can cause a decrease in body weight gain.[6]
- Potassium- Potassium requirement is seen to decrease with the increase in age.
- Sodium- Sodium requirement remains constant during the growth period. High salt intake elevates the blood pressure.[7]
- Copper- It is essential for maintaining an

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appropriate amount of haemoglobin in blood. Copper deficient mice develop anemia.

- Iron- It forms the core of the haemoglobin molecule & hence is important for O_2 transfer in blood. Low levels can cause sickle shaped RBCs[8].
- Zinc- It is important during pregnancy & for maintaining learning & memory function in rats.

MATERIALS & METHODS

Different methods were applied for the clinical estimation of there more & micronutrients. The data on which requirements are based were reported from many different laboratories that used colony management practices. The values have been converted to dietary content by assuming a dietary intake of 15g/rat/day for growing rats or adult rats for maintenance, 15 to 20g/rat/day during pregnancy.

RESULT & DISCUSSION

From the table we can say that for amounts of nutrients per kg body weight, is seen to increase during pregnancy. A high increase in the requirement of Zinc & Iron indicates their importance in maintaining the foetus & its developments.

Table 1- Amounts of nutrients per kg body weight

		Amount per kg diet		
Nutrient	**Unit**	**Maintenance**	**Growth**	**Reproduction (female)**
Fat	g	50	50	50
Protein	g	50[b]	150[b]	150[b]
Calcium	g	5	5	6.3
Magnesium	g	0.5	0.5	0.6
Phosphorus	g	3	3	3.7
Potassium	g	.6	3.6	3.6
Sodium	g	0.5	0.5	0.5
Copper	mg	5	5	8
Iron	mg	35	35	75
Zinc	mg	12	12	25

CONCLUSION

The amount of nutrients required to sustain maximum growth of young rats is different from the amounts needed to maximize enzymal activities. Male & female nutritional values differ considerably. A deficiency of these nutrients can cause various diseases such as insulin resistance, obesity, bone deformation or osteoporosis etc. For the maximal utility of laboratory rats, they must be well fed with proper amounts of nutrients & a stable diet.

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Status of Ichthyofauna in relation to hydrobiology of Ponds in Gaya, India

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Abstract: Ponds are good candidates of fishery and having a place of diversity of species. Each pond has to be characterized thoroughly to investigate the reason for availability of particular species. Physico-chemical parameters of pond water and seasonal variation influence the survival of certain species. The results are quiet fluctuating in the ponds investigated for one year. This is due to quality of water and biota which were reported of varying degree.

Key words: Biota, fishery, Physico-chemical, seasonal variations, Suryapohkhra

INTRODUCTION

Gaya is a good place for fish culture and found suitable for presence of diversity of fish species .The place has been observed a unique fauna in Bihar state[1]. The religiously important Gaya city is the district headquarters of Gaya district and divisional headquarters of Magadh Division of Bihar state of India. The Geographical location of Gaya is $24^0$46' 60 N Latitude and 85^0 01' Longitude. It is situated on the bank of river the Falgu. It has an area 50.17 Km^2 and population of 4, 63,544 as per 2011 census[2]. Besides the river 'Falgu' there are major ponds namely Suryapokhra, Digghi, Ramsagar, Vaitarni and Rukmini which are known for aquaculture practices. The present work is an attempt to record fish fauna at the above mentioned pond of Gaya, Bihar, India along with certain water quality parameters.

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Investigation was carried from June, 2018 to May, 2019. Identification of a number of fish species was done which is related with physico-chemical parameters of pond water[3].

Day (1889)[4] studied the fish available in Indian waters. Venkateswarlu (1972)[5] gave a consolidated account of the fishes occurring in Bihar state. With this in view, the fish fauna of Gaya has been studied and altogether 46 species of fishes belonging to 19 families have been collected and recorded. [6]

MATERIAL & METHODS

An investigation of fish species was conducted from June 2018 to May 2019 in 5 (five) ponds of Gaya namely (i) Suryapokhra (ii) Digghi (iii) Vaitarni (iv) Ramsagar & (v) Rukmini during the year 2018-2019. The study was carried to depict an overview of fish diversity as well as physico-chemical properties of water.

Sampling was done during pre-monsoon, monsoon and post-monsoon periods. A total of 46 fish species belonging to 8 orders 19 families were recorded from these ponds for physico-chemical parameters, water samples was collected on fortnightly basis. Water depths were measured by secchi disc. Dissolved oxygen was estimated by winkler method. Other physico-chemical parameters include air and water temperature, carbonate and bicarbonate were estimated in the laboratory at A.N. College, Patna.

Description of study area:

The district (Gaya) is divided into two regions demarcated by the river 'Falgu' Minimum temperature is recorded as 2.7^0 Celsius and maximum temperature is 45^0 Celsius. The average rainfall 1105 mm.[7] Pre monsoon exists from mid May and ends in last week of May. However monsoon comes in first week of June and ends in last week of June. Post monsoon starts from July and exists till last week of August. Monsoon and post monsoon period are favorable for breading of fishes[8].

Gaya lies on 120 m above sea level. The climate in Gaya is warm and temperate. The average temperature is 26.5^0 C. There are many ponds and reservoirs in this district. However 5 (five) are major ponds and investigated as varying degree of depth and physico-chemical properties.

The fishes were collected from the ponds during the periods June 2018 to May 2019 and fixed in 10% formalin and transferred to 80% formalin for preservation[9]. Sampling was done during monsoon, monsoon and post monsoon seasons. The areas surveyed include the localities in and around the pond. Certain physico-chemical parameters such as temperature (air and water), pH, total alkalinity, total hardness and dissolved oxygen were analyzed.

For the physico-chemical parameters, standard procedures (APHA, 2000)[10] were followed with the help of existing literatures and keys proposed by various authors. IUCN status of fish is recorded as per CAMP report (1998)[11], NBFGR (2010)[12]. Water depths of the ponds are recorded with the help of secchi disc. Dissolved oxygen was estimated following Winkler method[13].



Fig 1: Map of Gaya Town Surrounded by various area

Data analysis

All data were analyzed at A.N. College, Patna. Dissolved oxygen was estimated by using Winkler method. Total alkalinity and hardness of water were calculated by experimental methods.



Suryapokhra Pond



Ramsagar Pond



Digghi Pond



Rukmini Pond



Vaitarni Pond

Fig 2: Pictures of Ponds in Gaya

RESULTS

Altogether 46 species of fishes have been found which includes 8 orders and 19 families. A wide fluctuation of physico-chemical factors was observed which influence the availability of fish species. Some ponds like Suryapokhra and Vaitarni showed an adaptive condition due to appropriate amount of dissolved oxygen and pH.

Less productivity of fishes has been observed at Ramsagar, Digghi and Rukmini. This happens due to less death of water and disposal of domestic effluents. For overall development in terms of future productivity, a great innovation of technological aspects as well as mass level awareness must be needed.

General habitat ecology of the ponds

All the ponds show a wide fluctuation is terms of physico-chemical parameters. The ponds surveyed were highly adaptive for the reported species of fishes[14]. The data related to physico-chemical parameters were recorded over a period of one year (2018-19).

Monsoon data showed little variation in the physico-chemical parameters. Air temperature was slightly more than the water temperature. All the values showed fluctuation is different seasons like Dissolved Oxygen, pH, total alkalinity and total hardness[15].

Ichthyofauna:

46 fish species belonging to 8 orders and 19 families were recorded from these ponds. The fishes are categorized according to their habitat as supported by Nath (1987)[8] and Ao *et al.* (2008)[16]. The following lists mention the fishes collected from these ponds (Table-6). Valid names are given according to www.fishbase.org and www.aquabase.org. Identification was done with the help of Jayaram (1999)[17]. For systematic arrangement of the fishes, classification followed by Jayaram (1999)[17] was referred.

Table 1- Monthly variations in the physico-chemical characteristics of Suryapokhra pond water

	2018							2019				
	JUNE	JULY	AUG	SEP	OCT	NOV	DEC	JAN	FEB	MAR	APR	MAY
Water Depth[m]	4.3	5.5	6.3	6	5.8	5.5	5.3	5	4.7	4.5	4.3	4
Air Temp [^{0}C]	40	35	33	32	28	26	20	18	22	25	36	39
Water Temp [^{0}C]	35	31	29	28	24	22	17	15	18	21	32	35
Transparency [cm]	46	45	43	42	53	48	46	45	43	50	52	53
pH	7.8	7.6	8.3	8.2	8	7.9	8.1	7.8	7.7	8.2	8.1	7.2
DO (mgL^{-1})	6.5	8.2	8.5	8.8	9.2	9.5	12.2	11.5	11	9.5	9	7.8
Carbonate (mgL^{-1})	40.12	25.3	25.6	32	28.5	35.3	28.9	32.6	33	32	40.5	42.5
Bicarbonate (mgL^{-1})	120	125	123	105	109	115	133	120	130	139	145	122

Table 2. Monthly variations in the physico-chemical characteristics of Digghi pond water

	2018							2019				
	JUNE	JULY	AUG	SEP	OCT	NOV	DEC	JAN	FEB	MAR	APR	MAY
Water Depth[m]	2.5	2.8	3	3.1	3.2	3	3	2.9	2.8	2.7	2.6	2.5
Air Temp [^{0}C]	40	38	36	35	33	30	27	23	21	23	36	14
Water Temp [^{0}C]	34	33	32	30	28	25	23	18	17	18	30	36
Transparency [cm]	32	33	31	29	30	28	26	27	33	35	36	38
pH	7.3	7.5	7.6	7.5	7.8	7.9	8	8.1	8.2	7.9	7.8	7.7
DO (mgL^{-1})	6.4	6.2	6.3	6.5	6.8	6.9	7	7.2	7.4	7.2	6.9	6.5
Carbonate (mgL^{-1})	31.2	32.3	28.6	26.3	27.2	28.3	29.6	24.7	23.9	25.7	28.2	29.3
Bicarbonate (mgL^{-1})	120	118	105	102	98	101	105	102	103	99	102	104

Table 3. Monthly variations in the physico-chemical characteristics of Ramsagar pond water

	2018							2019				
	JUNE	JULY	AUG	SEP	OCT	NOV	DEC	JAN	FEB	MAR	APR	MAY
Water Depth[m]	2.5	3	3.1	3.2	3.8	3.6	3.9	3.5	3.3	2.7	2.6	2.5
Air Temp [^{0}C]	40	38	36	35	33	28	27	26	23	21	33	40
Water Temp [^{0}C]	35	33	32	31	29	24	22	22	19	17	29	35
Transparency [cm]	32	33	33	29	31	32	33	32	31	28	32	35
pH	7.6	7.7	7.9	8.2	8.3	7.9	7.8	8.3	7.6	7.7	7.8	8
DO (mgL^{-1})	6.4	6.2	6	6.3	6.2	6.8	7	7.2	7.3	7.4	6.8	6.5
Carbonate (mgL^{-1})	32.2	33.1	34.2	35.5	31.2	32.1	33.2	31.6	29.5	30.1	28.6	27.5
Bicarbonate (mgL^{-1})	102	101	102	98	100	105	102	103	112	110	108	106

Table 4. Monthly variations in the physico-chemical characteristics of Vaitarni pond water

	2018							2019				
	JUNE	JULY	AUG	SEP	OCT	NOV	DEC	JAN	FEB	MAR	APR	MAY
Water Depth[m]	5.2	6.2	6.5	6.4	6.3	6.2	6.1	5.9	5.8	5.7	5.6	5.1
Air Temp [^{0}C]	42	40	38	36	30	28	25	20	19	23	36	40
Water Temp [^{0}C]	36	35	33	32	25	23	20	15	14	16	30	35
Transparency [cm]	40	38	36	35	38	40	42	38	36	43	42	40
pH	7.6	7.8	7.4	8	8.5	8.3	8.2	8.2	8.6	8.3	8	7.6
DO (mgL^{-1})	6.3	6.7	6.8	7.1	7.3	7.8	8.3	8.5	8.6	8.4	7.2	6.9
Carbonate (mgL^{-1})	42.1	40	38.2	41.2	42.3	43.5	41.6	40.3	42.1	41.5	39.8	38.6
Bicarbonate (mgL^{-1})	123	120	118	117	125	130	132	128	126	124	125	122

Table 5. Monthly variations in the physico-chemical characteristics of Rukmini pond water

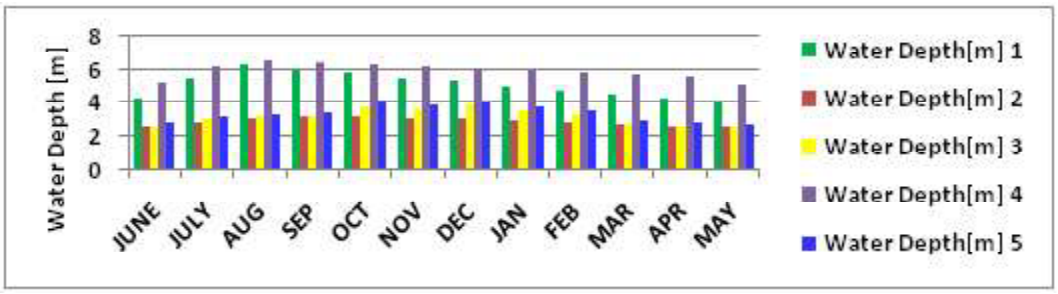
	2018							2019				
	JUNE	JULY	AUG	SEP	OCT	NOV	DEC	JAN	FEB	MAR	APR	MAY
Water Depth[m]	2.8	3.2	3.3	3.4	4.1	3.9	4	3.7	3.5	2.9	2.8	2.7
Air Temp [^{0}C]	40	39	37	34	32	29	26	22	20	22	35	41
Water Temp [^{0}C]	35	34	33	29	28	25	22	18	15	18	30	37
Transparency [cm]	43	42	38	39	42	41	39	44	45	43	42	40
pH	7.8	7.6	7.4	7.9	8.1	8.2	7.7	7.6	7.8	7.9	7.7	7.5
DO (mgL^{-1})	6.6	6.4	6.2	7	7.2	7.5	7.8	8	8.2	7.6	7.2	6.9
Carbonate (mgL^{-1})	33.1	32.2	35.3	31.2	33.3	32.2	33.4	35.2	33.2	31.2	30.2	30.3
Bicarbonate (mgL^{-1})	105	108	113	112	115	116	113	111	109	105	103	104

Tables : 6 List of fishes recorded

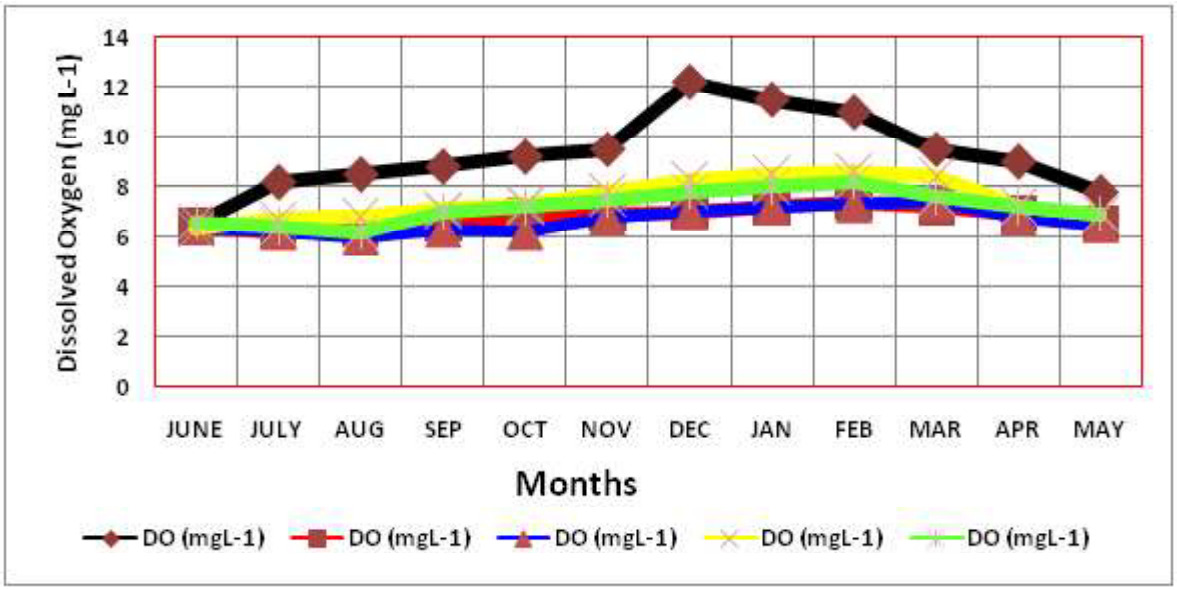
Systematic Position	Scientific Name	Local Name	Distribution in Ponds
Phylum-Chordata Sub phylum-Vertebrata Super class-Gnathostomata Series-Pisces Classs-Teleostomi Sub class-Actinopterygii Order-Clupeiformes Sub order- Clupeoidae			
1. Family-Clupeidae	1. *Gadusia chapra* (Ham.)	Suhia	1, 2, 4
2. Family-Engraulidae Sub order-Notopteroidei	2. *Setipinna phasa* (Ham.)	Phansi	1,3,4
3. Family-Notopteridae Order-Cypriniformes Division-Cyprini Sub order-Cyprinoidei	3. *Notopterus notopterus* (Ham.)		1,4
4. Family-Cyprinidae Sub family-Cyprinini	4. *Amblypharyngodon mola* (Ham.) 5. *Catla catla* 6. *Cirrhinus mrigala* 7. *Cirrhinus reba* (Ham.) 8. *Labeo rohita* (Ham.) 9. *Labeo calbasu* (Ham.) 10. *Puntius chola* (Ham.) 11. *Puntiu sarana* (Ham.) 12. *Puntius sophore* (Ham.) 13. *Puntius ticto* (Ham.)	Dhawai Catla Mrigal Reba Rohu Calbasu Pothia ” ”	1,3,4
Sub-family-Rasborinae	14. *Esomus donricus* (Ham.) 15. *Osteobrama*	” ”	
Sub-family-Abramidinae	16. *Oxygaster bacaila* (Ham.)	Gurda	

5. Family-Cobitidae Sub-family-Botini Sub-family- Cobitini Division-Siluri Sub-order-Siluroidei Super-Family-Siluroidae	1. *Botia iohachata* (Chaudhari) 2. *Lepidocephalichthys guntea* (Ham.)	Bagha Nakati	1,4
6. Family-Siluridae	3. *Ompok bimaculatus* (BI.) 4. *Wallago attu* (BI. & schn.)	Pabda Boari	2,3
7. Family-Bagridae	5. *Mystus cavasius* (Ham.) 6. *Mystus tengra* (Ham.) 7. *Mystus vittatus* (BI.) 8. *Mystus aor* (Ham.) 9. *Rita rita* (Ham.)	Tengra “ “ “ Rita	1,3,5
8. Family-Schilbeidae	10. *Aillia coila* (Ham.) 11. *Eutropichthys vacha* (Ham.)	Patasi Buchwa	1,2
9. Family-Saccobranchidae (Heteropneustidae) Order-Beloniformes Sub-order-Scomberesocoidei	12. *Heteropneustes fossilis*(BI.) 13. *Clarias batrachus* (Linn)	Singhi Mangur	2,3
10. Family- Belonidae Order-Mugiliformes	14. *Xenentodon cancila* (Ham.)	Kauwa	2,5
11. Family-Mugilidae Order-Ophiocephaliformes (Channidae)	15. *Sicamugil cascasia* (Ham.)	Lorhia	1,4
12. Family-Ophiocephalidae (Channidae)	16. *Channa gachua* (Ham.) 17. *Channa punctatus* (Bl.) 18. *Channa marulius* (Ham.) 19. *Channa striatus* (Bl.)	Chanaga Garai Saur Sour	2,3,5
13. Family-Amphipnoidae Order-Symbrachiformes	20. 36. *Amphipnous cuchia* (Ham.)	Aundha	1,3,4
14. Family-Centropomidae Super-family-Percoidae Order-perciformes Sub-order-Percoidei	21. *Chanda nama* (Ham.) 22. *Chanda ranga* (Ham.)	Chanari Chanari	1,3
15. Family-Sciaenidae	23. *Sciaena coitor* (Ham.)	Patharchatti	5
16. Family-Nandidate Sub-order-Anabantoidei	24. *Nandus nandus* (Ham.)	Dhebari	2,3
17. Family-Anabantidae Sub-order-Gobioidae	25. *Anabas testudineus* (Bl.) 26. *Colisa fasciatus* (Bl. & schn.)	Kawai Khosti	2,5
18. Family-Gobiidae Sub-family-Gobini Order-Mastacembelliformes	27. *Glossogabius giuris* (Ham.)	Bulla	1,4
19. Family-Mastacembelidae	28. *Macrognathus aculeatus* (Bl.) 29. *Mastacembelus armatus* (Lacep) 30. *Mastacembelus pancalus* (Ham.)	Baami	1,4,5

1. Suryapokhra 2. Digghi 3. Ramsagar 4. Vaitarni 5. Rukmini Pond



Graph 1: Water Depth of Ponds



Graph 2: Dissolved Oxygen

DISCUSSION

Higher temperature reduces oxygen solubility in water but can raise the oxygen and food intake demand of fish as their metabolic rates are raised[18]. Associated rises in gill ventilation rates can lead to increased uptake of aquatic pollutants, potentially rendering the flesh unfit for human consumption[19]. Higher water temperature can also favour the survival of parasites and bacteria[20]. Digghi Pond and Rukmini Pond showed less depth in compare to that of Suryapokhra, Vaitarni and Ramsagar. So less number of species is recorded in Digghi and Rukmini Pond. Water temperature also influenced the dissolved oxygen which is an important factor for survival of species[21]. All the response combines to potentially reduce fish survival, growth and reproductive success in aquaculture. Similarly many species in temperate regions have characteristics temperature ranges in which they live and breed and rises in temperature may result in species being less abundant[22].

The dry season is a period of great stress to many fish species arising from diminished feeding opportunities and water quality as well as elevated risk of predation or capture fish survival during this period[23].

pH and other factors depicted in table 1,2,3,4 & 5 are equally important for survival of a number of species. So physico-chemical parameters play an important role in diversity of fish species at the particular pond.

CONCLUSION

Gaya covers a wide range of hydriobiological fauna. Besides river Falgu, there are a large number of ponds and reservoirs where culture of food fishes (*Labeo rohita, Labeo calbasu, Catla catla, Cirrhinus mrigala* etc.) and live fishes (*Clarias batrachus*, *Heteropneustes fossilis*, *Channa punctatus* etc.) can be carried out successfully.

The ponds like Rukmini, Digghi and Ramsagar are showing less productivity, which in terms of fish culture truly reflects that there is over exploitation of our & resources at all levels, irrespective of catch age group, reproductive period and season. Another factor is breakdown of aquatic food chains at various levels on accounts of loss of one or other types of biotic components. Increased eutrophication also leads to less production. It occurs due to disposal of domestic effluents.

There must be a need to develop the public awareness towards maintenance of proper aquatic environment and fisheries potential. For developing more number of fish species, there should be a hatchery in Gaya town which can provide pure fish seed at cheap rate.

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Effect of nitrates & nitrites on pond water ecology of Madhepura, Bihar

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Abstract: Pond ecology must be stabilizing so that a dynamic equilibrium is maintained. Often, this equilibrium is a resultant of specific quantities of organic & inorganic elements. But whenever these quantities increase abruptly, the ecology of the ponds goes off balance, usually leading to collapse of a near to ideal ecosystem. Through necessary, nitrates if in excess quantity, has adverse affects on the pond water ecology. Excessive of nitrates in a pond usually denotes contamination by anthropological activities. To maintain water health, it is necessary to understand the effects of these inorganic elements.

Key words: Pond ecology, nitrates, nitrites, dynamic equilibrium

INTRODUCTION

Nitrates are derivatives of Nitrogen & are the forms of N_2 which is usually taken up by plants for their metabolic needs. Nitrogen is essential for all living things as it is a component of protein.[1] Increase in nitrates level is usually covered by overstocking & over feeding fishes, or more generally due to surface runoff of agricultural products Nitrogen is highly leachable, like fertilizers that contain high amounts of inorganic N_2.[2]

Fish waste usually produces Ammonia, which further is oxidized to form nitrates. Untreated human sewage can contribute to nitrate levels in surface & ground leaking or poor functioning septic systems are source of such nitrates. Presence of nitrates stimulates algal growth.[3,4] Whenever this algal growth is high, it reduces the O_2 levels in water causing fish death.

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Nitrates are undetectable without testing as it is colorless, odorless & tasteless. A low cost test can be done using dip tests.[5-7] Milky colour of pea soup is usually an indication of high concentration of algae. Excessive nitrogen, in either form (nitrate or nitrite) leads to massive death of fish, shrimps & eutrophication. Nitrite (NO_2^-) is toxic to aquatic organism due to the damage to hemoglobin. While nitrification couples the conversion of NH_4^+ -N to nitrate, de-nitrification reduces to nitrogen, further releasing nitrogen into the atmosphere.[8]

About 75% of nitrogen in feed ends up in water through ammonification of uneaten feed & excretion.All aquatic organisms excrete wastes and aquatic plants and organisms eventually die. These activities create ammonia. Some bacteria in the water change this ammonia to produce nitrite which is then converted by other bacteria to nitrate. Nitrates (NO_3^-) are an oxidized form of nitrogen and are formed by combining oxygen and nitrogen.[9]

The transformation capacity of de-nitrification done by de-nitrifiers, which require high carbon content for their growth, is limited due to high nitrate/nitrogen concentration. Freshwater ponds provide abundant food resources to humans, supplying tons of fishery production every year. Recently, high density and intensive farming models with superfluous feeding and fertilization have been used to achieve higher economic efficiency, resulting in nitrogen accumulation owing to excessive residual feed and excrement.

MATERIALS & METHODS

Site Description

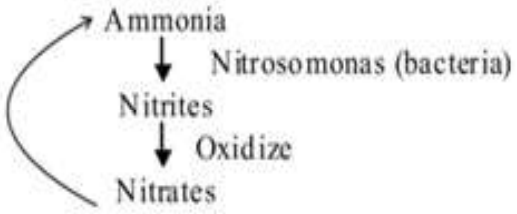
Random sample of pond water were collected from two shallow (<3 m) vernal ponds located on outskirts of Madhepura during the research work. Ponds are surrounded by growth woods, which act as a natural buffer strip on all sides. Ponds at these sites form within shallow depressions. They are replenished via irrigation ditches with water sourced from agricultural fields. All ponds at those sites are often filled in the late spring when irrigation water is abundant, but a lack of rain water later leads to gradual evaporation throughout the summer. Though some ponds may completely dry up at the end of the summer, others retain water, though at much lower levels.

Data Collection

Field campaigns were conducted to investigate pond nutrient limitation. We followed up on observed field patterns by collecting water samples from both previously surveyed ponds. During the study surface water samples were collected in triplicate weekly from both the ponds. All surface water samples were analyzed for total dissolved organic carbon (TDOC).

For ponds equally spaced samples were taken around the circumference.

RESULTS



Effects

- Excess nitrate concentration spoils the living environment of aquatic animals, therefore posing an ecological concern.
- Excessive nitrogen in drinking water causes methemoglobinemia or blue baby diseases.
- Elevated nitrate cause great problems in intensive culture of commercial fish species as well as ornamental fish species.
- Freshwater fish are hyper osmotic and therefore require an active uptake of ions across their gills, usually through their duct or via certain mechanisms associated with chloride cells of gills. Nitrite has affinities for these chloride cell uptake mechanisms & have tendency to accumulate in the cell.
- Nitrite accumulates in tissues such as gills, liver, brain & muscle.
- Blood appears as a primary target for nitrite. From blood plasma, nitrite diffuses into RBC, where it oxidizes Hb into methaemoglobin or ferri haemoglobin. This form of Hb, loses its capacity to bind to O_2 & changes the blood colour from red to brownish.
- When a fish with methaemo globinaemia is frightened or otherwise found to become active, it may die of Anoxia.
- Physiological disturbances are induced by nitrite.

CONCLUSION

Increasing use of pesticides or fertilizers with high N_2 concentrations, are highly leachable. This leaching process usually pollutes nearby water bodies. Increase in nitrate causes wild growth of algae leading to decrease in water's oxygen content (hypoxia). While increase in nitrate, is usually related with toxicity & anoxic condition.

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Six records of Reduviidae (Heteroptera: Hempitera) from Ranchi, Jharkhand

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Abstract: The paper presents the record of six species viz. *Polididus armatissimus* Stal, *Sycanus collaris* and *Rhynocoris costalis* (Fabricius) of subfamily Harpactorinae, *Acanthaspis fulvipes* of subfamily Reduviinae and *Ectomocoris ochropterus & Cleptocoris lepturoides* Wolff of subfamily Peiratinae of family Reduviidae from Ranchi, Jharkhand. Key to taxa and distributions of each species in India and abroad have been included.

Key words: Hemiptera, Reduviidae, record, Ranchi, Jharkhand

INTRODUCTION

The family Reduviidae is the largest and the most diverse successful group of true bugs. It belongs to the Superfamily Reduvioidea of the Suborder Heteroptera of the Order Hemiptera under the Division Exopterygota of Class Insecta. They are widely distributed in terrestrial ecosystems. Few members are blood suckers. Some of them are pathogenic and transmit various diseases to man and animals.

Members of this family are commonly known as "Assassin bugs". These are small to large, robust or elongated, somewhat flattened, smooth, hairy or spiny bugs, which may vary in size ranging from 2 mm (Holoptilus) to large and extremely hardy (Haema torrhophus) to 40 mm. The Reduviidae species have adapted to a wide range of terrestrial habitats and diversified in their prey choices while developing a wide repertoire of innovative prey capture strategies[1-5]. They are mostly common in tropical rainforest, but they are generally found in tropical rainforest, semi-arid zone, scrub jungle and agro-ecosystem[7]. This family is represented globally by more than 6878 species and subspecies under 981 genera belonging to 25 subfamilies[6]. Of which 465 species under 144 genera belonging to 14 subfamilies[7] are recorded from India[8]. Earlier 417 species under 133 genera belonging to 13 subfamilies are recorded from Indian region[9-10].

The classification of the family has been mainly adopted after Henry (2009)[6]. Distant (1904,1910)[9,10] recorded two species from Ranchi viz. *Coranus siva* Kirkaldy and *Acanthaspis quinquespinosa* Fabricius of Reduviinae from Ranchi. Present study deals with six new records from Ranchi viz.*Polididus armatissimus* Stal, *Sycanus collaris* and *Rhynocoris costalis* (Fabricius) of subfamily Harpactorinae, *Acanthaspis fulvipes* of subfamily Reduviinae and *Ectomocoris ochropterus* & *Cleptocoris lepturoides* Wolff of subfamily Peiratinae of

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family Reduviidae from Ranchi,Jharkhand.The work is based on the recent collections. Keys to various taxa, diagnostic characters, references, distribution in India and elsewhere under each species are given in the paper.

MATERIALS AND METHODS

The present study is based on the materials collected by hand picking method by using insect net & light trap. The specimens are deposited in the Zoological Survey of India, Hemiptera Section, Kolkata.

RESULTS AND DISCUSSION

SYSTEMATIC LIST

Suborder- HETEROPTERA
Infraorder- CIMICOMORPHA
Family- REDUVIIDAE

Subfamily I. HARPACTORINAE

Genus 1. *Polididus* Stal, 1858
1. *Polididus armatissimus* (Stal, 1859)
Genus 2. *Sycanus* (Amyot & Serville, 1843)
2. *Sycanus collaris* (Fabricius, 1781)
Genus 3. *Rhynocoris* (Kolenati,1857)
3. *Rhynocoris costalis*

Subfamily II. REDUVIINAE (Acanthaspidinae Distant, 1902)

Genus 4. *Acanthaspis* (Amyot and Serville, 1843)
4. *Acanthaspis fulvipes* (Dallas, 1850)

Subfamily III. PEIRATINAE

Genus 5. *Cleptocoris* Stål, 1866
5. *Cleptocoris lepturoides* (Wolff, 1804)
Genus 6. *Ectomocoris* Mayr, 1865
6. *Ectomocoris ochropterus* (Stål,1866)

Key to the subfamilies of the family Reduviidae

1. Pronotum with or without any constriction; hemelytra with a quadrangular areolet or cell at interior area of corium near base of membrane .....HARPACTORINAE

   Hemelytra with a quadrangular areolet or cell at interior area of corium near base of membrane ..... HARPACTORINAE

2. Pronotum with constriction behind the middle; hemelytra without a quadrangular areolet or cell at interior area of corium near base of membrane .....PEIRATINAE

   Pronotum constricted behind middle ....PEIRATINAE

3. Anterior tarsi 3-jointed .....REDUVIINAE

Subfamily HARPACTORINAE

Key to the genera of the subfamily Harpactorinae

1. Specimen always spinous in body and legs …… .....Polididus Stal

2. Head long and slender, pronotum unarmed ....Sycanus Amyot & Serville

   Abdomen on each side much dilated ....Sycanus Amyot. & Serville

3. Posterior lobe of pronotum not longitudinally impressed .....Rhynocoris Kolenati

   Specimen not spinous in body and legs …… .....Rhynocoris Kolenat

Genus 1. *Polididus* Stal, 1858

1858. *Polididus* Stal, Ofv. K. Vet. *Ak. Forh.,* **15:** 448.

1. *Polididus armatissimus* Stal, 1859 (Fig. 1)

1859. *Polididus armatissimus* Stal, Ofv. Vet.-*Ak. Forh.,* **16:** 376.
1904. *Polididus armatissimus* Distant, Fauna Brit. India, *Rhynchota*, **2:** 386.
2006. *Polididus armatissimus* Ambrose, Zoos' *Print Journ.,* **21(9):** 11.

Material examined: 1 ex., ICAR, plandu, Ranchi, Jharkhand, 16.VIII.2011.

Diagnostic character:-

This species is very unique among the congeners in showing wing dimorphism and is easily recognized by the pronotum dark brown to blackish with a pair of brownish yellow longitudinal stripes, setiferous spines on the pronotal humeri, scutellum and laterotergites IV to VII blackish in the basal half and brownish yellow in the apical half, the femur of each leg blackish in the apical fifth, the meso and metatibiae lacking setiferous spine, the metatibia with a blackish narrow annulation subbasally.

Length- 10 mm

Distribution:- Saudi Arabia, India, Sri Lanka, Myanmar, Vietnam, China, Korea, Japan, Philippines, Malaysia

(Malaya), Indonesia (Sumatra, Java, Bali), Hawaiian Islands, Caroline Islands.

Genus 2. *Sycanus* Amyot & Serville, 1843

1843. *Sycanus* Amy. & Serv., Hem.,: 360.

Key to the species of the Genus *Sycanus* Amyot & Serville

1. Apical half of corium pale stramineous..... .....collaris (Fabricius)
2. Apical half of corium golden yellow..... ..... croceovittatus Dohrn



Fig. 1- *Polididus armatissimus*

2. *Sycanus collaris* (Fabricius, 1785) (Fig. 2)

1785. *Reduvius collaris* Fabricius, Spec. Ins., **2:** 380.

1904. *Sycanus collaris*: Distant, Fauna Brit. India, *Rhynchota*, **2:** 351.

2006. *Sycanus collaris* : Ambrose, Zoos' *Print Journ*., **21(9):** 14.

Material examined: 4exs., CTR&TI, Piska Nagri, Ranchi, 15.viii.2005.

Diagnostic character:-

Body black in colour; pronotum piceous; apical half of corium excluding apical angle and basal half of membrane reddish brown; membrane bronzy; antennae black, basal and subapical annulations to first joint, subbasal annulation to second joint and apex of rostrum reddish brown; head about as long as pronotum and scutellum together; first joint of antennae subequal to anterior femora; scutellar spine long, obliquely erect, apex bifid; abdomen strongly dilated on each side especially at third and fourth segments, posterior angles of second and third segments acute.

Length: 22-25 mm.

Distribution: India: Punjab (Gurdaspur), Chhattisgarh, Tamil Nadu, Assam, West Bengal, Meghalaya, North Khasi hills.

Elsewhere: China, Malaysia, Philippines, Sri Lanka, Thailand, Malay peninsula and several islands of the malayan archipelago.

Genus 3. *Rhynocoris* Kolenati, 1857

1857. *Rhynocoris* Kolenati, Fascia Bulletin Moscou, **29:** 419-502.

Key to the species of the Genus *Rhynocoris* Kolenati

1. Posterior lobe of pronotum not, or very obscurely rugosely granulate ..... fuscipes (Fabr.)
2. Species sanguineous in colour; base of antennae sanguineous; disk of sternum, coxa, trochanters, anterior lobe of pronotum reddish ochraceous; posterior lobe of pronotum rugose without any excavation .....marginatus (Fabr.)
3. Species black in colour; antennae black; disk of sternum, coxae, trochanters and pronotum rugulose with central excavation ....squalus (Distant)



Fig. 2- *Sycanus collaris*

3. *Rhynocoris costalis* (Stal, 1866) (Fig.3)

1866. *Reduvius costalis* Stål, Ofv. Vet.- Ak. Forh., 285.

2006. *Rhynocoris costalis*: Ambrose, *Zoos'Print Joun*., **21(9):**11.

Material Examined-1ex.Biodiversity park, Ring Road, Ranchi, 11. iv. 2011.

Diagnostic character:-

Body black, lateral margin and basal area of posterior pronotallobe, lateral areas of corium, coxae, trochanters abdomen beneath, apex of scutellum coral red; anterior lobe of pronotum sculptured, centrally impressed posterior lobe not impressed; a spot between and a lateral spot behind eyes, coral red; head ventrally and spots to femora beneath brownish yellow. Habit and Habitat: Although this species occur in the vegetable gardens and other agricultural fields on vegetation especially on foliage, they may be seen in

the cotion fields predating upon cotton bugs *Dysdercus koenigii*.

Length-12-14mm

Distribution : India : Andhra Pradesh, Assam, Meghalaya, West Bengal.

Elsewhere: Malayan Peninsula, Mayanmar.

Subfamily II. REDUVIINAE

Key to the genera of the Subfamily REDUVIINAE

1. Eyes and ocelli large or very prominent; eyes extending transversely quite across the lateral areas of the head; pronotum broadly longitudinally impressed or sulcate .....Gerbelius Distant
2. Eyes and ocelli small or mediocre; eyes never extending transversely across the lateral areas of the head; pronotum is not broadly longitudinally impressed or sulcate .....Acanthaspis Amy. & Serv.

Genus 4. *Acanthaspis* Amyot & Serv., 1843

Key to the species of the Genus *Acanthaspis* Amyot & Serv, 1843

1. Posterior lobe of pronotum with two long discal spines.....quinquespinosa (Fabricius)
2. Pronotum unicolorous, unspotted; legs unicolorous .....siva Distant
3. Pronotum dark with pale spots and markings; legs annulate .....sexguttata (Fabricius)
4. Pronotum palely marked or spotted .... ûavipes Stål
5. Piceous in colour; central disc of abdomen reddish brown; ûrst joint of rostrum distinctly longer than second .....fulvipes (Dallas)
6. Purplish-brown in colour; abdomen beneath yellowish brown; ûrst joint of rostrum shorter than second .....luteipes Walker.



Fig. 3- *Rhinocoris costalis*

4. *Acanthaspis fulvipes* (Dallas, 1850) (fig.4)

1850. *Platymeris fulvipes* Dallas, Tr. Ent. Soc. f.3

1904. *Acanthaspis fulvipes* Distant, Fauna Brit. India, *Rhynchota*, **2:**259.

2006. *Acanthaspis fulvipes* Ambrose, Zoos' *Print Journal.,* **21(9):** 19.

Material examined: 1ex, ICAR,Plandu,Ranchi , 20. vii. 2001.

Diagnostic character:-

Pronotum with anterior lobe moderately sculptured and deeply centrally foveate on posterior margin; posterior lobe rugulose, its lateral angles acutely spinous, scutellar spine long, obliquely ascendant; antennae, rostrum, legs and central disk of abdomen reddish brown; a large spot behind middle of corium and spot to connexivum above and beneath ochraceous.

Length: 25-26 mm.

Distribution: India: Maharashtra, Chhattisgarh, Assam, Sikkim, and West Bengal.

Elsewhere: Bhutan and Bangladesh.

Subfamily III. PEIRATINAE

Key to the genera of the Subfamily Peiratinae

1. Head long, porrect; antennae inserted remotely from eyes; intermediate tibiae without a spongy furrow .....Sirthenea Spinola
2. Pronotum tuberculously granulate ....Catamiarus Amy. & Serv.
3. Spongy furrow occupying more than half of anterior tibiae.....Ectomocoris Mayr.
4. Spongy furrow occupying less or not more than half of anterior tibiae ..... Spilodermus Stal
5. Anterior femora spinose beneath ....Lestomerus Amy. & Serv.
6. Second joint of antennae as long as pronotum .....Spilodermus Stål
7. Second joint of antennae about as long as anterior lobe of pronotum .....Cleptocoris Stål

Genus 5. *Cleptocoris* Stål, 1866

1866. *Cleptocoris* Stål, Ofv. Vet.-Ak. Forh., p. 262.

Key to the species of the Genus *Cleptocoris* Stål

1. Membrane fuscous, unicolorous ....lepturoides (Wolff)
2. Membrane spotted with black ....atromaculatus Stål



Fig. 4- ***Acanthaspis fulvipes***

5. ***Cleptocoris lepturoides*** **Wolff., 1804 (fig.5)**

1804. *Reduvius lepturoides* Wolff, Ic., 4: 165.

1904. *Pirates lepturoides*: Distant, Fauna Brit. India, Rhynchota, 2: 301.

2006. *Cleptocoris lepturoides*: Ambrose, Zoos' Print Journ., 21(9): 15.

Material examined: 1ex., Birsa Munda Zoological Park, Ranchi, 8.vi.1999.

Diagnostic character:-

Black; corium brownish yellow in colour, a small spot at inner angle at base of membrane, the membrane, brownish yellow; connexivum above and beneath spotted with yellowish brown; anterior pronotal lobe centrally impressed; anterior femora unarmed beneath.

Length: 9-10.5 mm.

Distribution: India: Chhattisgarh (Surguja) Madhya Pradesh (Hoshangabad), Maharashtra.

Elsewhere: Sri Lanka, Myanmar, Indonesia.

Genus 5. ***Ectomocoris*** **Mayr, 1865**

1865. *Ectomocoris* Mayr,Verh.Z.-b.Ges.Wien., p. 438.

Key to the species of the Genus ***Ectomocoris*** **Mayr, 1865**

1. Hemelytra abbreviated .....simulans Distant
2. Hemeltra fully developed .....cordiger Stål

Genus 5. ***Ectomocoris*** **Mayr, 1865**

1865. *Ectomocoris Mayr*, Verh. *Zool. Bot. Ges. Wien.,* **15:** 438.

6. ***Ectomocoris ochropterus*** **Stal, 1866 (Fig. 6)**

1866. *Ectomocoris ochropterus* Stal*, Ofv. K. Vet. Ak. Forh*., **23:** 256.

1904. *Ectomocoris ochropterus, Distant,* Fauna Brit. India, *Rhynchota*, **2:** 294-295.

2006. *Ectomocoris ochropterus,* Ambrose, *Zoos'Print Journal,* **21(9):** 16.

Material examined: 1 ex., ICAR, Plandu, Ranchi, Jharkhand, 6.X.1964.

Diagnostic character:

Body black; first segment and basal area of second segment of antennae, rostrum, clavus, corium, basal third of membrane, lateral areas of abdomen and legs yellowish brown; inner margin of clavus, a linear fascia near lateral margin of corium and apex of membrane brownish yellow; second segment of antennae shorter than anterior lobe of pronotum and granulate; legs finely granulate.

Length: 19-20 mm.

Distribution: India: Jammu and Kashmir (Udhampur), Bihar, West Bengal. Elsewhere: Egypt, Eritrea, Ethiopia.



Fig. 5- ***Cleptocoris lepturoides***



Fig. 6- ***Ectomocoris ochropterus***

SUMMARY

This paper deals with 6 species of Reduviid bugs belonging to 6 genera under 3 subfamilies from Ranchi. Relevant literature references, diagnosis of each species, keys to various taxa, distribution of each species have been incorporated. It is also expected that the further study of unexplored area may yield many more species.

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Survey of types of malaria found in Bihar with serodiagnostic test analysis

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Abstract: Malaria is a parasitic disease caused by a protozoan of the genus *Plasmodium*. Despite the fact that it is a preventable disease, it ranks among the major health challenges for developing countries including India. This paper is based on a survey conducted for the analysis of condition of malaria during 2015- 2019 in Madhepura region of Bihar. During this survey it is found that malaria is a major cause of death in different regions of Bihar. Effective ways to manage malaria now exists but then also the number of malaria cases is still increasing, due to multiple factors. Proper diagnostic methods are essential for the management and control of malaria. In rural areas traditional methods for diagnosing malaria remain problematic; therefore, new technologies have been developed and introduced to overcome the limitations. This paper deals with the serodiagnostic test analysis for malaria.

Key words: *Plasmodium*, malaria, diagnosis, method, serodiagnostic test

INTRODUCTION

India is a developing nation which accounts for about 6% of the worlds malaria cases. It has a death rate of, 1.97 per 1,00,000 individuals. A great geo-ecological diversity, large population and a variety of distribution agents or vectors, makes malaria a complicated problem in India. Malaria is caused by protozoan parasite *Plasmodium* and its four different species namely *falciparum*, *malariae*, *ovale* and *vivax* are responsible for malaria in humans. It is transmitted by female anopheles mosquito. The disease has an incubation period of about seven days.

Symptoms include fever, chills, headache, vomiting, severity of the disease may cause coma, followed by death. Malaria caused by *Plasmodium falciparum* is prevalent in Madhepura region, accounting for approximately 65% of the total infections. *P. vivax* is ranked second in infections, accounting for 15-20% of the total infections. There are about 9 species of anopheles which act as vectors. Mosquitogenic conditions like stagnant water leads to increased potential of malaria transmission.

The life cycle of malarial parasite comprises of schizogony, an asexual phase occurring in humans and sporogony, a sexual phase occurring in mosquito. The cycle is completed in 7-15 days. Presence of foetal haemoglobin is found to be unfavourable for *P.falciparum* development. Maternal immumity, and low exposure play a vital role in infants defense against malaria.

Co-existence of different malarial species in the same human host is very common in areas where one species is endemic. Bihar has always dominated the disease map. 7 out of 38 districts in Bihar is endemic for malaria. The risk of acquiring this disease during monsoom is very high.

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Respective authorities have ensured to keep the antimalarial drugs amount suffient and prevent any kind of breeding ground due to water logging or drainage failure.

MATERIALS AND METHODS

Taking patients' statistics conservatively equal to the annual number of serological tests done in Bihar. A cross-sectional survey was conducted randomly in ten regions of Madhepura, Bihar. 4023 people were approached for the survey purpose and 963 families were also approached for a mosquito net-owing survey. Some hospitals were also surveyed about their medical facilities regarding the test of malaria. The data were analyzed and illustrative statics were used. Firstly the areas were divided into three categories viz. high, middle and low malarial cases based on the government data provided by health department. Everyone in the survey area was invited to take part in the survey but only 4023 people were interested in this survey.

During the survey various questionnaire regarding the treatments, diagnostic, report and precautions were developed. The details of report were as follows:-

Table 1- Data collected from the ten sites of Madhepura, Bihar

Sl.no.	Sites	No. of samples	Positive	%
1	S1	512	136	26.56
2	S2	421	102	24.22
3	S3	369	135	36.58
4	S4	412	165	40.04
5	S5	561	101	18.00
6	S6	357	98	27.45
7	S7	322	80	24.84
8	S8	385	79	20.51
9	S9	328	56	17.07
10	S10	356	69	19.38
Total		4023	1021	25.37

Table 2- Data from the health ministry department was collected for last 5 years.

Year	Number of malaria cases
2015	2043
2016	4020
2017	5205
2018	4006
2019	1085

After seeing these numbers, which seem to be fluctuating every year, we can say that environmental factors play a major role in spreading this disease. The year 2020 seems even more prone to malaria as monsoons seem to start much earlier. The humidity is high forming a favourable environment for their breeding.

CONCLUSION

To enable transmission the climate must be favourable for long enough for malarial parasite to persist. The extent of transmission depends upon the density and pathogenecity of the vector. In places where density of humans is high or there is lack of appropriate sanitation

the occurrence of malaria is high. The female anopheleses are usually active during dusk or dawn, a time when most people love to stay outdoors.

To prevent from malaria many methods have been listed by health officials as well as the government of India has developed various schemes to counteract the spread of this disease.

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Assessment of fish productivity of Gaya, Bihar, India

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Abstract: The present investigation describes the fish productivity of Gaya ponds, Bihar (India) pertaining to its geological, morphological, economical features, requires urgent modernization to increase the productivity. It is the fact the fisherman community is one of the poorest of the poor sector of our society who sustain their livelihood on fish production and marketing. The result obtained through this research may be useful for poor people of the area; those are depending on wetland resources. Diversity of fishes is significant for wetland study. Fisheries development from the open water is considered to be environment friendly, labour productive and from social and economic perspectives. Reservoirs that spread over the area can play a significant role in increasing the fish availability in the state and providing gainful employment to its rural people. It has long way to go towards optimum utilization of this important resources. The Gaya ponds have good potentiality of fishery including composite fish farming and integrated fish farming. However, socio-economic conditions of the people of this region may be changed if proper utilization of pond produce is made with caution. Thus, the process of fishery in the pond is very old and it requires urgent modernization to increase the productivity and enormous scope exists in this country to harvest the water economy. Taking into consideration of our water resources, there is need of serious developmental efforts, which should be directed towards increasing fish production.

Key words: Freshwater fishes, biodiversity, growth, pondecosystem, Gaya

INTRODUCTION

Bihar, one of the poorest states in our country where more than 70% population are sustained their livelihood on agriculture and allied activities like aquaculture, dairy, poultry etc. At present time, in our state fishermen community constitutes about 14% of total population.[1] The fishermen community is one of the poorest of the poor sector of our society who sustain their livelihood on fish production and marketing. Bihar has tremendous potential to be developed into a highly organized fish production centre. It is endowed with wet lands, water reservoir, several major rivers, many ponds, lakes etc. are the fishing operations. Only a small fraction of fish farming is presently under utilization.[2] The rank of Bihar regarding fisheries resources among all states is twelve and occupied third position in inland fish production. Fish production has been reported in the state of Bihar in year 2015-16, that is 50% of the total need. Fisheries play an important role in the economy of not only in Bihar but, in whole country. However, India is an agricultural based country

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in which contribution of fisheries in total Agricultural Gross Domestic Production is 4.2% (Source – Hindu Survey of India Agriculture 2004) fish protein (food) production and supply raising nutritional level, generating employment opportunity/revenue and earnings of foreign exchange.

Studies on wetlands of Bihar are inadequate and very fragmentary.[3,4] The result obtained through this investigation may be useful for poor people of the area those are depending on lake and wetland resources. The data may also be helpful in understanding the structure and function of the pond ecosystem. Central Agricultural Minister of India stated that fish production was exceeded 20% i.e. 327 lakh tones during 2016-2017. Now Bihar is the second largest inland fish producing state in India, contributing 11% to the national production of 1.7 million tonnes in 1991-92. The track record of the state in the aquaculture sector is quite impressive. Total nursery of Bihar area is 350 hac is the largest in India and its annual fish seed production of 348 million fish is the third highest. Taking into consideration of our water resources, there is need of serious developmental efforts which should be directed towards increasing fish production. Considering that Bihar is poor state with a very low per capita income and poor intake of protein. Fisheries development from the open water is considered to be environment friendly, labour productive and from social and economic perspectives. Reservoirs that spread over an area of nearly one lac hac.can play a significant role in increasing the fish availability in the state and providing gainful employment to its rural poor people. The Gaya pond has good potentiality of fishery including composite fish farming and integrated fish farming. On the other land, this lake has a wide scope of aquaculture which can multiply the income of local fisherman into many folds.

MATERIALS AND METHODS

The estimation of fish productivity, the records of fish production from pond and area was collected from local society of fisherman. Monthly production of fish was recorded between November 2017 and October 2019. The data were generated after the collection, preservation and identification of fishes. The samples were collected with the help of bag net and were preserved in 10% formalin. Identifications were done with the help of available literature of Jhingran (1991)[5]. The data of fish production estimated was obtained from fisheries department.

The fishermen have to take valid licenses from the fisheries department for operating their nets and tackles in the water reservoir. About 11 fishing gears and nets are applied in this region. With the scientific advancement there has been tremendous change in the nature. The application of gears varies with current, depth of water, nature of fish to be captured and local availability of raw construction materials. By and large the cotton and jute nets are being replaced by nylon nets. Application of bamboo reeds and strips for forming barriers (Bari) across the width are also very common. The gill net is the most common fishing gear used for commercial fishing. These nets vary in length from 80 to 100m, the width varying from 2.4 to 10.0 m. The mesh size operated almost throughout the year.

OBSERVATION AND RESULT

The pond has potentiality to support large number of fishes. The pond is dominated by weed fishes of small size which have very little economic importance. For estimation of productivity of fishes, the records of fish production from the pond area was collected from the local society of fishermen, which is well marked in the Table. Monthly production of fish was recorded during study period.

Annual fish production of the pond estimated and these data show that the pond has rich potentiality of fish production. There are many causes which are responsible for the higher production of fishes in the pond. Jhingran (1991)[5] states that shallow ponds with light penetration up to the bottom is highly suitable for the growth of fishes. But the water of the pond with higher alkalinity does not influence the fish productivity. In the pond investigated oxygen level drops below 5m/g which is unfavorable for fish production as also observed by Banerjee (1979)[6]. Low dissolved oxygen of the pond is unable to support the development of large fishes with high metabolic rate. Low dissolved oxygen level in water is responsible for abundance of small sized weed fishes with low metabolic rate. However, nitrate level in water was recorded always more than 0.2 mg/l which is favourable for fish production as also observed by Banerjee (1979)[6]. Soil of the pond where nitrogen level is in higher range may not be

considered good for fisher. Khanna (1988)[7] stated that lacking of phosphorus in the ecosystem is the main cause of low productivity of fishes. Thus, it is clear that certain properties of water and soil are favourable for fishes while others either do not effect or have inhibitory effect on the productivity of fishes.

Table 1-Monthly variation in production of fishes from Gaya pond, Bihar (Nov. 2017-October 2019)

Year	Months	Seasons	Production kg/month	Production kg/day	Seasons Average	Kg/ha/yr
2017	November	Winter	21008	700.27		
	December	Winter	26815	865.00		
2018	January	Winter	31260	1020.00	26677.75	
	February	Winter	27268	940.28		
	March	Summer	26146	843.42		
	April	Summer	24290	809.67		
	May	Summer	23684	764.00	24626.00	127.35
	June	Summer	22928	764.27		
	July	Monsoon	13175	425.00		
	August	Monsoon	7285	235.00		
	September	Monsoon	16170	539.00	13497.50	
	October	Monsoon	17360	560.00		
	November	Winter	20100	670.00		
	December	Winter	26040	840.00		
2019	January	Winter	35036	1130.19	27595.00	
	February	Winter	29204	1043.00		
	March	Summer	288830	930.00		
	April	Summer	25880	862.67		
	May	Summer	26630	859.03	24310.00	132.50
	June	Summer	15900	530.00		
	July	Monsoon	14266	460.19		
	August	Monsoon	9510	310.00		
	September	Monsoon	15900	530.00	15138.00	
	October	Monsoon	20776	670.19		

Table 2- Systematic list of fishes recorded from Gaya ponds, Bihar

Sl. no.	**Order**	**Family**	**Species**	**Local name**
1.	Cypriniformes	Cyprinidae	*Catla catla* (Ham)	Catla
2.	Cypriniformes	Cyprinidae	*Cirrhinus mrigala* (Ham)	Naini
3.	Cypriniformes	Cyprinidae	*Labeo rohita* (Ham)	Rohu
4.	Cypriniformes	Cyprinidae	*Puntius chola* (Ham)	Pothia
5.	Siluriformes	Bagridae	*Mystus aor* (Ham)	DaraiTengra
6.	Siluriformes	Clariidae	*Clarias batrachus* (Linn)	Mangur
7.	Siluriformes	Hetropneustidae	*Heteropneustes fossilis* (Bloch)	Singhi
8.	Channiformes	Channidae	*Channa gachua* (Ham)	Chenga
9.	Channiformes	Channidae	*Channa punctatus* (Bloch)	Garai
10.	Mastacembeliformes	Mastacembelidae	*Macrognathus aculeatus* (Bloch)	Gaichi
11.	Perciformes	Anabantidae	*Anabas testudineus* (Bloch)	Kabai

CONCLUSION

Landed fish fetches maximum price when sold in fresh condition worthy of human consumption, for several reasons including non-availability of a ready market, non-nearness of market, lock of facilities for fast transport or refrigerated transport etc. an appreciable amount of fish is preserved for long term storage. This preserved fish is also meant for human consumption as food. But there is yet other utilization of fish in which protein, fat and other useful contents in the body of fish are processed into a number of valuable products and by-products for a variety of usages. Important fish products are fish liver oil, fish meal and other uses like fish flour, fish silage etc. However, fat and vitamin in content of liver vary from fish to fish and from season to season. In general, fish liver oil when taken will ensure good growth of bone, teeth, skin and vision and will develop more resistance to counter bacterial attack and so forth. At present time, fishes play important role in the production of insulin, biscuits, bread, cakes, soup, sweets etc. However, the Gaya pond has good potentiality of fishery including composite fish farming and integrated fish farming. The pond has a wide scope of aquaculture, which can multiply the income of local fisherman into many folds. The process of fishery practiced in this pond is very old and it requires urgent modernization to increase the productivity. It is clear that certain properties of water and soil are favourable for good productivity of fishes while others either do not effect or have inhibitory effect on the productivity of fishes.

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Role of carbohydrate in male *Hydrophilious olivaceous* life processes

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Abstract: Carbohydrate metabolism in insects follows a more or less similar pattern as in vertebrates. In insects, like other animals, glucose has a central place in carbohydrate metabolism. But in majority of insect's species, the amount of free glucose is quite small. The most prominent sugar of haemolymph and other body tissues of insect is trehalose, a disaccharide of glucose. It has a glycocidic linkage between anomeric carbon atoms of both glucose molecules. It is a non reducing sugar. In most cases, the sugar contents in the haemolymph is relatively low level of glucose along with higher trehalose contents. In the present investigation on male *Hydrophilous olivaceous* it was observed that the copper sulphate had an initial carbohydrate lowering effect and latter elevating trend on the haemolymph, fat body, and testis of and male insects. Overall initial decline of the carbohydrate content in haemolymph, fat body, ovary and testes were observed in insects exposed to sub-lethal concentrations of copper sulphate.

Key words: Carbohydrate metabolism, *Hydrophilous olivaceous*, glucose, copper sulphate.

INTRODUCTION

The problem of pollution and its effect on cellular metabolism of an organism is so vital in present complex and civilized society that it has posed a serious threat to purposeful existence of humanity. It has been aptly remarked that during the last decade we have seen a marked increase in the use of toxicants, detergents, industrial effluents, petrochemical products, psychoactive drugs, productive insecticides and other chemicals in our society. It affects all metabolic processes including carbohydrate metabolism directly or indirectly.[1]

There is no doubt that carbohydrates are the most vital and versatile organic compounds of the living world. These are the most abundant among the major classes of bio-molecules. Chemically they are simple organic compounds that are aldehyde or ketone derivatives of the polyhydric alcohol or as compounds that yield these derivatives on hydrolysis. It is an important source of energy for vital activities. It has central role in cellular metabolism of all living organisms.

Carbohydrate metabolism in insects follows a more or less similar pattern as in vertebrates. In insects, like other animals, glucose has a central place in carbohydrate metabolism. But in majority of insect's species, the amount of free glucose is quite small. The most prominent sugar of haemolymph and other body tissues of insect is trehalose, a disaccharide of glucose. It has a glycocidic linkage between anomeric carbon atoms of both glucose molecules. It is a non reducing sugar. In most cases, the

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sugar contents in the haemolymph is relatively low level of glucose along with higher trehalose contents.[1-3]

Metamorphic changes in both exopterygote and endopterygote are usually accompanied by substancial depletion of their carbohydrate reserves. During this period glycogen and trehalose supply glucose which provides an energy source and the substrate for various metabolic processes. Wyatt gave a comprehensive account of this process in insects.

In the present investigation on male *Hydrophilous olivaceous* it was observed that the copper sulphate had an initial carbohydrate lowering effect and latter elevating trend on the haemolymph, fat body, and testis of and male insects. Overall initial decline of the carbohydrate content in haemolymph, fat body, ovary and testes were observed in insects exposed to sub-lethal concentrations of copper sulphate.

MATERIALS AND METHODS

Quantitative estimation of haemolymph, fat body, ovary or testis carbohydrates of treated and control insects were done by Anthrone reagent Method.

The Anthrone reagent is the basis of a rapid and convenient method for the determination of hexose, aldopentose, hexuronic acid, either free or present in polysaccharides. The blue green solution shows absorption of maximum at 620 nm although some carbohydrates may give another colour.

Reagents

Anthrone reagent: 0.2 % in concentrated H_2SO_4.

Mixture of glucose: 10 mg / 100 ml and glycogen 10 mg / 100 ml (as standard).

TECHNIQUE

(A) Technique of Haemolymph Carbohydrates:-

Insect heads were pricked with a needle and were kept inverted in centrifuge tubes, one insect in one centrifuge tube and were centrifuged at 1500 rpm to collect haemolymph. 0.1 ml haemolymph was taken in separate centrifuge tube and added 5 ml TCA to precipitate protein. It was centrifuged at 3000 rpm for supernatant; test for carbohydrates was carried out.

(B) Technique of Fat body, Testis Carbohydrates

Treated and control insects were dissected. Fat body and ovaries or testes were taken out and crushed in 2 ml of distilled water with the help of mortar and pestle separately. The crushed material was collected in a centrifuge tube and 4 ml of 10% TCA was added to obtain the precipitate of proteins. It was centrifuged at the 3,000 rpm. The supernatant was kept separately for performing quantitative assay of carbohydrate.

4 ml of Anthrone reagent was added to 1 ml of test solution and has rapidly mixed. Boiled in water bath, the absorption was read at 620 nm against the anthrone reagent as blank.

Calculations

The carbohydrate content was calculated as follows:-

$$\frac{\text{OD of Unknown Solution}}{\text{OD of Known Solution}} \times \text{Concentration of Solution} \times \text{Dilution of Known Solution}$$

Where OD = Optical Density.

RESULTS AND DISCUSSION

Observation of carbohydrate content in *H. olivaceous*. The total carbohydrate content in the haemolymph of mature female and male *H. olivaceous* were 600 ± 10 SD mg/100 ml and 605 ± 9 SD mg/100 ml respectively in control insect.

Carbohydrate Content in Mature Male *H. olivaceous*

Haemolymph Carbohydrate Content

Almost similar carbohydrate lowering effect on the haemolymph carbohydrate content was observed in mature male *H. olivaceous* on treatment with copper sulphate. At 24 hrs of treatment the carbohydrate content declined to 185 ± 6 SD mg/100 ml of haemolymph. Similar declining pattern was observed at 48 and 72 hrs after copper sulphate treatment that was 165 ± 4 SD mg/100 ml and 160 ± 4 SD mg/100 ml, showing a rising trend, 350 ± 5 SD mg/100 ml after 168 hrs of treatment. All differences were highly significant as compared it with that of control insects, (Table - 1).

Fat body Carbohydrate Content

The mature male *H. olivaceous* had a total carbohydrate content of 60 ± 4 SD mg/gram of fat body in untreated condition or in control insects, (Table - 1).

The carbohydrate content in the fat body of mature male *H. olivaceous* after different periods of treatment with copper sulphate were 35 ± 3.5 SD mg/gram at 24 hrs, 25 ±

2.5 SD mg/gram at 48 hrs, 16 ± 1.5 SD mg/gram at 72 hrs and 48.5 ± 4 SD mg/gram at 168 hrs. All these differences were highly significant (P < 0.001) as compared it with that of control insects, (Table-1).

Testicular Carbohydrate Content

The carbohydrate content in the testis of mature male *H. olivaceous* was 65 ± 5 SD mg/gram in control insects, (Table - 1).

Table-2 Carbohydrate content in haemolymph, fat body and testes of mature male *H. olivaceous* during control and after different periods of treatment with Copper sulphate

Periods of Treatment	Haemolymph mg/100ml	Fatbody mg/gram	Testis mg/gram
Control	605 ± 9	60 ± 4	65 ± 5
24 hrs	185 ± 6***	35 ± 3.5***	17 ± 1.7
48 hrs	165 ± 4***	25 ± 2.5***	15 ± 1.5
72 hrs	160 ± 4***	16 ± 1.5***	12 ± 1
168 hrs	350 ± 5***	48.5 ± 4***	35 ± 2.5

Mean ± SD SD = Standard Deviation
*= P < 0.05 (Significant) ** = P < 0.01 (Significant)
*** = P < 0.001 (Significant)

Copper sulphate had also a declining effect on the testis carbohydrate content in the mature male *H. olivaceous*. Carbohydrate content were 17 ± 1.7 (P < 0.001) SD mg/gram, 15 ± 1.5 (P < 0.001) SD mg/gram, 12 ± 1 (P < 0.001) SD mg/gram and 35 ± 2.5 (P < 0.001) SD mg/gram of ovary at 24 hrs, 48 hrs, 72 hrs. and 168 hrs respectively after copper sulphate treatment, (Table - 1).

Carbohydrate is structural and functional component of all insect tissues. It is found in nuclei, cytoplasm, and cell membrane as well as in extracellular fluid and supporting tissues. Carbohydrate with proteins and lipids form the principal classes of organic compounds that are found in insect and other organisms.

Insects have chitinous exoskeleton which is rich in amino polysaccharides. In addition, insects differ from vertebrates as trehalose, a disaccharide acts as the storage form of glucose in them. The synthesis and hydrolysis of trehalose is a central hub of carbohydrate metabolism of insects. The main site for the synthesis of trehalose from glucose is the fat body.[1] In most insects, Carbohydrates are present as glycogen and trehalogen which can readily be converted into glucose. In addition, various amounts of glycoproteins may be present especially in the haemolymph.

Sugar content of whole insect have been studied in some cases; Egorova and Smolin described a steady rise in trehalose level from egg to pupal stage followed by a decline in the subsequent development to the adult insect in *Antheraea pernyi.* The absorption of glucose through gut wall, its conversion into trehalose in fat cells and finally its distribution in different body tissues were first demonstrated by Kilby using radio isotopes in locust.

In *Bombyx mori*, Nandi reported increased glycogen concentration in the whole body, fat body and gut during the early period of pupal life which then decreased up to the emergence of the adult from the cocoon. In the gonads, the increasing trend of glycogen continued up to the latter half of the pupal life particularly into ovary.

Marked sex wise variation in the carbohydrate level has been reported for *Tribolim castaneum and Coregon cephalonica*[2] *Sphaerodema rusticum*[4], *H. olivaceous*[1], *L. maculatus*. The influence of pesticides and chemicals on the carbohydrate metabolism has been studied in different animals including insects.

In the present investigation on male *Hydrophilous olivaceous* it was observed that the copper sulphate had an initial carbohydrate lowering effect and latter elevating trend on the haemolymph, fat body, and testis of male insects. Overall initial decline of the carbohydrate content in haemolymph, fat body, ovary and testes were observed in insects exposed to sub-lethal concentrations of copper sulphate.

The carbohydrate content in the haemolymph, fat body and gonads of 25 ± 2 SD mg/gram in testes in control insects. It was 20 ± 2 SD mg/gram in testes at 24 hrs of treatment.

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Scenario of bottom fauna of Gaya pond, Bihar, India

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Abstract: This research paper represents the concept of bottom fauna of Gaya pond Bihar. Bottom fauna were collected with the help of mushroom shaped scooping bottom samples. The bottom contents will be identified in living condition and quantitative estimation has been done by using of sample of constant volume of bottom. The total volume of organisms has been determined by standard displacement method. Zoo plankton, macro invertebrate and fishes were studied. It is seen that zooplankton community showed its peak during February and December. There was variation in the dominance of Rotifers and Copepods. Its population declined during May and represented by copepods during January and represented by various species. Altogether many species of zooplanktons have been reported from pond out of which Protozoa, Rotifer, Copepoda, Ostracoda are important. However, the information gathered from two years studies may provide insight into the ways and means of conservation and propagation of certain commercially important fishes and other organisms from pond to augment regular supply of raw materials for food and other commercial purposes so that cottage industries may run properly contributing sustainable economic wealth and giving employment to thousands of people Gaya region. Besides this, exploitation of pond resources through scientific methods on sustainable basis may provide an idea about the biology of pond ecosystem.

Key words: Freshwater organisms, bottom fauna, Arthropoda, Mollusca, Protozoa, Rotifera, Gaya.

INTRODUCTION

Bottom fauna is a group of mutually adjusted living organisms in habiting in a particular area. Odum (1963)[1] has described it is an assemblage of population living in a prescribed area, habitat and it is an organized unit to the extent that it has characteristics additional to it individual and population components and functions as unit through coupled metabolic transformation. Thus, benthic community has specific trophic structure and energy flow pattern with functional and compositional unity. Benthic organisms provided valuable indication of past and present water quality conditions because of their long life history and central position in the food chain Cairns (1997) claims that diversity is the numerical expression that can be used to make comparison between communities and it is the best single means, for assessing biological integrity in fresh water streams. The communities of organisms living on the bottom of a water body form an important link in the food web of fishes apart from their biological role in the mud water exchange of nutrients. A number of studies have

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indicated that production of benthos was dependent upon the primary production. Never the less, species composition and dominance of benthic invertebrates were not related to the trophic status in some lakes at New Zealand, measured the total benthic production by calculating g/m^2 per year (dry weight) individual genera production such as *Chaoborus sp* and *Hexagenia* were also analyzed individually by applying the same method. There are evidences to suggest that seasonal changes of benthic invertebrates are profound. In some water bodies a maximum peak was observed in monsoon whereas most of the organisms occurred in maximum densities during the period of January through April in other water bodies showed an unimodel pattern of distribution during the year with a single broad peak occurring from January to March and single through during autumn (September to October). Further distribution of benthic invertebrates in a pond system was found to depend upon their morphometery. However, Michael showed the dominance of molluses in the shore zone and oligochaetes in the middle zone of a pond. Gupta observed greater abundance of molluscs in shallower depths, whereas the oligochaetes were observed at little bit shallower regions.

Special attention has been given to the study of community structure of different water bodies in India. Important contributions are those of Jana (1973)[3] who studied seasonal variation in plankton of a pond in West Bengal. Many have studied the benthic community of different water bodies of India.[4,5] Sharma have studied molluscan fauna of Bhagalpur district. However, Bilgrami and Munshi (1985)[6] have studied the benthic fauna of river Ganga.

MATERIALS & METHODS

The surface and subsurface water sample for physicochemical analysis were collected from sampler once in a month. The collection time is generally in the morning hour. Surface sampling were performed at two different sites were pooled together to get an average for each month of the year. For this purpose bottom fauna was collected with the help of Mushroom shaped scooping bottom sampler. The bottom content was identified in living conditions and quantitative estimation has been done by using sample of constant volume (200cc) of bottom. The total volume of organisms was determined by standard displacement method. Samples were collected in quarterly period during (2017-2019) at every two m. depth. Upto 6 m using Ekman's dredge and taking three candomhavels. The special type of hand grab sampler following the method described by Jhingran *et al.* (1969)[7]. Sediment samples, after collection were passed through a sieve (No.40) using liberal quantities of water in order to separate animals from the detritus. The bottom fauna was preserved in 10% formalin and then enumerated and identified.

RESULT AND DISCUSSION

The Gaya pond is qualitatively and quantitatively rich in flora and fauna. The phytoplankton diversity, Periphyton diversity, Macrophytic vegetation and fish population is well known. However benthic macro-invertebrate comprised *Melania striatella tuberculata, Bellamya bengalensis, Lymnaea piscicida, Indoplanorbis exustus, Gyraulus hirsutus, Unio* sp. and *Lamellidens marginalis* amongst Mollusca (*Chironomus* sp) Diptera; *Ephemerella* sp. Ephemeroptera (*Laccotrephes maculatus*) Hemiptera (*Hydracarima* sp) and (*Tubifex tubifex*) Annelids *Chironomu s*sp. were throughout dominant in almost all seasons in both the sites A & B.

Altogether thirteen species of macro-invertebrate were recorded from the pond. Annelids were represented by two families that is Tubificidae and Nainidae. Three species were collected belonging to the family Tubificidae and five species were from Nainidae. *Chaoborus* is the most important group among all the benthic invertebrates collected from Site B of Gaya pond. Pond Oligochaete was represented by *Tubifex tubitex.* However, Ostracod was mainly represented by *Cypris indica*. The benthic fauna recorded were of two types that is filter feeder and deposit feeders. The filter feeders lie *Unio*, clam and other bivalves in absence of their mouth depend on the water flow carrying small organisms which can pass through the gill sieve system. The deposit feeders like snails (gastropods) with the help of their mouth consume deposited organic matter from the bottom. Freshwater mud crab was abundantly collected from bottom from bank of pond. The huge populations of mud crabs are collected for the trade and for eating purposes by the local people.

Altogether thirteen species of macro invertebrates were recorded from pond they belong to three major groups among which arthropods were dominant. Eleven molluscs

species, annelids species, were observed. The population of Annelids shows increasing trends in winter and decreasing trend in summer.

Molluscs were represented in the pond only by Gastropods. Altogether fourteen species of gastropods were recorded. The population was encountered more of gastropods were recorded. The population was encountered more in monsoon followed by winter (Table -1,3).

The community structures of benthic macro-invertebrate were recorded during summers. Bottom biota included organisms belonging to Trichoptera, Diptera, Hemiptera, Zygoptera and Mollusca. Population of the bottom biota are listed as Table 1,2 & 3.

Table 1. Bottom fauna molluses collected from Gaya pond.

Sl. no.	Gastropods
1.	*Bellamya bengalensis* (Lamarck)
2.	*Bellamya bengalensis* F. typical (Lamarck)
3.	*Bellamya dissimilis* (Muller)
4.	*Pila globosa* (Swainson)
5.	*Thiara scabra* (Muller)
6.	*Thiara tubercidate* (Muller)
7.	*Thiara lineate* (Gray)
8.	*Broia costala* (Rafinesque)
9.	*Lymnaea accuminata*
10.	*Lymnaea accuminata* & *rufescens* (Gray)
11.	*Indoplinorbis exustus* (Deshayes)
12.	*Lamellidens marginolis* (Lamarck)
13.	*Lamellidens corrianus*
14.	*Parreysia favidens*

Table 2.Macro benithic biodiversity of Gaya pond.

Species	Site- A	Site- B
Melania striatella tuberculata	+	+
Bellamya bengalensis	+	+
Indoplanorbis exustus	-	+
Lymnaea piscicida	-	-
Lamellidens marginalis	+	-
Gyraulus hirsutus	-	-
Unio sp.	-	+
Tubifex tubifex	+	+
Chironomus sp.	+	-
Ephemerella sp.	-	-
Laccotrephes maculatus	-	-
Hydra carcina sp.	-	-
Paratelphusa spinigera (Mud crab)	+	+

Table 3.Seasonal variation in the occurrence of zooplankton at Gaya pond.

Species	Winter	Summer	Monsoon
PROTOZOA			
Arcella disco ides	+	+	-
Centropyxis aculeata	-	++	+
ROTIFERA			
Polyarthra sp. Platyias	+	-	-
Platylus monostyla sp.	-	+	-
Bracbionus forficula	-	+	-
Filinia longiseta	-	-	+
Pompbolyxsulcata	-	+	-
COPEPODA			
Phyllodiaptoimus sp.	-	-	+
Spicodiaptovius sp.	-	-	+
Mesocyclops sp.	-	-	
Cyclops sp.	-	+	++
*Diaptomus*sp.	-	-	+
Bbinediaptomus indicus	+	-	-
Nauplius larvae	++	+	++
CLADOCERA			
Diapbanosomaexcisum	++	+	++
OSTRACDA			
*Cypris*sp.	-	-	+

CONCLUSION

It is seen that water of Gaya pond is clear and semitransparent. During the present investigation pH was recorded slightly acidic which fluctuates between 5.5 and 7.5. The presence of rich biodiversity which constitute the zooplankton, Macrophytes along with macro invertebrates that is Molluscs as well as fish population represent significant potentially of biodiversity. However, zooplankton showed maximum density during summer as also observed by Michael in freshwater ponds. Summer increase may be favourable water condition and high phytoplanktonic density upon which they depend for their food. Among macro invertebrates arthropods were the most dominant group followed by molluscs and annelids as also reported by Singh and Roy (1992)[8] in the pond. Nutrients are the major biotic component influencing density and diversity of aquatic plants and animals. The nutrients absorbed by plants are utilized by themselves and are also transferred to the other trophic levels. Thus, bottom biota included organisms belonging to different orders, classes and phyla of the animal kingdom, which are beneficial for

the proper growth of fishes and it is the essential for our society. A good bottom is one that provides correct soil from the point of view of pond productivity. The flora and fauna are useful for fish feeding.

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Role of aquatic insect diversity in monitoring water pollution in Hatma pond, Morabadi, Ranchi, Jharkhand

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Abstract: Aquatic insects are very versatile biotic components of water bodies in the lithosphere. They play important role in monitoring water pollution. Unfortunately, very little attention has been given to these creatures in understanding their vital role related to the dynamics of water bodies. In general, these are helpful in management of health and water quality through their qualitative and quantitative existence in terms of species richness and abundance which are the integral components of species diversity. In other words species diversity of these aquatic insects has been found to be directly related with the health and hygiene of water bodies. In the present paper, the results of statistical computation of species richness and abundance have been inferentially discussed by presence absence (+,-) checkerboard of eight aquatic insects with eleven physic-chemical of the perennial water body called Hatma pond situated behind Ranchi College Campus Building presently known as Dr. Shyama Prasad Mukherjee University, Morabadi, Ranchi, Jharkhand (India).

Keywords: Hatma Pond, aquatic insects, richness, abundance, pollution monitoring

INTRODUCTION

Water is one among the prime necessities of life required for growth and activity of all living beings on globe. Only small amount of water that occurs in fresh water rivers, streams, lakes and tanks is available for the terrestrial life. Insects are the most diverse group of organisms in freshwater. Insects being an important biotic component of aquatic ecosystems are very abundant and diverse group that inhabits a variety of water bodies.[1] Aquatic insects are therefore used as bioindicators frequently helpful in biological assessment of water quality. The number and type of species of aquatic insects present in certain water sampled the quality of the water body site from which they have been collected (Metcalfe, 1989)[2]. Some studies reported that aquatic insects are very good bioindicators in detecting water quality and useful degree of anthropogenic disturbance (Clarke *et al.* 2002)[3].

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Most of the water bodies around the world are gradually drying out due to over exploitation. The situation is particularly aggravated by the high level of pollution resulting from various human activities which has further led to destruction or degradation of the aquatic ecosystem by various exotic species.

The use of aquatic insects for biomonitoring the health of aquatic ecosystems has many advantages. Aquatic macro invertebrates constitute an important component of an aquatic ecosystem and they exhibit differential tolerance to changes in environmental condition. They include

insects, annelids, oligochaetes, crustaceans, molluscs and gastropods. Of all these groups macro invertebrates the number and diversity of aquatic insects are greater than other animals that share the same habitat with them. The distribution of most aquatic insects is nevertheless influenced by several biotic and physico-chemical factors in both lentic and lotic ecosystem.

MATERIALS AND METHODS

Study area and Sampling site

Hatma Pond (23°23'27"N 85°19'13"E) located behind Ranchi College campus, Morabadi, Ranchi district of Jharkhand. The diverse eco-systems of Hatma Pond support various aquatic species. Aquatic insects and water samples in were collected seasonally during 2018-19 by different methods such as "all out search" method, "a nylon pond net" method etc.

Aquatic insects were identified using insect identification books.[4-6] All aquatic insects were identified until family level.

Sampling procedures

Aquatic insects were sampled once in every fortnight during the study period using a stratified random sampling design. The reservoir was divided into four parts: which were east, west, north & south. Several methods were used to collect the aquatic insects which include hand picking technique, stone rubbing technique etc.

Table 1:- Observation on physico chemical parameters of water samples from three stations of Hatma Pond, Morabadi, Ranchi, Jharkhand

Sl. No.	Water Parameters	Sampling Station- I	Sampling Station- II	Sampling Station- III
1.	Temperature	28.84±0.212	24.15±0.021	24.46±0.006
2.	pH	6.38±0.035	5.47±0.015	6.46±0.040
3.	Dissolved oxygen (mg/L)	5.62±0.015	7.98±0.006	8.16±0.030
4.	Turbidity (NTU)	2.15±0.285	3.35±0.249	2.18±0.017
5.	Biological oxygen demand(mg/L)	1.63±0.015	1.01±0.012	2.37±0.200
6.	Conductivity (ms/cm)	0.036±0.000	0.030±0.000	0.022±0.000
7.	Total dissolved oxygen (mg/L)	0.026±0.000	0.019±0.000	0.020±0.000
8.	Total suspended solid (mg/L)	0.005±0.005	0.006±0.005	2.31±1.995
9.	Velocity (s)	17.16±0.763	45.33±9.386	25.90±2.128
10.	Depth (m)	0.314±0.036	0.691±0.102	0.472±0.033
11.	Width (m)	0.161±0.005	0.638±0.024	0.390±0.015

Table 2: Seasonal presence absence checkerboard of different species of aquatic insects in Hatma Pond

Sl.No.	Scientific name	Common name	Family	Order	Status			
					Summer	Rainy	Winter	Autumn
1	*Gerris tristan*	Water strider	Gerridae	Hemiptera	+	+	+	+
2	*Metrocoris indicus*	Water siphoner	Gerridae	Hemiptera	+	+	+	+
3	*Onychotrechus rhexenor*	Water cricket	Gerridae	Hemiptera	-	-	+	-
4	*Hydrometra greeni*	Water stick	Hydrometrideae	Hemiptera	+	+	-	-
5	*Notonecta glauca*	Back swimmers	Notonectidae	Hemiptera	+	+	+	+
6	*Nepa cinerea*	Water scorpion	Nepidae	Hemiptera	-	+	-	+
7	*Hydrophilus olivaceous*	Water beetle	Hydrophilidae	Coleoptera	-	-	+	-



Fig. 1. ***Gerris tristan*** **(Gerridae)**
Water strider/ Pond Skater



Fig. 2. ***Metrocoris indicus*****(Gerridae)**
Water siphoner



Fig. 3 ***Onychotrechus rhexenor*** **(Gerridae)**
Water cricket



Fig.4. ***Hydrometra greeni*** **(Hydrometrideae)**
Water stick



Fig. 5. ***Notonecta glauca*** **(Notonectidae)**
Back swimmers



Fig. 6. ***Nepa cinerea*** **(Nepidae)**
Water Scorpion



Fig. 7. ***Hydrophilus olivaceous*** **(Hydrophilidae)**
Water Beetle

DISCUSSION

Observation table 2 reflects that species no. 1, 2 & 5 are very abundant & effective in all the seasons to act as pollution indicator with respect to their predatory and feeding functions in the pond. Species no. 3 & 7 is represented only in winter season hence exhibiting low profile participation in the control of the pollution of water body, Taxonomically family Gereidae represented by three different species is the most efficient group in pollution monitoring of the water body.

CONCLUSION

The study revealed the presence of 7 species of aquatic insects, 6 belonging to 4 different families of order Hemiptera and 1 belonging to family Hydrophilidae of order Coleoptera (Table 2). In summer season 4 species belonging same order Hemiptera were encountered while 5 species of Hemiptera were also recorded in monsoon season. In winter also, 5 species were recorded and those were of two orders i.e., Hemiptera & Coleoptera.

Aquatic insects are important for biological control. These insects play an important role in transmission of human and animal diseases. They play an important ecological role in keeping freshwater ecosystems functioning properly. Therefore ecological study on aquatic insects is essential and can provide information about ecology of insects in an area for any decision making policy for the management of water body. In recent years the conservation of natural resources and biodiversity has become urgent issue. For attaining an environmentally sustainable future we should conserve natural resources. The improvement and development of existing and new biomonitoring tools using aquatic insects are a major effort among aquatic entomologists. Hence a long term monitoring programme and use of diversity and biotic indices may throw light on the management strategies of the fresh water bodies that can also influence government policies of conservation.

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Prevalence of *Escherichia coli* in urinary tract infections in the people of Hajipur (rural area)

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Abstract: *Escherichia coli* (*E. coli*) is one of the most common pathogen causing 60 to 90% of Urinary tract infection. UTI is one of the health problems in the rural area in all ages and both sexes. This study focused on the prevalence of *E.coli* in different gender and age presenting with signs and symptoms of UTI in the rural area of Hajipur (Bihar). In 82 urine samples 55 from females and 27 from males, the found cases showed that *E.coli* was prevalent in UTI of the both females and males patients. It was significantly occurred and more frequently in females (66%) than males (34%). It was found that the *E.coli* and then *Klebsiella* sp. was the most common gram negative and *Streptococcus* was the most frequent gram positive isolates. There was significant increase in the UTI of *E.coli*, *Klebsiella*, *Staphylococcus*, *Pseudomons* and *Proteus* spp; was found more in females than males due to anatomical, poor hygiene and sanitary conditions in females.All urine sample was- fulfill the criteria for significant bacteriuria (>10^5 colony-forming units (CFU)/ ml.)

Keywords: *Escherichia coli*, urinary tract infection, females and males, Hajipur

INTRODUCTION

Urinary tract infection (UTI) is broadly defined as an infection of the urinary system, and may involve the lower urinary tract or both the lower and upper urinary tracts.[1] Anatomically, UTI can be classified into lower urinary tract infection involving the bladder and urethra and upper urinary tract infection involving the kidney, pelvis and ureter. The majority of the UTI occurs due to ascending infection.[2, 3] The infections can affect several parts of the urinary tract, but the most common type is a bladder infection (cystitis).

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UTIs are among the most common bacterial infections in humans both in the females and males, and they occurin all age groups. Most common cause of urinary tract infections is Escherichia coli, which uses panoply of virulence factors to cling to epithelial cells and cause inflammation. Other frequently encountered urinary pathogens are *Klebsiella*, *Staphylococcus*, *Pseudomonas* spp and *Proteus* spp., *E. coli* often gains entry into the urinary tract via stool. Females are more prone to UTIs than males because their urethra is much shorter and closer to the anus,[4] giving the bacteria easier access to the bladder, where the majority of UTI occur.

Urinary tract infections may be asymptomatic, symptomatic, acute and chronic in both male and females.

Asymptomatic infection can be diagnosed by culture. Asymptomatic bacteriuria may be defined as the "presence of actively multiplying bacteria within the urinary tract excluding the distal urethra", at a time when the patient has no urinary tract symptoms.[5,6] The definition of a symptomatic UTI generally requires the presence of urinary tract-specific symptoms in the setting of significant bacteriuria with a quantitative count of e"105 colony forming units of bacteria per milliliter (CFU/ml) in one urine specimen.[6]

Bacteriuria is defined as the presence of bacteria in urine.Acute UTI is more frequently seen in female of all ages; these patients are usually treated on an outpatient basis and are rarely admitted to hospital. Chronic UTI in both males and females of all ages is usually associated with underlying diseases (e.g. pyelonephritis, prostatic disease, or congenital anomalies of the genitourinary tract) and these are most often hospitalized.

Against this background, as well as the paucity of reports of urinary tract infections in our rural community (surrounding Hajipur), this study was undertaken.The aim of this study is to determine the prevalence of *E.coli* in urinary tract infection in rural community as well as the effect of gender and age on its prevalence.

MATERIALS & METHODS

This study was conducted in some neighboring rural communities (villages) Hajipur. After giving questionnaires about their lifestyle two hundred and sixty-seven (267) morning clean catch midstream urine samples were randomly collected from peoples of different rural areas in Hajipur.

Significant bacteriuria was observed in 82/267 (30.8%). A "clean catch" midstream urine sample was collected from each patient into sterile screw–capped universal container, containing few crystals of boric acid as preservative. The specimen was mixed, labeled and transported to the laboratory for processing.

A total of 82 patient's sample from said rural areas with signs and symptoms of UTI. They consisted of 55 females & 27 from males with age ranging from 18 to 70 years. All urine sample was- fulfill the criteria for significant bacteriuria (>10^5 cfu/ml), sample inoculation, bacterial isolation and identification, was performed by standard methods, using by Vandepitte *et al*. (2003) Cheesbrough (2002).[7,4] Identify the bacteria by colony morphology and biochemical test.[4]

RESULTS & DISCUSSION

A total of 82/267 (30.8%) samples 55 from females and 27 from males cases, *Escherichia coli* strain were 50 (60.9%) among all isolated bacteria with UTI during the study period (Table 1).This is followed by *Klebsiella* spp (14.6%) *Staphylococcus* (12.1%), *Pseudomonas* spp (8.5%) and *Proteus* spp (3.6%). Isolated bacterial distribution from females and males cases was, females 33 (66%) and males 17 (34%). (Table 1) It was found of that *E. coli* was the most prevalent in UTI of both male and female patients. It was significantly occurred and more frequently in females (70.9%) than males (40.7%). (Table.1).

The prevalence of UTI was highest within the 28-37 years age group (30.9%), and was least above the 68 years (6.27%) in females but in male highest in above 68 years age group and least within the 18- 27 years age group. (Table 2) .

E. coli often gains entry into the urinary tract via stool. Women are particularly at risk for UTIs because their urethra sits close to the anus, where *E. coli* is present. It's also shorter than a man's, giving the bacteria easier access to the bladder, where the majority of UTIs occur, and the rest of the urinary tract.In the very old and the very young females, symptoms may be vague or non-specific.[8] There was significant increase in UTI with increasing age of male patients due to prostatic hypertrophy causing obstruction to the normal flow of urine leads to high UTI.

In this study evaluation was conducted on bacterial distribution of UTIs isolates. *E. coli* (60.9%) was the prevalent isolate isolated from the urine specimen followed by *Klebsiella*, *Staphylococcus*, *Pseudomonas* and *Proteus* spp. *E.coli* was higher in all isolates, and females are more affected than males. Our result is further supported by different authors .[9, 10] Similar pattern was seen in study done by other studies.The second most common uropathogen in this study was *Klebsiella*, followed by *Pseudomonas* and *Proteus*spp.[11]

A similar percentage of *E.coli* has been obtained in Bettiah (Bihar)[12] 59.26%, Rohtas (Bihar)[13] 64.57%, Nepal (55%),[14]. But a different percentage of *E.coli* has been reported in Kathmandu Valley[15] (49%).

There was significant increase in UTI was higher in females than males due to poor hygiene and sanitary conditions in females and also due to the shortness of the female urethra. Malnutrition, poor hygiene, low socio-economic statuses are associated with urinary tract infections and these factors are usually found in rural settings.[16]

CONCLUSION

To sum up, *E. coli*is most prevalent uropathogen in the cases of both female and male patients, but it is more prevalent in cases of female due to anatomical structure i.e, the shortness of the female urethra. Hygiene & sanitary condition are also responsible for causing more cases of *E.coli,* so in order to manage these infections the hygiene and sanitary issues should be addressed sincerely. From this, we suggest that Gram stain of centrifuged urine is a very sensitive screening method to detect bacteriuria.

Table 1. Isolated bacterial distribution from UTI among females and males.

Isolated bacteria	No. of isolates and %age	No. of isolates & % in Female	No. of isolates & % in Male
E.coli	50 (60.9)	39 (70.9)	11 (40.7)
Klebsiella spp	12 (14.6)	8 (14.6)	4 (14.8)
Staphylococcus	10 (12.19)	7 (12.7)	3 (11.1)
Pseudomonas spp	7 (8.5)	5 (9.09)	2 (7.4)
Proteus spp	3 (3.6)	2 (3.6)	1 (3.7)

Table-2 Distribution of UTI patients according to age group and gender

Age groups in year	No. of females infected (%)	No. of males infected (%)
18-27	9 (16.3%)	0
28-37	17 (30.9%)	0
38-47	12 (21.8%)	2 (7.4%)
48-57	7 (12.7%)	4 (14.8%)
58-67	6 (10.9%)	9 (33.4%)
>68	4 (7.27%)	12 (44.5%)

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Effect of mixture of *Eichhornia, Ipomoea* and *Acorus* leaf powder on adult mortality of *Rhizopertha dominica (fab)* infesting rice in different ratio 2:1:2

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Abstract*:* - Mixture of plant products like *Eichhornia, Ipomoea* and *Acorus* leaf powder in different ratio 2:1:2 Showed remarkable effect on Mortality, Survival & growth of adult *R.dominica* in store rice. Rice was weighted 100 gm. separately taken in 6" x 2" glass container. Standard *Eichhornia, Ipomoea and Acorus* leaves powder where taken from laboratory stock in 2:1:2 ratio & mixed with grains separately & shaken for 15 minutes. Five pairs of fleshly emerged test insect were also taken & released into each replicate. All experiments were set at same temperature and relative humidity 60-70% with the equal degree of ventilation. Six doses (0.5gm,1gm,1.5gm,2gm, 2.5gm & 3gm) such treatment while there replicates with untreated grains were installed for control. The observation showed on adults was remarkable effect in Mortality & delayed emergence.

Key words: - ***Rhizopertha dominica*** **(F.), Adults mortality, Petroleum ether extract,** ***Eichhornia, Ipomoea*** **and** ***Acorus*** **leaf.**

INTRODUCTION

The present study is confined to "comparative study of effectiveness of some plant products as protectant against *R. dominica* infesting rice on its adult, larval and egg population."

A lot of work has been done on the insect infestation in different food grains, such as wheat, jawar, maize, pulses. But very little or no work has been found about the effectiveness of *Eichhornia, Ipomoea*, and *Acorus* were used as pesticides and their effect were observed.

Recent years have seen a rapid advancement in the field of storage of food grain. There is a lack of common storage production even in given agricultural area. Comprehensive principle of storage practices have to be developed keeping a view on its eco-friendly nature & human health. The practice of use of plants products as protectants against certain pest may certainly help the farmers immensely.

Men have created problems for themselves by interfering in the nature & disturbing the ecosystem. To maximize the yield of crop, men have used several devices to control the out break of pests. Indians have fallen serious victims to pesticides poisoning. In India, There are two law to regulate the pesticides in food. *"preservation of food Alternation Act. 1954 and Insecticide Act 1968*." It has appreciably contributed towards improving general economic conditions. However, there is growing awareness among planners, Manufactures & researchers about

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introducing newer & safer Methods of pest control with greater pest specificity in order to keep the environment free from pollution hazard.

Yield of crop depends not only on the genetic heritage but also on susceptibility of their natural pest. However Rao (1983)[1] reported heavy loss of crops due to pest in storage. The loss of food grains in storage have been estimated to be about 6.8% in our country by Dr. Pansi committee appointed in 1966 by Govt. of India. The maximum loss in storage is caused by insects 2.55%, by rodent 2.50% & by birds 0.85% according Ghosh MR (1989).[2]

Yadav (1983)[3] observed that green gram, cow pea, Bengal gram, horse gram and pea are common hosts of *C.maculatus* & rice, wheat, Maize are common host of *R.dominica* in India.

To minimize the loss and heavy use of agrochemical are in practice. The effect of these chemicals is well known as there is world wide debate on discoursing such use. India recognized this problem early as in 1967 in "*Thakar Report"* A high level of DDT & BHC residue have been detected in sample of wheat, pulses, & bean, due to direct mixing of these pesticides during storage (Report of ICAR special committee on harmful effects of pesticide). Analysis of cereal, pulse, milk, egg, meat, vegetable, showed that more than 50% of sample tested contained pesticide residue and in more than 30% residue exceeded tolerance limit prescribed by WHO.

The present work is being undertaken to examine use of *Eichhornia*, *Ipomoea and Acorus* leaves powder on most common coleopteron pest – *R. dominica,* & find out the ideal plant products to check loss of grains through mortality of attacking pest.

Oryza Sativa is the latin name, In English called *rice*, In Hindi called *chaval, Dhan*. It belongs to family *Gramineae*.It is mainly used as food but it is also used for straw board, paper, mats from its straw. Rice bran oil is used in soap, cosmetics, anti corrosion oil. Rice husk is used as fuel, soil conditioner, animal feed & preparing activated carbon, sodium silicate & silicon. Our country is a major producer of rice. According to Government of India, about $1/10^{th}$ of total production of rice in our country is lost due to faulty storage. Besides, the infestation of pest affect adversely economy of country and health of people.

According to Kaura *et al.* (1972)[4] :- Insect infestation caused loss of weight in variety of storage grains in Egypt. The average weight loss was 32.64% in rice.

***Eichhornia crassipes* :-** Its common name is water hyacinth. It belongs to family Araceae. It can be used in making compost, fuel, paper, for generating CH_4 & removing nutrients & toxic chemicals from water. Also observed in reduction of metallic pollution lead from water in Orissa[5]. Petroleum ether extracts of water hyacinth was evaluated for its biological activity against pests. It retarded development and causes mortality high . The extract did not showed any toxicity on *S.orizae*.[6]

Ipomoea cornea:- It is universal in distribution and more abundant in warm regions. Roots are advantageous and haustorial penetrating into most tissue. It is cultivated as hedge plant. It has large genus with 400 species. Several species are used in medicines on account of latex.

***Acorus calamus*:-**It is herbaceous flag like plant commonly called *sweat flag* in English & *safe buch or ghora buch* in Hindi. It belongs to family *Araceae*., mainly used as medicines & insecticides. *Asarone* (Obtained from rhizome of *A. calamus*) were tested as insecticides against an egg & adult of *C. chinensis* reduced by 20-60% (Panji HR *et al.*1998)[7]

Rhizopertha dominca:- Commonly called lesser grain borer or Australian wheat weevil. It belongs to Class *Insecta*, order- *Coleoptera* & family *Bostrichidae* Both larvae & adults form are destructive in nature.

MATERIALS & METHOD

1. Procurement of plant products food grains & insects.

Eichhornia is abundantly found in Bihar a type of water weeds. Leaves were collected & dried in sunlight for 7 days continuously further grinding for power. Leaves of *Ipomoea* were thoroughly washed in running water & dried in 8 hours at 60^0 C. Then grinding further over dried for 24 hours at 50^0 C. *Acorus* leaves were obtained from pond. They were kept in glass container for 3 days 40-50^0C. Rice was collected from agricultural farm Pusa, Muz. and considered free from bacterial, fungus, infestation.

After verification, Rice was kept in incubator to check the moisture content. Temp. of incubator was 30-40^0 C.

2. Treatment with plant product :-

Rice were weighted 100 gm. separately & taken in same size glass. Leaves powder of plant products were taken in ratio 2E:1I:2A. They were mixed with test grains separately & shaken for 15 minutes from laboratory culture. Five pairs fleshly emerged test insects were taken and released into each replicates. Besides, three replications were installed for control. All replication were set at same temp. & relative humidity with equal degree of ventilation. Room temp. was 30+2^{0}C & humidity 60-70%. Out of these ratios,the different ratio (0.5gm, 1gm, 1.5gm, 2gm, 2.5gm, 3gm) were measured & used for treatment.

OBSERVATIONS & RESULTS

Mixture of *Eichhornia, Ipomoea, Acorus* in 2:1:2 ratio used & result showed that ratio was quite effective in no. of adult weevils.

0.5gm. of this mixture eliminated 33% weevils within 24 hours. But addition of 3% mortality was noticed on 3rd days, On 4th days mortality rate was 53%, On 5th days 60% & On the 10th days ware 66%.

1gm. dose pushed the mortality rate ahead 40% on 2nd days, 56% on 3rd days & 4th day, 60% on 5th days, & 80% on 10th days. Only 20% weevils were surviving but most inactive conditions.

1.5gm. treatment showed the mortality 66% on 3rd days, 73% on 4th days, 80% on 5th days and 100% on 10th days.

Bigger doses of mixture had phenomenal effect on weevils 60% insect died on 2nd days. The rate further moved to absolute mortality in sequence of 73% , 90%, & 100% on 3rd, 4th & 5th days respectively.

2.5gm. mixture had a considerable toxic effect on adult weevils & showed 100% mortality on 4th days. The highest dose of 3gm.had killed 100% on 3rd days.

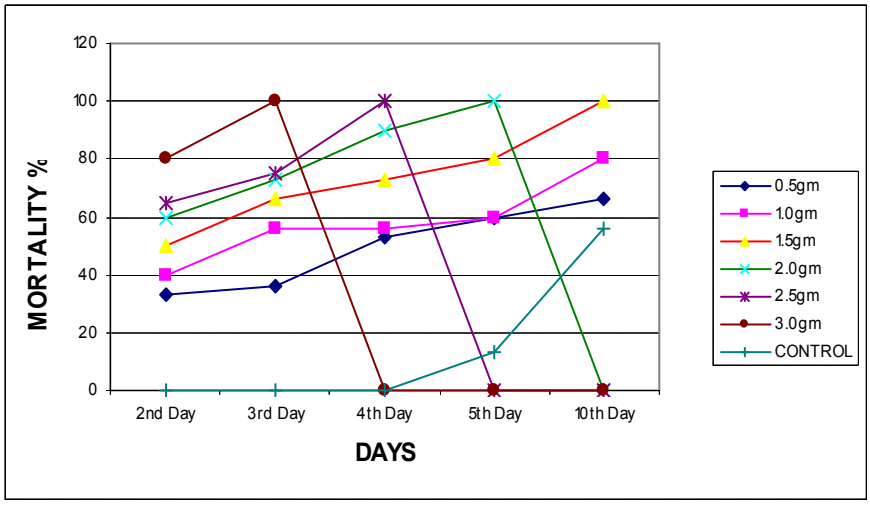
The control mortality was noticed on 5th day at 13% reaching up to 56% on 10th day.

Result showed that quicker more mortality depends upon doses and duration. The mixture of plant products in ratio 2:1:2 has strong insecticidal effect & bigger dose caused quicker death.

Effect of mixture of *Eichhornea, Ipomoea & Acorus* leaf powder on *R. dominica*. Infesting rice in different ratio (2:1:2)

DOSE	ADULT MORTALITY %				
DURATION	2nd day	3rd day	4th day	5th day	10th day
0.5gm	33	36	53	60	66
1gm	40	56	56	60	80
1.5gm	50	66	73	80	100
2gm	60	73	90	100	——
2.5gm	65	75	100	——	——
3gm	80	100	——	——	——
CONTROL	——	——	——	13	56

Effect of mixture of *Eichhornia*, *Ipomoea* and *Acorus* leaf powder on the percentage mortality of *R.dominica* infesting rice in different ratio (2:1:2)



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Effect of phosphate concentration on Sporulation of Blue-green Algae

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Received : 20th December, 2019 ; Revised : 17th February, 2020

Abstract: Five local isolates of Blue-green Algae, *Nostoc commune, N. linckia, Anabaena circinalis, Anabaena macrospora, Anabaena lapponica* were selected to assess the effect of Phosphate cane on Akinete formation. It was observed that increases in Phosphate conc. Reduced Akinate formation.

Key words: Akinetes, ***Anabaena circinalis, A. macrospora, A. lapponica, Nostoc commune, N. linckia.***

INTRODUCTION

Blue green Algae are prokaryotic in nature. Sexual reproduction is totally absent in them, however genetic recombination has been reported.[1] They reproduce vegetative and asexually. Most of the unicellular and colonial forms reproduce by cell division. In members of Oscillatoriaceae hormogonia are produced, which are the main source of reproduction.[2,3] In Nostocaeceae and Rivulariaceae Kinetes are the principal means of reproduction. Akinetes are produced adjacent to Heterocyst, Akinetes are thick walled large spores, with two layered wall.[4,5] Outer wall is thick, yellow or brown in colour. Akinetes are resistant to temperature and desiccation. Akinete formation is dependent on several factors like light quality, pH, Nitrogen & Phosphorous. Dificiency of Ca, Mg, Fe and S led to decrease in number of Akinetes. Increased concentration of Nitrogen enhances akinete formation while increased concentration of Phosphorus decreases akinete formation.[6,7]

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MATERIAL & METHODS

Blue-green Algae collected from local area were identified on the basis of their morphology with the help of Standard monograph (Desikachari T.V.). Five local strains were selected to assess the effect of Phosphate concentration on their sporulation, selected strains were *Nostoc commune, Nostoc linckia, Anabaena circinalis, Anabaena macrospora* and *Anabaena lapponica.*

Each strain was cultured in BG II medium. Axenic culture was prepared from each strain. To elevate effect of Phosphate concentration 10 different concentration of Dispotassium hydrogen phosphate was prepared (1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9% and 10%) in BG 11 medium. One ml. of each strain was inoculated in each concentration. BG II medium without Dipotassium hydrogen phosphate was considered as control. After 15 days akinete frequency was calculated as percent of total cell population under light microscope at least 1000 cells were counted for each reading.

RESULT

Number of akinetes increased in the control in which Phosphate was absent, while the number of akinetes

decreased by increasing conc. of Phosphate. In a conc. of Phosphate above 4% no any akinete was produced in both species of *Nostoc*.

In *Anabaena circinalis* No. of akinetes were equal to control in 1% Phosphate concentration but increased conc. of Phosphate showed decrease in number of akinetes. At a conc. of 7% no any akinete produced. Akinete formation in *Anabaena macrospora* and *Anabaena lapponica* gradually decreased of 6% and above there was no any akinete development.

Table 1- Name of species and no. of Akinetes in different Phosphate conc. of after 15 days

Sl. No.	Name of species	No. of akinetes	No. of akinetes in different Phosphate conc. of after 15 days									
		in control after 15 days	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1.	*Nostoc commune*	8	6	4	3	2	0	0	0	0	0	0
2.	*Nostoc linckia*	7	5	3	2	1	0	0	0	0	0	0
3.	*Anabaena macrospora*	11	9	7	5	4	2	0	0	0	0	0
4.	*Anabaena circinalis*	9	9	7	6	5	3	2	0	0	0	0
5.	*Anabaena lapponica*	7	6	4	3	3	2	1	0	0	0	0

CONCLUSION

Phosphate concentration is an important factor in sporulation of Blue-green Algae. As the conc. of Phosphate increases sporulation is decreased. In higher conc. of Phosphate sporulation is completely blocked.

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The occurrence of *Alternaria triticina* associated with infected *Triticum aestivum* L. from fields of the Madhepura, Bihar

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Received : 12th December, 2019 ; Revised : 11th February, 2020

Abstract: Wheat (*Triticum aestivum)* is an important food crop of Asia and is grown on more land area than any other food crop. Yields can be negatively influenced by various factors, including disease, particularly those caused by fungi which create problems in both production and storage. Foliar diseases of wheat such as *Alternaria* leaf blight may cause significant yield losses, with reductions in plant health and seed quality. The disease is caused by a fungal pathogen and causes necrotic leaf lesions and in severe cases shriveled. The work reported here determined the incidence of *Alternaria* species infecting wheat seeds grown in the Madhepura, Bihar. This work confirms the identity of a major source of wheat leaf blight in Bihar which will aid in formulating effective disease management strategies.

Key words: *Triticum aestivum*, *Alternaria triticina*, yields, occurence, infection

INTRODUCTION

Wheat (*Triticum aestivum* L.) is one of the oldest and most important food crops, widely cultivated for its seed, a cereal grain which is a worldwide staple food. According to a survey it is the second most-produced cereals throughout the world after maize. World trade in wheat is greater than for all other crops combined.[1] Cultivation and repeated harvesting and sowing of the grains of wild grasses led to the creation of domestic demands, as mutant forms of wheat were preferentially chosen by farmers. In domesticated wheat, grains are larger, and the seeds remain attached to the ear by a toughened spine during harvesting.[2] In traditional agricultural systems wheat populations maintain high levels of morphological diversity. Major breeding objectives include high grain yield, good quality, disease and insect resistance and tolerance to abiotic stresses, including mineral, moisture and heat tolerance. The major diseases in temperate environments include the foliar diseases.

Numerous microorganisms, especially fungi, pose a challenge to both wheat production and storage. Wheat agriculture faces numerous disease problems, root rot, leaf blight, leaf spot etc. However, wheat is also susceptible to leaf blight caused by *Alternaria triticina*. Leaf diseases of wheat, such as leaf blight diseases caused by *Alternaria triticina* result in substantial loss in yield and deterioration in quality and vigor of seed. Leaf blight of wheat via *Alternaria triticina* is "one of the most important foliar diseases of wheat in India".[3]

Alternaria triticina is the causal agent of *Alternaria* leaf blight of wheat, initially described from India. It can cause major damage to susceptible wheat varieties under

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wet or humid conditions. This is one of several species in the genus that have been isolated from wheat leaves; it is demonstrated to be pathogenic, while others appear to be primarily saprophytes. The species have been reported from different hosts and different places also but its presence is mainly reported from south and southwestern Asia.[4,5]

Symptoms of Alternaria leaf blight include the formation of round to irregular spots of up to 10 mm diameter. Alternaria species are dispersed from region to region through various pathways. The aim of the work described in this paper was to determine the *Alternaria* species associated with wheat seeds grown in the Madhepura, Bihar as an example of the source of yield losses affecting production of this important crop in South Asia. To the best of our knowledge, no such studies have been carried out till date.

MATERIALS & METHODS

Symptoms like lesions (with either perithecia or without the dark pycnidia), burnt appearances occur on the leaf, leaf sheath, awns & glumes. These symptoms comprise of the initial stage of detection of the disease. When the fields are heavily infected, they can be spotted in form of burnt appearances from a distance (Prabhu & Prasada, 1966)[6]. For final confirmation in the seed, the agar plate is used (Mathur & kongsdal, 2003)[7].

Diagnosis

A. triticina is isolated by surface sterilization of infected plant parts like, stems, leaves or seeds using the methods & keys of Simmons (2007)[4] the isolates should be transferred into a carrot agar medium & then observed in alternating day & night cycle.

Differences in *A. triticina* & other *Alternaria* species residing in the leaves of infected can be done by PCR or other DNA sequencing methods. The culture filtrates of *A. triticina* were found to be toxic to wheat seedlings. The sequences of ITS region of rDNA of these isolates are preserved in Gen Bank.

Pathogenicity test

A standard nutrient agar was made using a mixture of 1.3g K_2HPO_4, 1.06g Na_2CO_3, 5g $MgSO_4.47H_2O$, 5g of dextrose, 1g asparagine & 20g agar in1 liter of distilled water. This mixture is thought to be best for sporulation (Prabhu & Prasada 1966), though its time is dependent on temperature & humidity. When temperature is high, conidiophores appear single or in bundles of 2-10.

For inoculation, plants are sprayed with water containing thick conidial suspension & are then covered with plastic bags to maintain high humidity for first 24 hours. These plants are then placed in a humid chamber for next 36 hours. After 7 days of inoculation, the symptoms appear. For the disease to appear a concentration of 40,000 spores/me is essential. Plants were observed for either the presence or absence of irregularly shaped spots from dark brown to gray in colour.

RESULT

The disease severity (D.S) was calculated using revised rating system.

0 = no symptoms
1 = 0-25 % infection of leaves
2 = 25-50 % infection of leaves
3 = 50-75 % infection of leaves
4 = 75-100 % infection of leaves (plant is dead)
Total no. of plants = 40

Zhao *et al*. rating	No. of Plants
0	2
1	12
2	5
3	15
4	6

CONCLUSION

It was found that only small specks occur in a resistant cultivar, pointing to the facts that increase in the age of plant; increase the susceptibility & density of the disease. Disease presents in the seeds of cultivar, can become a source of disease outbreak in the entire field, if the environment at that instant is favorable for it. An infected seed not only gives rise to an infected plant, but also infects the soil & the infection seeps through the entire cultivar. Such infection can lead to crop failure. By deciding the infection or disease severity rate, one can decide whether or not to keep the crops for the entire season.

If the disease severity is between 0-1, one can take appropriate measures to control the infection. However if the DS rate 2-4, the chances of crop failure are very high. In such circumstances process & get rid of the entire

infected crop by methods like burning. Early method of disease detection can always save money as well as man power.

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Seed germination behaviour and medicinal uses of *Boerhaavia diffusa* Linn. in adjoining areas of Patna (Bihar)

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Abstract: *Boerhaavia diffusa* Linn. (Purnarnawa Sag) is distributed in subtropical and tropical regions of the world. Mostly species of *Boerhaavia* are adapted in base dry sandy soils. The initiation period of seed germination may take place in 1 to 2 weeks. The vegetative growth of *Boerhaavia diffusa* Linn. was found to increase with the application of NAA, IAA, GA_3 and NPK mixture. People use this part of leafy vegetable in their daily diet. It contains a large number of compounds as flavonoids, alkaloids, steroids, triterpenoids, lipids, lignins, carbohydrates, proteins etc. It is rich in vitamins and minerals. It is nutritionally and medicinal used in daily diet on the basis of traditional knowledge. The plant resources are also used against various diseases like oedema, heart problems, low blood pressure, anaemia, urinary disorders, abdominal pain, asthma, cholera, cough, constipation, fever, gonorrhea, jaundice, kidney disorders etc.

Key words: ***Boerhaavia diffusa*** **Linn. habitats, Seed germination behaviour and Medicinal uses.**

INTRODUCTION

Boerhaavia is a genus of 40 species, almost all of which are widely distributed in subropical and tropical regions of the world.[1,2] The plants are usually spreading in bare sandy areas, side of the canal, railways, road, ditches etc. The seed germination takes place in the first rain and continues to the rainy season. The best average temperature to grow the plant is 18-24^{0}C (64-75^{0}F). The physical factors are soil, moisture, range of temperature and light intensity. Leafy vegetable are very rich in vitamins and minerals which consists of important chemical compounds flavonoids, alkaloids, steroids, triterpenoids, lipids, lignins, carbohydrates, proteins and glycoproteins, ursolic acid etc. These vegetables are consumed by the tribal and indigenous persons in their daily diet.[3] Commonly this vegetable is called 'Purnarnawa' but locally it is called as 'khapara sag'. Many of these plants are used as medicines against different kinds of diseases like as anemia, asthma, blood purification, cancer, child birth, cholera, edema etc.[4]

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MATERIAL AND METHODS

The field study was conducted at an agricultural farm of Mokamah and Diara land of Hajipur. The soil analysis of Mokamah and Diara land of Hajipur was done[5] and reported in Table-1.

Table-1. Soil Analysis Features of Mokamah and Diara Land of Hajipur

Farm	Soil Type	Soil Colour	Moisture Content	Organic Carbon	pH
Mokamah, Patna (Bihar)	Gangetic Alluvium	Light Grey	11.5	0.70	7.20
Hajipur, Vaishali (Bihar)	Sandy Loam	Light	10.5	0.47	7.35

In the month of October and November each year the matured and ripped seeds were collected from the plants of *Boerhaavia diffusa* Linn. and dried in natural sun-light. Stock solution of 100ppm each of Naphthalene, acetic acid and IAA, 200 ppm of GA were prepared in glass redistilled water. The soaked seeds were also sown in the field according to the concentration of each hormone and NPK mixture to study the growth, flower initiation, flowering, fruiting, seed production (Fig.).



Fig. 1- Morphological Stage of *Boerhaavia diffusa*

RESULT AND DISCUSSION

There are important chemical compounds are found in root, stem and leaves as flavonoids, alkaloids, steroids, triterpenoids, lipids, lignins, carbohydrates, proteins, potassium nitrate, ursoilic acid, punarnavoside[6], boeravinone A-F[7], Kaemferol[8], boeravinone G. and H.[9], ursolic acid[10] etc.

The *Boerhaavia diffusa* Linn. is used in medicine as abdominal pain, anemia, asthma, blood purification, cancer, childbirth, cholera, constipation, cough, digestive sluggishness, dropsy, edema, eye problems, fever, gonorrhea, heart ailments, heart disease, internal parasites, jaundice, kidney disorders, kidney stones, lactation aid, liver disorders, liver support, menstrual disorders, renal insufficiency, snake bitting, spleen (enlarged), urinary disorders, weakness etc.[11]

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Effect of callus initiation of *Murraya koenigii* Spreng. of different concentrations of 2,4-D, IBA and IAA

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Received : 15th December, 2019 ; Revised : 19th February, 2020

Abstract: Different concentration of phytohormones affected callus formation of *Murraya koenigii* was cultured in MS medium supplemented with different concentration of IAA, IBA. Medium supplemented with various phytohormones for organogenesis culture were kept on 25+2° temperature and 16 hr photoperiod white callus was observed on different concentration of auxin along in combination. Most suitable medium for callus formation form shoot tips was that MS+2,4-D (0.3mg/l) + IBA (1.5 mg/lt. And Ms+2, 4-D (3.0mg /l) + IAA (1.5mg/l).

Key words: *Murraya koenigii,* organogenesis, auxin, IAA, IBM

INTRODUCTION

Murraya koenigii (L.) spreng is known as miracle plant, belongs to family Rutaceae. The main stem is green to brownish, with numerous dots on it. The girth of the main stem is 16 c.m. it is a multifunctional Ayurvedic herb[1] the essential oils form the leaves of *Murraya koenigii* (Spreng), *Murraya exotica* (Linn)[2,3]. Diversity in Australia populations of *Murraya paniculata* (Rutaceae). On the basis of differences in habitat, habit leaf morphology, pedicel length and volatile leaf oils are present[4]. Among 14 global species belongs to the genus of *Murraya* only *Murraya koenigii* (Spreng) and *Murraya Paniculata* (Linn) Jack is available in India.[5]

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Taxonomic Status

Kingdom	:	Plantea
Sub kingdom	:	Tracheobionta
Division	:	Magnoliophyta
Class	:	Magnoliopsida
Sub Class	:	Rosidae
Order	:	Sapindales
Family	:	Rutaceal
Genus	:	*Murraya*
Species	:	*koenigii*

Vernacular names in Indian languages.

Bengali	:	Barsunga
Gujarati	:	Mitho Lindo
Hindi	:	Meetha Neem, Karipatha Kathnim, Bursunga
Kannada	:	Karibevu
Marathi	:	Karhinimb, Jhirang

Oreya	:	Bansago
Malyalam	:	Kariveppilei, kareapela, karipat, Karhi

MATERIAL & METHODS

Green young explants *Murraya koenigii* were collected from our garden. They were washed with running top water and 1-2 drop of savlon for 2 minutes, they were surface sterilized in 70%, ethanol for 30 seconds and immersed in 0.1% Hgcl for 1/2 minutes. Then rinsed with autoclaved distilled water (5 times). Small pats of plant were inoculated in test tubes containing Ms basal medium.

Culture Medium

Solid Ms Medium containing 4% sucrose with varying concentration of IBA and IAA. The pH of the media was adjusted 5.8 before gelling with agar (0.8% w/ V) and autoclaved. The small parts of plant were inoculated into incubated culture room.

Culture condition

Callus were incubated and 25+ 2°C under cool fluorescent light (1500-2000 Lus) with 16/h/ 8h / light dark cycle. Each treatment consisted of minimum 100 explants and all experiments were repeated at least 5 times.

RESULTS & DISCUSSION

In this investigation explants taken from young plans gave better response callus was obtained from the shoot tip of the plant parts of *Murraya koenigii* Spreng. The best callus growth from shoot tip of *Murraya koenigii* Spreng was observed in (Ms+2,4-D+-0.3 mg/l+IBA 1.5 mg/l) Here 80% callus induction was observed. The best callus growth from MS+2,4. D (3.0mg/l) +IAA 1.5 mg/l) Here 60% callus induction was observed. As tissue culture technologies may provide methods for large scale propagation of tree species.[6,7,8] This work was under taken to establish efficient and reproducible methods for *in- vitro* mass propagation of these species from mature trees. Regeneration from adult materials will be useful since the phenotype and desirability of the selected plants are known.

OBSERVATION

Callus induction of *Murraya koenigii* Spreng. was achieved.

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Table 1- Effect of deferent concentrations of 2,4-D with IAA, IBA incorporated in MS medium showing callus induction from shoot tip of *Murraya koenigii* Spreng.

Concentration of Phytohormones in .mg/l	% culture showing nature of callus after 5 weeks
MS+2,4-D (0.1 mg/l + IBA (0.5mg/l)	27
MS+2,4-D)0.2mg/l+IBA (1.0mg/l	55
MS+2,4-D (0.3mg/l) + IBA (1.5mg/l	**80**
MS+2,4-D (0.4mg/l + IBA (2.0mg/l)	60
MS+2,4-D(0.5mg/l) +IBA (2.5mg/l)	45
MS+2,4-D(1.0mg/l) +IAA (0.5mg/l)	20
MS+2,4—D (2.0mg/l+ IAA (1.0mg(/l)	40
MS+2,4-D (3.0mg/l + IAA (1.5mg/l)	**60**
MS+2,4-D (4.0mg/l + IAA (2.0mg/l)	55
MS+2,4-D (5.0mg/l +IAA (2.5mg/l)	42

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Effect of 9-aminoacridine hydrochloride hemihydrate on the aerobic degradation of sucrose to citric acid by *Aspergillus candidus* NCIM-883

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Abstract: Citric acid is the most important commercial product which is produced by using various sugar substrates in the terrestrial environment. The present study is an attempt to show the effect of 9-aminoacridine hydrochloride hemihydrate on the degradation of sucrose to citric acid by the fungal strain *Aspergillus candidus*. The aerobic degradation of sucrose to citric acid can decrease sucrose yield and affect end product quality. It has been found that mutagen of 9-aminoacridine hydrochloride hemihydrates has stimulatory effect on the novel method of citric acid production by *Aspergillus candidus* NCIM-883 and yield of citric acid to some extent.

Key words: 9-aminoacridine hydrochloride hemihydrates, mutagen, aerobic degradation, sucrose, Citric acid, *A.candidus*

INTRODUCTION

Citric acid has high economic potential because of its numerous applications. It is usually produced by microbial fermentation using *Aspergillus candidus*. There is always a need for the discovery and development of better production techniques and solutions to improve production yields and the efficiency of product recovery due to its higher demand and growing markets.[1] For its enormous scale of production it is necessary and important for the production process to be environmental friendly by utilizing readily available and inexpensive agro-industrial waste products at the same time high production yields also. 9-aminoacridine hydrochloride is a hydrochloride salt resulting from the reaction of equimolar amounts of 9-aminoacridine and hydrogen chloride.[2,3] It has a role as an anti infective agent, an antiseptic drug and a mutagen. It contains a 9-aminoacridine (1+). This article reviews the biochemistry of citric acid formation, fermentation strategies, the effects of various fermentation conditions.[4]

In general prediction of mutagenic activity cannot be made on the basis of chemical structure along some compounds. The mutagenic chemical's effect the molar concentration and production of citric acid has been enhanced but order of citric acid productivity is reverse in respect to increasing molecular concentration of 9-aminoacridine hydrochloride hemihydrates.[5] However the aerobic degradation of sugar to citric acid by *Aspergillus candidus* NCIM-883 under the influence of each

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concentration of 9-aminoacridine hydrochloride hemihydrates used has been stimulating and the yield of citric acid has been found greater. The influence of different chemical mutagens in different fermentations process has been investigated by a number of workers. Chemical mutagens are required for genetic manipulation and exploitation especially for citric acid fermentation and in view of this, the influence of 9-aminoacridine hydrochloride hemihydrates on aerobic degradation of sucrose to citric acid by *Aspergillus candidus* NCIM-883 have been studied.[6]

MATERIALS & METHODS

The influence of 9-aminoacridine hydrochloride hemihydrates on aerobic degradation of sucrose to citric acid by *Aspergillus candidus* NCIM-883. For the degradation of sucrose to citric acid by *Aspergillus candidus* NCIM-883, a composition of the production medium was prepared with following components, Sucrose-1485gm (15%), $MgSO_4.7H_2O$-34.65 gm (0.35%), pH-2.2, NH_4NO_3-24.75gm (0.25%), KH_2PO_4-24.75g (0.25%). The pH of the production medium was adjusted to 2.2 by adding requisite amount of KCl-HCI buffer solution and pH was determined by a pH meter. A total of 10 such fermented flasks were prepared. After preparing the fermentor flasks, the 9-aminoacridine hydrochloride hemihydrates were prepared from the mutagenic solution and were added to mutagenic chemical. Now the total volume in each fermentor flasks was made up to 100 ml by adding required amount of distilled water. The fermented flasks were sterilized, cooled, inoculated and incubated at 28^OC and analyzed after 6, 12 and 18 days for sucrose sugar left unfermented and citric acid formed.

RESULTS & DISCUSSION

9-aminoacridine hydrochloride hemihydrates have been found stimulatory for the aerobic degradation of sucrose to citric acid by *Aspergillus candidus* NCIM-883. It is found that 9-aminoacridine hydrochloride hemihydrate influences the citric acid fermentation process in different phases. The main characteristics of the 9-aminoacridine hydrochloride hemihydrates are as follows:-

The molar concentration of 9-aminoacridine hydrochloride hemihydrate influences the yield of citric acid in a regular manner enhancing the yield.

CONCLUSION

From the above findings it is clear that various chemical mutagens and some other mutagenic agents are used to produce mutants. If a microbial population exposed to the effect of mutagens differs in cultural properties then these mutations may be differentiated according to size, shape, structure and colour of the colonies. Mutations of biochemical properties are revealed by means of minimal media containing only salts and carbohydrates. Thus it is concluded that a large number of mutagens have been employed to generate the mutants of different microbes but still there are some chemical mutagens whose influence on citric acid fermentation by species of fungus have not been well studied and established.

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